Where Do Protests Win or Lose Votes? Exposure and Participation in the George Floyd Movement

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Abstract

This paper asks how protests affect voting in the neighborhoods where they take place and in those where their participants live. In the 2020 George Floyd movement, I find that two effects coexist and pull in opposite directions: neighborhoods near protest sites shifted toward the Republican candidate—a backlash—while those where participants lived shifted toward the Democratic candidate—a mobilization. To separate them, I combine 7,920 geocoded protest events with mobile phone location data and measure for each block group in American metropolitan areas its exposure to nearby protests and its residents' participation. I then exploit differences in exposure and participation across block groups within the same zip code to estimate the effect of each on the change in vote shares between the 2016 and 2020 presidential elections. The backlash decays with distance and its point estimates are larger near violent protests, while the mobilization comes mostly from participation in nonviolent protests. Both results hold in a difference-in-differences design that follows the same neighborhoods across the presidential elections from 2012 to 2024, with no detectable pre-trend before the protests. Taken at face value, my estimates imply that participation moved about 137,000 votes toward the Democrats across the sample, while exposure within three kilometers of protest sites cost them about 87,000; adding the effect of exposure farther away, the net is not distinguishable from zero.

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1 Introduction

In democracies, protests are an important way to express political preferences outside of elections. The protests that followed the murder of George Floyd in May 2020 were among the largest such expressions in the history of the United States: according to the New York Times, between 15 and 26 million people took part in their first weeks (Buchanan et al., 2020). Whether protests on this scale also change how people vote is less clear, and the evidence points in different directions. Madestam et al. (2013) find that the Tea Party rallies of 2009 won votes for the Republican Party in the following election. The violent protests of the 1960s cost the civil rights movement votes (Wasow, 2020). Gethin and Pons (2026) find that, of fourteen waves of protest in the United States since 2017, only the George Floyd protests moved votes. How can these contradictory results be reconciled? Do protests win votes in some settings and lose them in others, or can both happen within the same movement? And if they coexist, which of the two effects prevails?

Most existing studies estimate the effect of protests on votes by comparing counties or other large areas with more and less protest activity. At that level, the votes that a movement wins cannot be isolated from those that it loses and only their sum can be observed. I address this challenge by assembling a novel dataset that combines the geocoded location of protests with smartphone location data, which allows me to identify the neighborhoods where their participants live. With these data, I treat protests as spatial events that reach the residents of two kinds of neighborhoods: those where they take place, whether the residents want it or not, and those where their participants come from. Exposure to a protest nearby might generate a backlash against the movement, or sympathy for its cause. Participation in a protest might generate mobilization for the movement, or wear down support for its cause.

This paper estimates how the George Floyd protests of 2020 affected voting in the neighborhoods where they took place and in those where their participants lived. I measure exposure to and participation in these protests for each census block group in the metropolitan areas of the United States. Because the protests spread across many venues within the same city, I can compare the change in vote shares between the 2016 and 2020 presidential elections across block groups of the same zip code that differ in exposure or in participation. This comparison nets out the effect of the protests through any channel that is common to all block groups of a zip code, such as the media or party politics, and isolates their direct local effect through exposure and participation. The two effects coexist within this movement: neighborhoods near protest sites shifted toward the Republican candidate—a backlash against the movement—while neighborhoods whose residents attended protests shifted toward the Democratic candidate—a mobilization for it. Their net effect is not distinguishable from zero, and even in the states where the 2020 presidential election was decided by the narrowest margins, the direct local effects of the George Floyd protests fall short of those margins.

The George Floyd protest movement of 2020 is a good setting to separate backlash from mobilization

for four reasons. First, the wave was large and spread out. Between 27 May and 2 November 2020, 7,920 protests took place at 4,815 venues in 368 metropolitan areas, and more than half of the neighborhoods where they took place had seen no protest in the three preceding years. Second, the 2020 presidential election came five months after the protests began, so that a change in opinion could be expressed at the polls. Third, the movement was nationally salient and its cause was the same in every city. Fourth, the cause quickly took a partisan side, so that a reaction to the protests could be traced to a vote for one party or the other.[1]

My dataset is at the level of the census block group, which covers a few city blocks and is the smallest unit for which the Census publishes demographic estimates. The dataset combines three types of information: protest venues, protest participation, and voting in presidential elections. The venues are the locations of the George Floyd protests recorded by the Crowd Counting Consortium, which I geocode to the street address from the media reports behind each entry. Participation is measured from the real-time location pings of 12.9 million mobile devices in 2020: a resident of a block group is counted as attending a protest when their device is observed at a venue while a protest is held there. To my knowledge, this is the first paper to use these data to measure protest participation neighborhood by neighborhood in a months-long movement at the national scale.[2] The voting data are the block-group returns of the presidential elections of 2012, 2016, 2020 and 2024, allocated from precinct returns in proportion to population.

The George Floyd protests did not take place in a few central locations, but at multiple venues within the same city. As a result, many neighborhoods were close to a protest, and the people who attended came from all over the city. I exploit this variation in exposure and participation to study the effects of protests on voting in the neighborhoods that received a protest and in those that sent participants to them. For exposure, I compare block groups that had a protest nearby with block groups of the same zip code that did not. For participation, I compare block groups where protest participants lived with block groups of the same zip code where none did.

When each is estimated in its own regression, both effects are present, depending on whether a neighborhood was exposed to a protest or sent participants to one. Exposure generated backlash against the movement.[3] Block groups with at least one protest within one kilometer of their centroid shifted 0.19 percentage points more toward the Republican candidate than block groups of the same zip code without one, holding fixed whether a block group had protests nearby before the George Floyd wave. The rise comes mainly through a fall in Democratic votes. Near violent protests—those with reported arrests, injuries, property damage or tear gas—the point estimates are larger, as

[1] In June 2020, 67 percent of American adults supported the Black Lives Matter movement, 91 percent of Democrats and 40 percent of Republicans; by September, support among Republicans had fallen to 19 percent while support among Democrats was little changed, at 88 percent (Pew Research Center, 2020a,b).

[2] Van Dijcke and Wright (2020) used device traces to detect where and when these protests took place, not who attended them. Rotman and Shalev (2022) used mobile phone location data to measure participation in a six-week protest campaign in Israel in 2011.

[3] "Backlash" means a shift in the vote toward the Republican candidate rather than a measured change in voters' preferences.

Wasow (2020) found for the protests of the 1960s, and the shift is detectable out to three kilometers rather than two. The backlash was still present in 2024, most clearly near violent protests.

Participation generated mobilization for the movement. Block groups where at least one resident was observed attending a protest shifted 0.07 percentage points more toward the Democratic candidate than block groups of the same zip code that sent none, holding fixed how often their residents visited the protest venues on ordinary days and how mobile their residents were before the George Floyd protest wave. The shift comes mostly from participation in nonviolent protests and from residents who did not regularly visit the venue. Heavier attendance at violent protests, and attendance by residents who regularly visited the venue, moved the vote away from the Democratic candidate instead. This pattern is consistent with a backlash to violence entering through participation as well. The mobilization appears in the extensive margin of participation—whether anyone from a block group went—and the reaction to violence in the intensive margin—how many went. The mobilization effect did not subside by 2024.

The identifying assumption of both designs is that the vote in the block groups near a George Floyd protest, or in those where protest participants lived, would have moved like the rest of their zip code had the protests not taken place, given their demographic characteristics before the wave. The assumption does not require that exposure or participation be random. In fact, neither is: protests do not take place in random locations, and residents choose whether to attend. What it requires is that these block groups were not already on a different voting trend from their neighboring block groups. Two tests support this assumption. First, a placebo exercise separately estimates the effect of exposure and of participation on the change in vote between the 2012 and 2016 presidential elections, before the protests took place: neither coefficient is distinguishable from zero.[4] Second, two difference-in-differences regressions follow the same block groups across the four presidential elections from 2012 to 2024 and ask whether the vote share of the block groups near a protest, or of those where protest participants lived, moved differently from the rest of their zip code in each election relative to 2016. In both regressions the two groups show no detectable pre-trend between 2012 and 2016 and diverge in 2020 and 2024.[5]

While these tests support the design, they cannot rule out that between 2016 and 2020 some confounding factor unrelated to the protests moved the vote of the block groups near protest venues, or of those where protest participants lived, but did not move the vote of the rest of their zip code. For example, the polarization associated with the ongoing realignment of voters by education level might have moved the vote of the block groups where participants in the George Floyd protests come from, independently of the protests (Longuet-Marx, 2026; Kuziemko et al., 2026). The block groups where protest participants lived are indeed more educated than the rest of their zip code,

[4]The test, however, has limited power as the placebo coefficients carry standard errors about half the size of the estimated effects, meaning that a 2012–2016 shift of that size would not have been detected.

[5]The difference-in-differences regressions contain a placebo test of their own: their 2012 coefficient, measured relative to 2016, is a pre-period contrast of the same kind as the placebo test, with the same limited power. What they add is the path of the effect across the four elections in a single regression.

and those near a protest venue less educated, so that the realignment alone would have moved both in the direction of the results. However, the estimates of both the exposure and the participation effects hold fixed the share of residents with a bachelor's degree or higher, on its own and interacted with race, so the shift in the vote that follows that share is netted out. The part of the realignment that these controls do not capture would then have to do several things at once. First, it would have to move toward the Democratic candidate the block groups where participants in nonviolent protests lived, but away from it those that sent more residents to violent protests or whose residents regularly visited the venue, although these are more educated too. Second, it would have to be strongest in the block groups near violent protests, which are no more educated than the rest of their zip code.[6] Third, it would have to be absent between 2012 and 2016, when the realignment was already under way, and present again in 2024. It is unlikely that this explanation holds, because it would have to be so narrowly confined in space and time.[7] I find more plausible the interpretation of this paper: taking part in a nonviolent protest mobilized the neighborhoods that sent participants, while a protest nearby, above all a violent one, pushed its neighbors away from the cause, as did heavier attendance at protests that turned violent.

Estimated together in one regression, with the net margin as the outcome, the backlash and mobilization effects coexist and pull in opposite directions. However, they differ in reach. Per neighborhood, the backlash effect is larger, but it is confined to the relatively small number of neighborhoods within a few kilometers of a protest, while the mobilization effect is smaller but spread across many neighborhoods, because most of them sent people to the George Floyd protests. Their magnitude can be compared with each other, and with other shocks to voting, through their persuasion rates, which I calculate following DellaVigna and Gentzkow (2010).[8] Because this rate accounts for how many residents each treatment reached, the rate of the mobilization effect is at the lower end of the range estimated for media interventions, while the rate of the backlash effect is on the order of a candidate's whole television advertising campaign and below the arrival of Fox News on local cable (Spenkuch and Toniatti, 2018; DellaVigna and Kaplan, 2007).

Participation moved about 137,000 votes toward the Democrats and exposure within three kilometers of a protest site about 87,000 toward the Republicans, both with intervals clear of zero in the main specification. Exposure beyond three kilometers adds 54,000 votes toward the Democrats, but this estimate is not distinguishable from zero. The net of the three—about 104,000 votes toward the Democrats, one tenth of a percent of the votes cast in the sample—is therefore not distinguishable

[6] Against the rest of their zip code, the share of residents with a bachelor's degree or higher is 2.4 percentage points higher in the block groups where protest participants lived, 1.9 for violent protests and 1.4 for regular visitors of the venues. It is 1.3 points lower within one kilometer of a protest and not distinguishable from zero near a violent one.

[7] The alternative explanation that a change in the residents who live near the protest venues drives the results was tested with turnover measured before the wave. The block groups near a venue do have higher residential turnover than the rest of their zip code, but adding Census measures of turnover to the specification leaves the exposure and participation estimates within their confidence intervals.

[8] A persuasion rate is the standard measure for comparing media and campaign interventions that reach different shares of their audience. The rate of the mobilization effect depends on how many residents of a sending block group the protest is assumed to have reached. It is at the lower end of the media range if only the residents detected at a venue were reached, and below the rates of the backlash effect if all residents were.

from zero.[9] Of the six states in the sample where the 2020 presidential election was decided by the narrowest margins, Georgia is the only one where the estimated shift is distinguishable from zero at the 5 percent level and on the scale of the final margin, though short of it.[10]

The George Floyd protest movement differed from recent protest movements in the United States in its high level of violence, which motivates a counterfactual exercise. I simulate what the effects of protest exposure would have been at different levels of violence. Had no protest turned violent, the net shift of the two effects would have been about twice as large as observed. The difference is not distinguishable from zero, but the point estimates suggest that violence may have cost the movement about half of the local gains it would otherwise have made.[11] However, on the point estimates, one protest in four would have had to turn violent for the backlash effect to outweigh the gains of the movement, far above the observed 10.4 percent.[12]

Finally, North Carolina's individual voter files, which record party registration and turnout but not how people voted, show two changes in the electorate near the protests. Near violent protests, the voter roll became less Democratic, a decline significant only at the 10 percent level that comes from new registrants less Democratic than the roll they joined and cannot be separated from the residential turnover of those neighborhoods, while the turnout of those already registered did not detectably change. Near nonviolent protests, turnout fell by 1.5 percentage points within one kilometer. The mobilization effect is estimated too imprecisely in the state to distinguish an effect as large as the national one from zero.

Related literature. This paper contributes to four streams of literature in political economy. First, the paper builds on work on the effects of protests on public opinion, voting and policy (Madestam et al., 2013; Wasow, 2020; Hungerman and Moorthy, 2023; Gethin and Pons, 2026; Klein Teeselink and Melios, 2025; Gillion and Soule, 2018; Gillion, 2020; Reny and Newman, 2021; Caren et al., 2025; Mazumder, 2018; González and Larreboure, 2021; Wallace et al., 2014; at finer levels, Enos et al., 2019; Caprettini et al., 2026; Leon-Ablan and John, 2024). Most of this work operates at coarse geographic levels, like counties or congressional districts, where the effects of protests through mass media, party politics and direct exposure are bundled together. I contribute to this literature by separating the direct effect of protests on voting in the neighborhoods where protests take place and in those where their participants come from.

Three of these studies look at the effect of the George Floyd protests on voting at the county level and find that the protests increased support for the Democratic Party (Gethin and Pons, 2026; Klein Teeselink and Melios, 2025; Caren et al., 2025). My results are complementary to theirs: by

[9] Net of the mean under random placement of the venues (Section 4.7), the 87,000 falls to about 29,000 votes and neither it nor the net of the three is distinguishable from zero.

[10] In Georgia the direct local effects come to about half of the margin, with a 95 percent interval from 16 to 82 percent of it.

[11] The difference falls from 0.10 to 0.07 percentage points if attendance at violent protests also counts as nonviolent, to 0.03 if only exposure within three kilometers counts, and to about zero with both changes.

[12] The 95 percent confidence interval of that crossing runs from −20 to 70 percent and includes the observed share, so the crossing is a landmark on the point path rather than an estimated threshold (Figure 11).

comparing neighborhoods within the same zip code, I separate the backlash in the neighborhoods near a protest from the mobilization in those that sent participants, and show that their sum hides a backlash concentrated near the protest venues. A positive county-level effect beside a net effect within zip codes that is not distinguishable from zero is consistent with the difference being the effect common to all neighborhoods of a zip code, such as through the media and national politics, which my comparison nets out.[13]

Closer to my approach, Caprettini et al. (2026) also treat protests as spatial events and find that Nazi marches in 1932 Hamburg persuaded voters in some neighborhoods along their route while repelling them in others. My paper departs from theirs in three ways. First, I study protests whose organization was largely decentralized, while they focus on marches organized by a party as campaign propaganda. Second, I measure participation at the block group level and estimate its effect on voting, while they study only exposure along the routes of the marches. Third, mine covers hundreds of metropolitan areas in a contemporary setting, while theirs covers one city. To my knowledge, my paper is the first to separate the effects of exposure and participation on voting at the neighborhood level, across thousands of protests and hundreds of metropolitan areas.[14]

Second, the paper speaks to the literature on protest participation and collective action (surveyed by Cantoni et al., 2024). This work has studied the decision to protest as a strategic choice (Cantoni et al., 2019; Chenoweth et al., 2022), the role of social media, mobile phones and social networks in coordinating protests (Enikolopov et al., 2020; Manacorda and Tesei, 2020; González, 2020), and the persistence of engagement after a first protest (Bursztyn et al., 2021). The closest to mine in data is Sonin et al. (2023), who use device traces to profile who traveled to the Capitol on 6 January 2021. In these studies participation is either the outcome or an individual treatment. Instead, I treat participation as a neighborhood-level treatment and estimate its effect on the vote where participants live.

Third, the paper relates to the literature, surveyed by DellaVigna and Gentzkow (2010), studying how media (DellaVigna and Kaplan, 2007; Enikolopov et al., 2011; Adena et al., 2015; Martin and Yurukoglu, 2017; Spenkuch and Toniatti, 2018) and in-person contact (Gerber and Green, 2000; Pons, 2018; Kalla and Broockman, 2018) move votes. I contribute to this stream of work by adding protests as a further local, in-person shock to the vote.

Fourth, this paper adds to the evidence that where people live shapes how they vote, which typically rests on differences in institutions and economic shocks between states, counties or commuting zones (Cantoni and Pons, 2022; Autor et al., 2020). I show that a political event in the neighborhood, like

[13] Gethin and Pons (2026) also find that protests cause no backlash unless they turn violent; within neighborhoods, a backlash appears near nonviolent protests too, with larger point estimates near violent ones. Anderson et al. (2025) find that the nonviolent protests of the 1968 Poor People's Campaign reduced the Democratic vote share and turnout in the South.

[14] Leon-Ablan and John (2024) separate the same two channels for the 2011 London riots, using arrest records rather than measured participation to identify the neighborhoods the rioters came from. Their study compares wards near a riot site with those where charged rioters lived, and finds that both moved the vote in the same direction.

a protest, moves the vote of its residents.[15] The paper also relates to the literature on the urban economics of unrest (DiPasquale and Glaeser, 1998; Collins and Margo, 2007; Brooks et al., 2026), which has studied why riots occur and their economic cost to the cities where they take place. I extend this evidence to the political consequences of unrest in the neighborhoods around protest venues and in those where participants live.

The rest of the paper is organized as follows. Section 2 describes the data and Section 3 where the George Floyd protests took place. Sections 4 and 5 estimate the effects of exposure and of participation. Section 6 puts the two effects together and estimates their net effect on the 2020 presidential election. Section 7 uses North Carolina's voter files to ask through which channel each shift arrived. Section 8 concludes.

2 Data

I put together a dataset at the level of the census block group, which links three sources: the locations and dates of the George Floyd protests, geocoded from the record of the Crowd Counting Consortium; the participation of each block group's residents in those protests, measured from the real-time location data of mobile devices; and the returns of the presidential elections from 2012 to 2024, reported by precinct and allocated to block groups. A block group covers a few city blocks and holds between 600 and 3,000 residents by the Census Bureau's design. This paper uses the terms block group and neighborhood interchangeably. A fourth source, North Carolina's voter files, records who was registered and who voted, separately from the vote itself; Section 7 uses it to explore the channels behind the main effects on vote shares. Appendix A describes how each source was compiled, and Appendix B validates the participation measure.

2.1 Protest events and venues

The protest events come from the Crowd Counting Consortium (Chenoweth et al., 2025). The Consortium finds events mainly in the press and publishes only the records its research assistants can confirm. Each record carries the date, the location, when the reports gave one, the type of action, the organizers, a summary of the participants' claims, the crowd size, and whether arrests, injuries, property damage or the police's use of chemical agents were reported.

I use the Consortium's compiled file for 2017 to 2020. I define the George Floyd wave as the events dated between 27 May and 2 November 2020 whose claims concern Black Lives Matter, racial justice or police violence; counter-protests are excluded. The Consortium's own coordinates locate only the city or town, so I geocode each event's location. I keep the events whose coordinate is precise to a street address (63 percent of the events kept), a named place (11 percent) or a street (26 percent), and drop those located only to a town or a district. The selection, reported step by step

[15] Ang and Tebes (2024) use a similar approach and compare census blocks within their block group to show that a police killing raises registration and turnout among the residents nearby.

in Appendix A.1, leaves 7,920 events at 4,815 venues, in 4,025 block groups of 368 metropolitan areas. The same geocoder, applied to all protests of January 2017 to 26 May 2020 whatever their claims, gives the 36,974 earlier protests that the estimates use to control for the places that had hosted protests before.

An event is violent when the Consortium's record reports any arrest, injury to participants or to police, property damage, or the use of tear gas or other chemical agents by the police. That is 823 of the 7,920 events, 10.4 percent.[16] This classification is used wherever the paper splits protests by violence.

2.2 The device panel and the participation measure

Participation is measured from the real-time location pings of mobile devices in the United States, from 31 December 2019 to 31 December 2020.[17] Each device is homed to the block group where it spent most of its nights in September and October 2020, after the wave's peak and before the election. The panel holds 12.9 million homed devices in 173,544 block groups of the 368 metropolitan areas.[18] Table A1 reports the panel, the attendance it detects and the voting sample. The panel's coverage per resident rises with the white share and falls with the renter share and density (Figure A1). The block groups it covers match the resident population on seven of eleven characteristics (Table A2).

I define a resident of a block group as attending a protest when a device homed there is observed within 200 m of a venue, between noon and midnight local time, on a day that venue hosted a George Floyd protest, counted once per device and day. Presence requires at least two pings inside that radius spanning at least ten minutes, so that a device passing by does not count; pings with a reported location accuracy worse than 100 m are dropped. Summed over the wave, the panel records 385,891 such attendances at 200 m, from 108,679 block groups. The participation measure of a block group has two parts: whether any of its residents attended, the extensive margin of participation, and the number of attendances as a share of the devices homed in the block group, the intensive margin. Residents who were regulars at those venues, present there on days without a protest for work or other reasons, are counted as attending; they are about an eighth of the attendances the measure counts, and Section 5.3 estimates their attendance separately from the others'. The ordinary presence of the block group's residents at the same venues between 4 and 26 May, the three weeks before the first event, on days the venues hosted nothing, enters every participation regression as a control.

[16] Arrests are the most common marker, reported at 555 of the 823 events, and at 263 of them the only one. The markers are not meant to assign the unrest to the crowd or to the police, but to identify the events at which people nearby perceived unrest associated with the protest, as they would when seeing the police make arrests. Section 4.7 reports the estimates with those 263 events recoded as nonviolent.

[17] Attendance at the protests of 2019, and presence at the venues in the same weeks of 2019, come from a separate export of the panel for that year, homed on its own September and October nights.

[18] A device whose owner moved between the wave and September carries its attendance to the new home block group. If the block groups they moved to were already trending toward the Democrats, this overstates the estimate on the extensive margin. The median block group holds 53 homed devices.

The aggregates are built under the disclosure rules stated in the note to Table A1. The participation measure and the ordinary-presence control are period totals, computed before the day-level suppression, which therefore does not affect them. The measure is a relative index of participation across block groups and not a count of participants (Figure A2).

2.3 Election returns and block-group characteristics

The electoral outcomes are the returns of the presidential elections of 2012, 2016, 2020 and 2024 at the block-group level. For 2016, 2020 and 2024 they are precinct returns that I allocate to block groups in proportion to the population of each piece of the intersection of a precinct with the block groups. Appendix A.3 describes the method. The 2016 and 2020 returns are the precinct files of the Voting and Election Science Team (Voting and Election Science Team, 2018, 2020). The 2024 returns are the precinct files of the New York Times (The New York Times, 2025), completed with the Redistricting Data Hub files for four states. The 2012 returns are the presidential results of the panel of Longuet-Marx (2026), which reports them at the block-group level.[19] They cover 42 states. Table A3 reports the coverage by year and source.

The outcome throughout the paper is the change between two elections in a party's share of all votes cast for president in the block group, in percentage points. The exposure sections use the change in the Republican share and the participation sections the change in the Democratic share.

The block-group characteristics come from the 2014–2018 American Community Survey five-year estimates: the shares of residents who are white,[20] Black, Hispanic and Asian, median age, median household income, the unemployment rate, the share of residents aged 25 and over with an associate degree or higher, and the renter share. Population and density are the 2010 census counts. To these I add the distance to the central business district. They enter the regressions as controls and are the dimensions along which I examine whether the effects vary. Every regression also carries the share of residents with a bachelor's degree or higher, on its own and multiplied by each of the four racial shares, to net out the realignment of voters by education discussed in Section 1.

The participation estimates of Section 5 are fitted on the block groups of the 368 metropolitan areas with vote returns in 2012, 2016 and 2020 and a device panel: 135,691 block groups in 333 metropolitan areas, of which 85,399 sent at least one resident to a protest. The sample is urban and leans Democratic: its block groups cast 54.2 percent of their 2020 presidential votes for the Democratic candidate, against 51.3 percent nationally. The ballots cast in the sample in 2020 are 68.5 percent of all ballots cast nationally that year.

[19] Those returns were allocated to block groups through the census blocks inside each precinct rather than by the method of Appendix A.3, so the 2012–2016 placebo compares returns allocated in two ways; the two allocations agree closely (Appendix A.3).

[20] White enters twice: non-Hispanic white, and white of any origin. The interactions use the first.

2.4 The North Carolina voter file

North Carolina's State Board of Elections publishes the state's voter registration file and its voter history file. I use the registration file as it stood on the day of each presidential election from 2012 to 2024. Each record is one registered voter, with a permanent identifier that follows the person across snapshots, the registration status, the party of registration, the residential address, and the person's age, sex, race and ethnicity. The history file lists, for each registered voter, the elections in which a ballot was cast. It begins with the general election of 2016. Neither file records how anyone voted.

A person is on the roll at a snapshot when registered with an active or inactive status. From the two files I build three measures at the block-group level: turnout, the composition of the roll, and the movements that change that composition. Turnout is measured over a window between two elections, for the people who were on the roll at both of them. At each election it is the share of those people who cast a ballot, so the change in turnout is not moved by who joined or left the roll. The composition of the roll is the share of registered voters in each of four classes—Democratic, Republican, unaffiliated, and a residual of the other parties—at each election. The movements are the five ways in which that composition changes between two elections: people newly register, leave the roll, move in from elsewhere in the state, move out, or change the party they are registered with. The five add up exactly to the change in a party's share. Appendix A.4 defines the movements in full and the sorting of voters by their record before the protests.

The North Carolina analysis uses the North Carolina block groups of the national samples: 3,655 in the exposure specification and 3,649 in the participation specification.[21] The individual-level estimates use the 3,606,754 people registered at both the 2016 and the 2020 snapshots, aged 18 to 110 in 2020 and living in 2016 in one of the 3,649 block groups of the participation specification. It is neither the national sample nor a random sample of the state, and block groups outside it are unobserved rather than untreated.

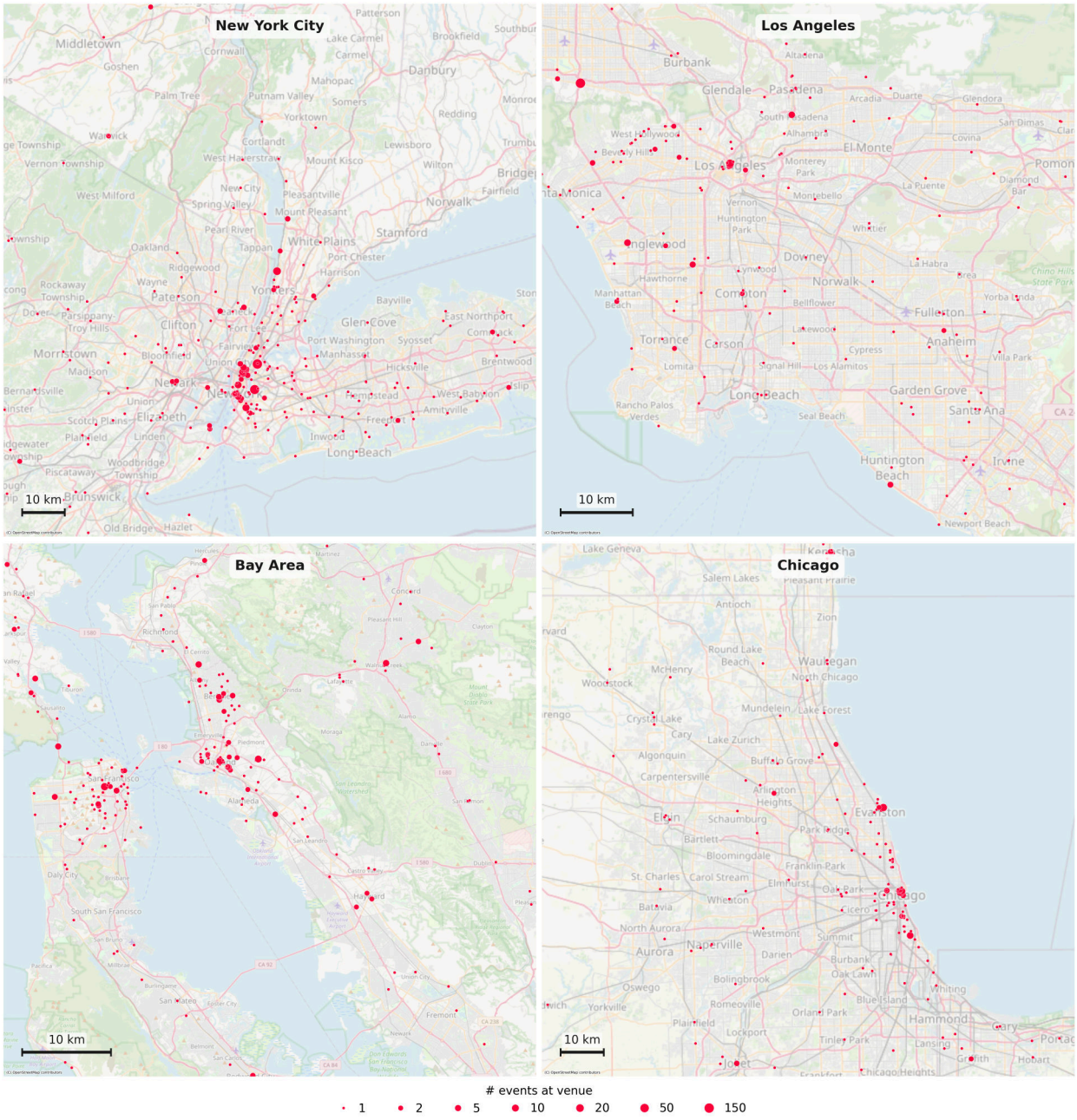
3 Protest Venues and Spatial Fragmentation

The George Floyd protests did not gather in a few central squares. Figure 1 maps the venues of four cities over the wave, with each marker sized by the number of events it hosted. In each city a few large markers stand out and many small ones are scattered through the towns and neighborhoods around them; most venues hosted a single event. Reported attendance was concentrated in the same way: in each of the four cities a single venue accounts for between a tenth and just over a third of the total crowd reported, while the rest of the reported crowd was spread across dozens to hundreds of other venues.

Los Angeles shows the pattern in detail. Its 178 venues hosted 400 events. The busiest venue of the

[21] The state holds 2.7 percent of the national sample, so an effect on the vote share of the national size would not be distinguishable from zero there.

Figure 1: Protest venues by number of events in four cities



Notes: Each panel maps the protest venues of one metropolitan area around its central city. A marker is one venue, sized by the logarithm of the number of George Floyd protest events recorded there. The record holds 291 venues and 660 events in the New York metropolitan area, 178 and 400 in the Los Angeles one, 241 and 377 in the San Francisco Bay Area and 171 and 228 in the Chicago one; each panel shows the city and its surroundings, so some venues of the metropolitan area lie outside the panel. The busiest venue of the Bay Area, with 20 events, is one of those. The venue with the largest total reported crowd over the wave drew 23,200 in the New York metropolitan area, 30,000 in Los Angeles, 13,200 in the Bay Area and 27,024 in Chicago, out of metropolitan totals of 203,463, 119,172, 65,486 and 78,069. The legend below the panels is shared by the four; each panel carries a 10 km scale bar. Basemap © OpenStreetMap contributors.

metropolitan area, with 154 events, is not downtown but in Sherman Oaks, a residential area of the San Fernando Valley; the largest crowd gathered on Hollywood Boulevard, and that single venue accounts for a quarter of the attendance reported in the metropolitan area. Around these two, the map shows single-event venues across the region, from Santa Monica and Pasadena through the South Bay and Long Beach to the cities of Orange County.

The empirical strategy relies on the fact that the protest venues were numerous and scattered across space. Even within the same zip code some block groups had a protest close by and others did not: of the 13,609 block groups with a protest within one kilometer, 92.5 percent share their zip code with a block group that had none at that distance. The same fragmentation shapes participation: 85,399 of the 135,691 block groups of the participation sample sent at least one resident to a protest, and within a zip code protest participants lived in some block groups and not in others, which is the comparison of Section 5.

Table A4 places the George Floyd wave beside the protests of the three preceding years and beside the other protests of the same months. Its 7,920 events took place in 4,025 block groups, 1.97 events per block group against 2.63 for the protests of January 2017 to May 2020 and 1.71 for the other protests of the same months. The mean reported crowd is 464 people, against 831 at the earlier protests. What set the wave apart is violence. The Crowd Counting Consortium codes 10.4 percent of its events as violent, against 2.2 percent of the earlier protests and 2.7 percent of the other protests of the same months. Among the block groups with a protest nearby, some were therefore near a violent one and most were not: 2,494 block groups had a violent protest within one kilometer, and 92.7 percent of them share their zip code with a block group that had none. Of the block groups with a violent event, 59.5 percent had hosted a protest before, against 44.8 percent of all the wave's block groups (Table A5).

The block groups that hosted George Floyd protests also differ in their demographic characteristics. Against both comparison groups, these events took place in block groups that have a higher Black population share, lower median income and a lower share of residents with an associate degree or higher (Table A6). The designs of Sections 4 and 5 do not require them to resemble the others; Sections 4.1 and 5.1 compare the block groups each design treats with the rest of their zip code (Tables 1 and 2).

4 Protest Exposure

This section asks whether the vote of a block group changed when a protest took place nearby. Exposure to protests could in principle move the vote in either direction. Votes moving toward the Republican candidate would suggest a backlash against the movement, while votes moving toward the Democratic candidate would suggest growing sympathy with the movement.

4.1 Specification

The question is whether block groups near a protest venue changed their vote differently, between the 2016 and the 2020 presidential elections, from block groups of the same zip code that were farther away, holding fixed their demographic characteristics and their proximity to earlier protests.[22] Distance to protests is measured in ten rings of one kilometer around the block group's population-weighted centroid (Butts, 2023). A block group is exposed in a ring when at least one George Floyd protest had its venue there, and a block group with protests in several rings is counted in each of them.[23] The average block group of the sample had 16.5 protests within 10 km of its centroid, of which 1.71 were violent.

The estimating equation is

$$\Delta Y_i = \sum_{r=1}^{10} \beta_r \, D_{ir} + \sum_{r=1}^{10} \gamma_r \, P_{ir} + X_i' \delta + \alpha_{z(i)} + \varepsilon_i, \tag{1}$$

where ΔY_i is the change in the Republican share of all votes cast in block group i between the 2016 and 2020 presidential elections, in percentage points; D_{ir} equals one when at least one George Floyd protest had its venue in ring r around the centroid of block group i, ring r spanning the distances from $r-1$ to r kilometers; P_{ir} is the same indicator for the protests of January 2017 to May 2020 and holds fixed whether the block group was already close to a usual protest site; X_i are the demographic controls of Section 2.3, measured before the wave; and $\alpha_{z(i)}$ is a fixed effect for the zip code of block group i. The regression is weighted by votes cast in 2020, and standard errors are clustered by metropolitan area. The coefficients of interest are the β_r: each is the shift in the Republican share associated with a protest in that ring, relative to block groups of the same zip code with no protest in it, holding the other rings fixed. The same equation with the 2012–2016 change as the outcome is the pre-period placebo test: if the block groups near a venue were already moving toward the Republicans before the wave, the β_r of that regression would show it.

The sample is the block groups of the metropolitan areas with at least one protest venue that have vote returns in 2012, 2016 and 2020 and every control, resulting in 136,644 block groups in 344 metropolitan areas and 11,457 zip codes—about twelve block groups per zip code.[24]

The block groups near a protest were not selected at random. The design does not need them to be: the identifying assumption is that, conditional on the controls, they would not have been on a different voting trend from the rest of their zip code without the protests. Table 1 shows that, within their zip code, the block groups with a protest within one kilometer differ from the rest on

[22] Zip codes are the Census Bureau's 2010 zip code tabulation areas. A block group is assigned to the zip code that contains its population-weighted centroid; the 0.4 percent whose centroid falls outside every zip code are excluded.

[23] Under the nearest-protest definition most block groups sit in the first three rings and the outer rings are thin, because a block group with a protest nearby is never counted in an outer ring even when it also has protests there; the definition used here keeps every protest within 10 km.

[24] Even in the outermost ring, three in four exposed block groups share their zip code with an unexposed one, so the comparison within zip codes exists at every distance.

fourteen of fifteen characteristics. Their 2012–2016 change in the Republican and in the Democratic share is not distinguishable from the rest of the zip code's, before or after the controls enter. These comparisons look back before the wave, so they do not speak to a shock of 2020 itself, such as pandemic-era moves or changes in turnout that differed within a zip code.

4.2 Exposure and the vote

Figure 2 plots the ten ring coefficients for exposure to any protest, with the pre-period placebo on the same axes; Table A8 reports the coefficients of the first five rings. Block groups with a protest within one kilometer shifted 0.19 percentage points more toward the Republican candidate than block groups of the same zip code with no protest in that ring, and block groups with a protest between one and two kilometers away shifted 0.17 points more; both are significant at the 1 percent level, and the five inner rings jointly reject zero ($p < 0.01$).[25] Standard errors that allow the errors of block groups up to 50 km apart to be correlated (Conley, 1999) leave every coefficient of the first five rings at the same significance level (the bracketed rows of Table A8). Beyond two kilometers the intervals cover zero, so the shift is detectable out to about two kilometers.

The pre-period placebo, plotted on the same axes, shows no detectable shift within 7 km. Beyond 7 km two of the three placebo coefficients are distinguishable from zero.[26] The block groups near a venue show no detectable pre-trend toward the Republicans before the protests.

The rise in the Republican share comes mainly through a fall in Democratic votes. Table A11 estimates the same specification on four other outcomes. Both the Democratic share and the logarithm of Democratic votes fall in the 0–1 km ring. The logarithm of Republican votes rises by 0.8 percent, significant only at the 10 percent level. Turnout, measured against the voting-age population, falls as well, by 0.4 percentage points, and by 0.8 near violent protests (Table A11, Panel B).[27] Whether the missing Democratic votes are abstentions, a change in who lives in the block group, or voters switching is not identified here; Section 7 asks that question with the North Carolina voter file.

4.3 Exposure by protest violence

Figure 3 repeats the estimation with exposure to a violent protest as the treatment; Table A9 reports the first five rings. The point estimates are larger at every inner ring: 0.34 percentage points within one kilometer, 0.27 between one and two kilometers and 0.16 between two and three, each significant at the 5 percent level. The shift is therefore detectable out to three kilometers rather than two, although the five inner rings are not jointly distinguishable from zero ($p = 0.22$, Table A13). The

[25] Within zip codes, the standard deviation of the 2016–2020 change in the Republican share, weighted by votes, is 2.4 percentage points, so the shift is about a twelfth of it.

[26] Isolated far-ring coefficients distinguishable from zero, in the estimates and in the placebo alike, are not read as effects of the protests, because nothing in the design predicts a shift confined to one far ring. The same holds for the 4–5 km coefficient in Table A10, negative and distinguishable from zero in five of its eight columns.

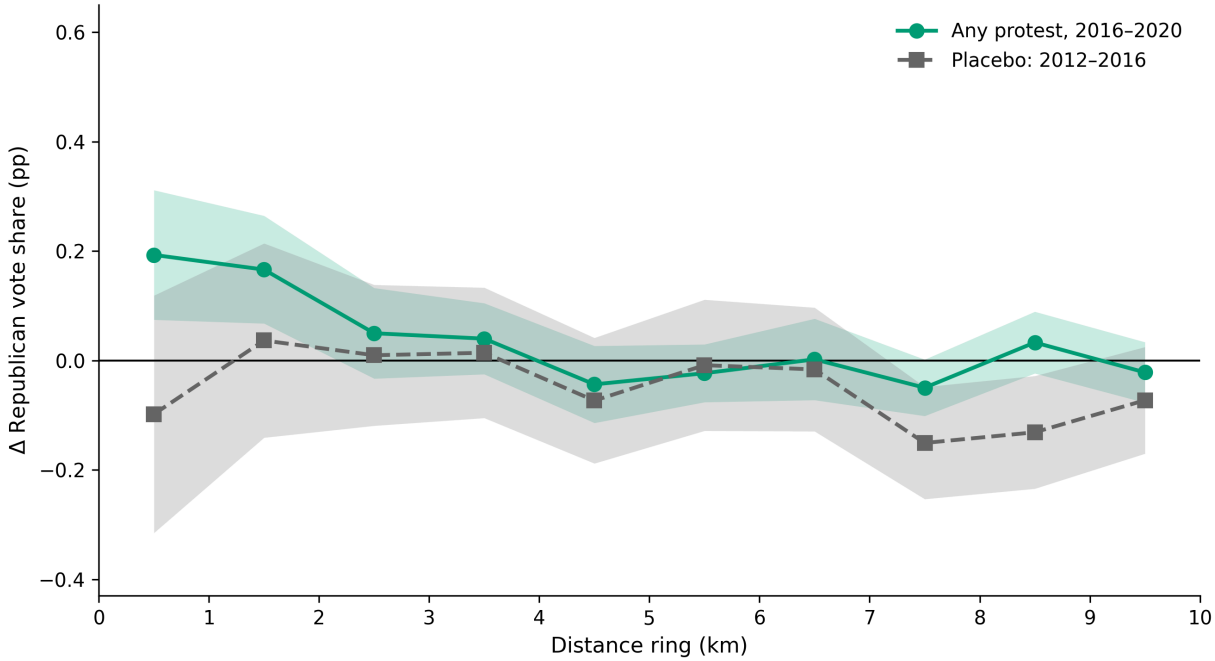
[27] Turnout is the least reliable of the five outcomes: its denominator is a survey estimate of the voting-age population, and the block groups where that estimate puts turnout above 100 percent in either year are excluded.

Table 1: Block groups with a protest within one kilometer against the rest of their zip code

	Exposed mean	Unexposed mean	Diff. (within zip code)
% White	50.9	60.6	-2.7***
			(0.5)
% Black	16.5	12.4	1.0***
			(0.3)
% Hispanic	20.3	18.8	1.6***
			(0.3)
% Asian	8.6	5.2	-0.2
			(0.1)
Median age	37.2	40.2	-2.0***
			(0.2)
Median household income	\$66,851	\$71,451	-\$11,841***
			(\$696)
Unemployment rate (%)	7.1	6.1	0.5***
			(0.1)
% associate or higher	46.4	40.2	-1.6***
			(0.4)
% bachelor's or higher	39.7	31.9	-1.3***
			(0.4)
% Renter	57.1	34.0	12.3***
			(0.8)
Population	1,292	1,539	-157***
			(23)
Density (per km^2)	7,482	2,196	821***
			(147)
Distance to city center (km)	16.4	24.8	-0.6***
			(0.1)
Republican vote share 2016 (%)	23.0	41.7	-3.4***
			(0.4)
Democratic vote share 2016 (%)	71.3	52.7	3.1***
			(0.3)
Δ Republican share 2012–2016 (pre-trend, pp)	**-2.93**	**-1.73**	**-0.10**
			(0.09)
Δ Democratic share 2012–2016 (pre-trend, pp)	**-0.63**	**-2.15**	**0.14**
			(0.09)
Observations	13,609	123,035	136,644
Joint-balance F (within zip code)			24.80
p-value			0.000

Notes: The table compares the block groups with at least one George Floyd protest within one kilometer of their centroid with the other block groups of the exposure estimation sample of Section 4.1, 13,609 against 123,035. Columns 1 and 2 are the unweighted means of each characteristic in the two groups. Column 3 is the within-zip-code difference: the coefficient from a regression of the characteristic on the exposure indicator with zip code fixed effects, weighted by votes cast in 2020, so that every comparison is between block groups of the same zip code, estimated on the 2,590 zip codes that hold both kinds of block group; standard errors in parentheses, clustered by metropolitan area; 344 clusters. The joint F tests the fifteen characteristics above the two pre-trend rows at once. The two pre-trend rows, set apart by a midline, are the 2012–2016 change in the Republican and in the Democratic share, the outcomes of the placebo tests of Sections 4.2 and 5.2. In these two rows column 3 is conditional on the controls: it is the same regression with the ten earlier-protest ring indicators and the demographic controls of equation (1) added, and without the other nine George Floyd rings. With those nine rings in, as in Figure 2's placebo, the Republican row keeps its point estimate and its standard error is 0.11. Units: the shares are percentages of residents (or, for renters, of housing units); median household income is in dollars; population is residents and density residents per square kilometer; distance to city center is the distance from the block group to the central business district, in kilometers; the 2016 vote shares and the two 2012–2016 changes are shares of all votes cast for president, in percent and in percentage points. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure 2: Exposure to any protest and the change in the Republican vote share, by distance ring

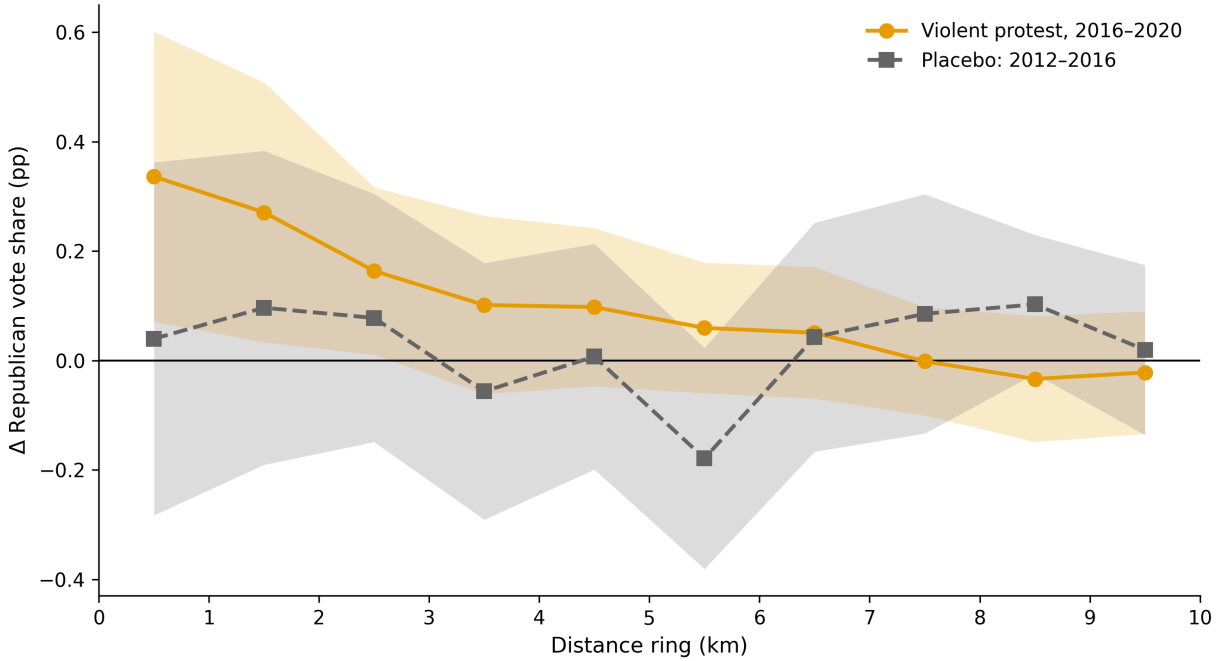


Notes: The figure plots, for each of ten one-kilometer distance rings around a block group, the estimated shift in the Republican vote share associated with at least one protest in that ring. Circles are the coefficients for the 2016–2020 change; squares are the same coefficients with the 2012–2016 change as the outcome, the placebo; shaded bands are 95% confidence intervals; markers sit at the midpoint of each ring. The regression is of the change in the Republican share of all votes cast between the 2016 and 2020 presidential elections, in percentage points, on ten indicators, entered jointly, for at least one George Floyd protest with its venue in each ring, with ten indicators for a protest of January 2017 to May 2020 in each ring, the demographic controls, and zip code fixed effects, so that every comparison is between block groups in the same zip code; the regression is weighted by votes cast in 2020. The demographic controls are population, density, racial composition, median age, median income, the share with an associate degree or higher, unemployment, renter share, distance to the central business district, and the share with a bachelor's degree or higher, on its own and interacted with racial composition. The sample is the block groups of metropolitan areas with a protest venue that have vote returns in 2012, 2016 and 2020; $N = 136{,}644$. Standard errors are clustered by metropolitan area; 344 clusters. The coefficients and standard errors for the first five rings are in Table A8.

same results hold under spatial standard errors (Table A9). The placebo is not distinguishable from zero at the 5 percent level at any ring. The shift is again a fall in Democratic votes: in Panel B of Table A11 the logarithm of Democratic votes falls, and the logarithm of Republican votes is not distinguishable from zero.

The backlash is present for all protests, including nonviolent events. Figure A4 plots the violent and the nonviolent profiles side by side: exposure to nonviolent protests shifts the vote toward the Republicans in the two inner rings, and the estimates for violent protests are larger at every inner ring. The two profiles are not statistically distinguishable at the 5 percent level: estimated in one regression, the joint test that they are equal over the five inner rings gives $p = 0.06$. The violent estimates are less precise, since 2,494 block groups have a violent protest within one kilometer against 12,792 with a nonviolent protest. What the estimates establish is a backlash near protests, with larger point estimates near those that turned violent; the split does not separate violence from

Figure 3: Exposure to a violent protest and the change in the Republican vote share, by distance ring



Notes: The figure repeats Figure 2 with exposure to a violent protest as the treatment: circles are the coefficients on ten one-kilometer ring indicators, each equal to one when at least one violent George Floyd protest had its venue in the ring, for the 2016–2020 change in the Republican vote share, squares the same coefficients for the 2012–2016 change (the placebo), shaded bands 95% confidence intervals. The outcome, the demographic controls, the zip code fixed effects and the weights are those of Figure 2, and the earlier-protest ring indicators are matched to the family: they count violent earlier protests. The sample is that of Figure 2, $N = 136{,}644$. Standard errors are clustered by metropolitan area; 344 clusters. The point estimates are larger than those for any protest at every inner ring; the violent estimates rest on far fewer block groups (2,494 within one kilometer against 13,609) and are less precise, and the difference between the two is not tested. Section 4.3 and Figure A4 compare the violent with the nonviolent profile. The first five rings are in Table A9.

other features of the events that turned violent, such as their size or duration.

4.4 Difference-in-differences approach

A difference-in-differences regression follows the same block groups across four presidential elections, from 2012 to 2024, and asks whether the block groups within 2.5 kilometers of a protest venue moved differently from all other block groups of the same zip code in each election relative to 2016 (Linden and Rockoff, 2008). While the ring design's placebo re-estimates the same equation on the 2012–2016 change, the goal of this specification is to estimate the pre-period inside one regression, so that a pre-trend would appear as a 2012 coefficient different from zero. The regression is estimated separately for exposure to a violent protest and for exposure to a nonviolent protest.

The estimating equation is

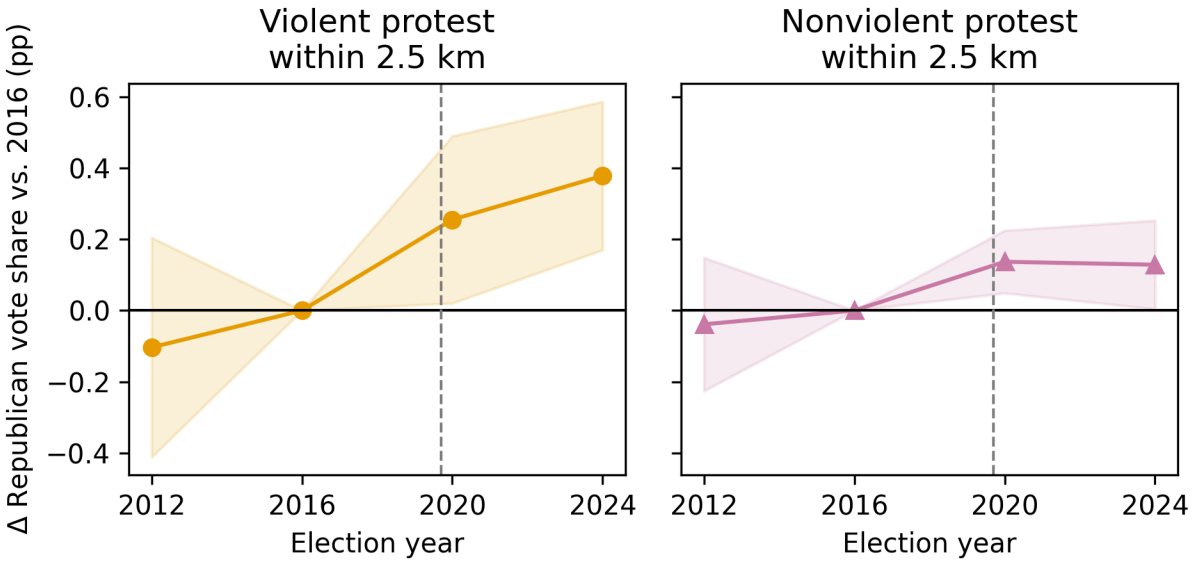
$$Y_{it} = \sum_{t \neq 2016} \beta_t \left(E_i \times \mathbf{1}[\text{year} = t] \right) + \sum_{t \neq 2016} \gamma_t' \left(C_i \times \mathbf{1}[\text{year} = t] \right) + \alpha_i + \delta_{z(i),t} + \varepsilon_{it}, \tag{2}$$

where Y_{it} is the Republican share of all votes cast in block group i in election year t, one of 2012, 2016, 2020 and 2024, in percentage points; E_i equals one when at least one George Floyd protest had its venue within 2.5 km of the block group's population-weighted centroid; $\mathbf{1}[\text{year} = t]$ is an indicator for election year t; C_i is the same set of controls as in equation (1), standardized within the panel; α_i is a block group fixed effect; and $\delta_{z(i),t}$ is a zip code by election year fixed effect. The coefficients of interest are the β_t, the difference between exposed and unexposed block groups in year t relative to the same difference in 2016; β_{2012} is the pre-trend test. The panel is balanced and includes 135,572 block groups with a Republican share in all four elections. The violent comparison is identified within the 898 zip codes that hold both exposed and unexposed block groups, while the nonviolent comparison is identified within 3,373 of them. The regression is weighted by votes cast in 2020 and standard errors are clustered by metropolitan area. Because exposure here is one indicator for a protest within 2.5 km, the comparison is all other block groups and the sample is the balanced panel, the coefficients are not comparable in size with the ring estimates. The identifying assumption is the same: exposed and unexposed block groups of a zip code would have moved alike without the protests.

Figure 4 plots the coefficients and Table A12 reports them, with two further columns in which the comparison group is restricted to the block groups whose nearest protest lies between 2.5 and 10 km. The difference-in-differences results show no detectable pre-trend before the protests and a divergence after them. The 2012 coefficient is not distinguishable from zero for both violent and nonviolent exposure, and with all other block groups as the comparison the 2020 and 2024 coefficients are positive and significant at the 5 percent level. With the comparison restricted to the block groups between 2.5 and 10 km, the violent 2020 coefficient is significant only at the 10 percent level and the nonviolent 2024 coefficient is not distinguishable from zero. Estimated in one regression, the violent and the nonviolent coefficients are not distinguishable in 2020 and differ at the 5 percent level in 2024 (Table A12, column 5). The 2.5 km cutoff lies within the two to three kilometers over which Figures 2 and 3 detect the shift. Figure A5 moves it from 0.5 to 5 km. Between 1 and 3 km the 2020 estimate stays between 0.21 and 0.26 percentage points for violent exposure and between 0.09 and 0.19 for nonviolent exposure. At the widest cutoffs, and for violent exposure at 0.5 km, the estimates are not distinguishable from zero. The 2012 placebo is not distinguishable from zero at any cutoff for both violent and nonviolent exposure. A sensitivity analysis following Rambachan and Roth (2023) asks how large a departure from parallel trends after 2016 would overturn the estimates.[28] The benchmark is the 2012 difference between exposed and

[28] The departure after 2016 is bounded by a multiple of the 2012 difference, and the multiple reported is the smallest

Figure 4: Event study of the Republican vote share around exposure within 2.5 km, 2012–2024



Notes: The figure is an event study of the Republican vote share in block groups within 2.5 km of a protest against all other block groups, across four presidential elections. The left panel is exposure to a violent protest and the right panel exposure to a nonviolent protest; the two are separate regressions. A marker is the coefficient for an election year relative to 2016, which is normalized to zero, so that a value above zero is a shift toward the Republican candidate relative to the unexposed block groups of the same zip code; bands are 95% confidence intervals; the dashed line marks the George Floyd protests. The regression is of the Republican share of all votes cast in the block group in each of the four elections, in percentage points, on the exposure indicator interacted with an indicator for each election year other than 2016 and on the controls interacted with the same indicators, with block group fixed effects and zip code by election year fixed effects, so that exposed and unexposed block groups are compared in the same zip code and election. Exposure equals one when at least one George Floyd protest of the stated family (violent, or nonviolent) had its venue within 2.5 km of the block group. The controls are the demographic controls of Figure 2, log population, log density and distance to the central business district, and the ten earlier-protest ring indicators of that figure, matched to the family. The panel is balanced: block groups with a Republican share in all four elections, 135,572 block groups and 542,288 block group by year observations. The regression is weighted by votes cast in 2020; standard errors are clustered by metropolitan area; 339 clusters. The 2012 coefficient is the pre-period test. Controls here are standardized within the panel. These interaction terms are not comparable in magnitude with the cross-sectional tables, which measure exposure in distance rings on a different sample. The plotted coefficients are columns (1) and (3) of Table A12.

unexposed block groups, the only pre-period estimate and an imprecise one. For violent exposure, the 2020 coefficient survives departures up to a quarter of that benchmark and the 2024 coefficient up to half. For nonviolent exposure the bounds are 0.60 and 0.10.

4.5 Exposure across block-group characteristics

Does the backlash differ by the kind of block group? The question is asked for two reasons: whether the shift is a general response to a protest nearby or is carried by a few kinds of block group, and

at which the robust confidence set covers zero. For violent exposure a quarter of the 2012 difference is 0.03 percentage points, a tenth of the 2020 coefficient. A multiple of one would be a departure as large as the 2012 difference, and every bound here is below one. With one election before the protests, the path cannot distinguish a break in 2020 from a trend that began before 2012.

whether the block groups near a venue were selected along race, the one dimension this movement was about. For each of six characteristics, block groups are split into four quartiles of the characteristic measured relative to the other block groups of their zip code. The ring indicators of the specification are interacted with the quartiles. Figure A6 reports the 0–1 km coefficient by quartile, with its placebo, for any protest and for violent protests, one panel per characteristic. Whether the four coefficients differ is decided by a joint test of the quartile interactions.

Income is the one characteristic that carries a pattern. For violent protests the backlash is largest in the quarter of block groups whose income lies furthest above their zip code's mean. The joint test rejects ($p < 0.01$) while its placebo does not (Figure A6); for any protest the four estimates are alike and the joint test does not reject. This is the opposite of what a story of polarization along education or income would predict. In the ongoing realignment, richer and more educated neighborhoods have been moving toward the Democratic candidate; here the shift toward the Republican candidate near violent protests is largest in the richest block groups of their zip code.

Race is the dimension along which selection into exposure would likely show, given the nature of this protest movement.[29] The wave's venues sat in block groups with a higher Black population share than the venues of earlier protests (Table A6), and if the block groups near a venue were already on a different electoral path for reasons tied to this movement, the estimate could be expected to differ across the Black share more than across any other characteristic; a trend shared by every quartile would not show in this test. Figure A6 splits block groups by their Black population share relative to their zip code: the backlash is positive in every quartile under both treatments, the placebo is not distinguishable from zero in any of them, the point estimates do not rise with the Black share, and the joint tests do not reject ($p = 0.60$ for any protest, 0.62 for violent protests). The figure is uninformative about a gradient: it neither establishes one nor rules one out. Education, density, age and the Hispanic share carry no detectable pattern either (Figure A6): the point estimates are positive in every quartile and none of the joint tests of the 2016–2020 coefficients rejects, although for education under any protest the placebo is not clean, its joint test rejecting at $p = 0.04$.[30]

4.6 Persistence to 2024

Does the backlash persist? Table A13 re-estimates the ring profiles with the 2020–2024 change and the cumulative 2016–2024 change as outcomes, on the 135,572 block groups with a 2024 return. The comparison stays within zip codes, so a shift in 2024 is relative to the block group's neighbors and not to the swing of the 2024 election. For violent exposure the cumulative point estimates at the four innermost rings are larger than in 2016–2020. The 2020–2024 coefficients on their own are positive at the four innermost rings and not distinguishable from zero. For nonviolent exposure the

[29] Proximity to an out-group has been found to move attitudes against it (Enos, 2014, 2016, 2017), and exposure to the 2006 immigrant rallies to move policy preferences (Branton et al., 2015); the Black-share panel is also where that frame would predict a gradient.

[30] Sharpened q-values, which control the expected share of false discoveries among the coefficients called significant, computed over the 50 coefficients of this section's heterogeneity family change no verdict at the 5 percent level; the joint tests of equality across quartiles are not adjusted.

cumulative coefficient is close to its 2016–2020 size at 1–2 km and smaller at 0–1 km, where it is not distinguishable from zero, while the 2020–2024 coefficients sit near zero. Consistent with the difference-in-differences estimates of Section 4.4, where the 2024 coefficient is positive for exposure to both violent and nonviolent protests, the backlash was still present in 2024. For violent exposure the point estimates in both designs are consistent with an effect that grew rather than persisted.[31]

4.7 Robustness

This section summarizes several robustness exercises that address four potential concerns with the main analysis. Each check is reported in full in Appendix D.

The first is whether the estimate picks up the places where protests of any kind take place. The earlier-protest controls are entered for all earlier protests and then separately by political orientation; every ring coefficient stays inside its confidence interval (Table A10).[32] Within zip codes, the block groups near a violent protest and those near only nonviolent ones are not distinguishable in their 2016 vote shares or in their 2012–2016 trend, although they differ on several demographic characteristics and the joint balance test rejects (Table A7). The 4,606 other protests of the same months, entered beside the George Floyd ones with their own ten rings, leave the George Floyd estimate within one kilometer at 0.19 percentage points (Table A15). Entered alone the other protests carry 0.09 and beside the George Floyd rings 0.04, neither distinguishable from zero, though their ten rings are jointly distinguishable from zero and not from one another (Table A15). Proximity to right-wing protests of the same months moves the vote the other way, with block groups within a kilometer shifting their vote toward the Democrats by 0.21 when the George Floyd rings are held fixed, with a placebo not distinguishable from zero at that ring, although the ten right-wing rings are not jointly distinguishable from zero (Table A15).

The second is whether the estimate depends on the controls. The coefficient on exposure to any protest falls to about two-fifths of its size when the demographic controls enter and moves little after that, including when the block group's own 2012–2016 change is added (Table A11); the violent estimate is not distinguishable from zero without fixed effects and is significant and of similar size in every column once the demographic controls enter (Table A11, Panel B).[33] Allowing the Hispanic share one slope per metropolitan area, because Hispanic neighborhoods moved toward the Republicans in 2020 by amounts that differed across cities, or controlling for the retail share of jobs in the block group, leaves the estimates inside their confidence intervals. Adding Census measures of residential turnover leaves every headline inside its confidence interval (Table A16). The places themselves changed, though: between the 2014–2018 and 2019–2023 Census surveys, block groups

[31] Growth over two elections is also what residential sorting after the protests, or a habit of abstention once formed, would produce; the two designs do not separate these from a persisting reaction.

[32] Rebuilding those controls without the 5,955 earlier protests located only to a town or a district leaves the estimates at 0.19 for any protest and 0.34 for a violent one.

[33] Following Oster (2019), selection on unobserved characteristics would have to be 2.10 times as strong as selection on the controls to bring the 0–1 km estimate to zero, and 3.88 times for the violent one; with equally strong selection the estimates would be 0.10 and 0.25.

within one kilometer of a protest gained population, renters and recent movers relative to the rest of their zip code, and near violent protests only the renter share rose.[34]

The third is whether the estimate depends on how the rings are built. In Table A14, the venues are relocated at random within their metropolitan areas. Random placement does not replicate the headline, and the observed decay across rings is steeper than the decay that random geometry itself produces. Net of the mean under random placement, the 0–1 km estimate is 0.11 for any protest and 0.23 for a violent one.[35] Comparing block groups of the same zip code and density decile leaves the estimates at 0.19 and 0.35. Letting the effect vary by density decile cuts them to 0.10 and 0.18, with permutation p-values of 0.08 and 0.16 (Table A14). The rings are redrawn so that every ring covers the same area as the innermost one. The 0–1 km estimates stay positive and significant for violent and for nonviolent exposure. Each of the 42 states and each of the 30 largest metropolitan areas is dropped in turn (Figure A7): none of the 72 drops moves the violent 0–1 km estimate outside the full-sample confidence interval. Recoding as nonviolent the 263 violent events whose only marker is arrests leaves the violent 0–1 km estimate at 0.34, significant only at the 10 percent level, and its placebo rises from 0.04 to 0.30, not distinguishable from zero but close to the estimate in size. The violent result is therefore weaker under that coding (Table A17).

The fourth is whether the estimate depends on how the vote is measured. Allocating precinct returns to block groups through the population of their census blocks, rather than of the block group as a whole, leaves the 0–1 km estimates at 0.18 for any protest and 0.31 for a violent one, with placebos not distinguishable from zero. Weighting by votes cast in 2016 rather than in 2020, or restricting the sample to the block groups that lie within a single precinct, leaves the any-protest estimate inside its confidence interval (Table A17). On the single-precinct subsample, which includes only about a fifth of the block groups, the violent estimate is too imprecise to interpret and the placebo is not clean. Adding the eight states and the District of Columbia that lack 2012 returns, for the 2016–2020 estimate alone, leaves the estimates at 0.19 and 0.34.

5 Protest Participation

This section asks what sending residents to a protest did to the vote of the block groups they came from. Participation could in principle move the vote in either direction. Votes moving toward the Democratic candidate would suggest a mobilization of the neighborhoods whose residents took part, while votes moving toward the Republican candidate would suggest that taking part wore down support for the movement's cause in those same neighborhoods.

[34]The 2019–2023 survey reaches three years past the election, and its recent-mover category spans three years against two in the earlier one. In North Carolina, moves in and out near violent protests do not detectably change the Democratic registration share (Figure A11).

[35]The 0–1 km estimates should be judged by the permutation p-values because the conventional t-test is not reliable here: under random placement the null profile is not centered at zero, so that test reports significance at 0–1 km on a large share of draws for any protest and on a smaller share for violent protests (Borusyak and Hull, 2023).

5.1 Specification

The question is whether block groups whose residents went to a George Floyd protest changed their vote differently, between the 2016 and the 2020 presidential elections, from block groups of the same zip code whose residents did not go, or went in smaller numbers, holding fixed their demographic characteristics and their residents' ordinary presence at the protest venues. Participation enters in two parts: whether any resident of the block group was observed attending, and how many did, measured as attendances per device homed in the block group, in logs. The outcome is the vote of the whole block group, so the estimate captures both the direct effect on the residents who attended and any effect on their neighbors; the data cannot separate the two.

The estimating equation is

$$\Delta Y_i = \beta_D \, D_i + \beta_L \, (L_i - c) \, D_i + A_i'\theta + X_i'\delta + M_i'\lambda + S_i'\kappa + \alpha_{z(i)} + \varepsilon_i, \tag{3}$$

where ΔY_i is the change in the Democratic share of all votes cast in block group i between the 2016 and 2020 presidential elections, in percentage points; D_i equals one when at least one resident of block group i attended a George Floyd protest, the extensive margin of protest participation; $L_i = \ln(a_i/n_i)$ is the intensive margin, the log of the participation rate, where a_i counts the attendances of block group i over the wave and n_i is the number of panel devices homed in the block group; c is the mean of L_i among the block groups that sent anyone, so that β_D is read at the typical participation rate of a sending block group rather than at a rate of one. A_i controls for the residents' ordinary presence at the same venues, measured between 4 and 26 May 2020 on days those venues hosted nothing and entered in the same two-part form as the treatment; X_i are the demographic controls of Section 2.3, measured before the wave; M_i are four measures of how far residents' devices ranged on an ordinary day before the pandemic; S_i are indicators for deciles of the number of panel devices homed in the block group; and $\alpha_{z(i)}$ is a fixed effect for the zip code of block group i. The regression is weighted by votes cast in 2020, and standard errors are clustered by metropolitan area. The coefficients of interest are β_D, the shift in the Democratic share associated with sending anyone, relative to block groups of the same zip code that sent nobody, and β_L, the further shift associated with sending more among the block groups that sent anyone. The same equation with the 2012–2016 change as the outcome is the pre-period placebo test: if the block groups where protest participants lived were already moving toward the Democrats before them, β_D in that regression would show it.

The sample is the block groups of the metropolitan areas with at least one protest venue that have vote returns in 2012, 2016 and 2020, every control and a device panel, resulting in 135,691 block groups in 333 metropolitan areas and 13,603 zip codes. Of these, 85,399 block groups sent at least one of their residents to a protest. Among them participation is thin: the median sending block group recorded 2 attendances across all the protests, the mean 3.5 and the ninetieth percentile 7.[36]

[36] An attendance is one panel device seen at a venue during a protest. The panel is a sample of residents: the crowd-size comparison of Figure A2 implies about 25 attendees per detected device.

The block groups where protest participants lived were not selected at random. The design does not need them to be: the identifying assumption is that, conditional on the controls, they would not have been on a different voting trend from the rest of their zip code had nobody gone. Table 2 shows that, within their zip code, these block groups differ from the rest on eleven of fifteen characteristics. Unlike for exposure, their 2012–2016 change in the Republican and in the Democratic share also differs from the rest of the zip code's before the controls enter (Table A22). Once the controls enter, the difference is no longer detectable (Table 2), and the estimate moves little when the block group's own 2012–2016 change is held fixed (Table A22).

5.2 Participation and the vote

Figure 5 plots the estimate, with two placebos on the same axes; Table A18 reports the coefficients. Block groups whose residents turned up at George Floyd protests, holding fixed their residents' ordinary presence at the protest venues before the wave, shifted 0.07 percentage points more Democratic in presidential vote share between 2016 and 2020 than otherwise-similar block groups in the same zip code that sent nobody; sending more added no detectable shift.[37]

The pre-period placebo, plotted on the same axes, is not distinguishable from zero: with the 2012–2016 change as the outcome, the interval of the extensive margin covers zero. With the controls in place, the block groups where protest participants lived show no detectable pre-trend.[38] The second placebo replaces the treatment with attendance, measured in the same way, at the protests of 2019 and of early 2020 held at venues that later hosted a George Floyd protest. That attendance predicts no detectable shift in the 2020 vote: the shift goes with participation in the George Floyd protests, not with earlier protest-going at the same venues.

Table A23 estimates the specification on four other outcomes. The Republican share moves in mirror image, and the two shares are not exact mirrors only because the denominator is all votes cast. Turnout, Democratic votes and Republican votes per voting-age person show no detectable change, although for Republican votes the placebo of the intensive margin is significant at the 5 percent level. Whether the shift came from more Democratic votes, from fewer Republican ones, or from a change in who lives and votes in the block group is not identified.

5.3 Participation by protest violence and by habitual attendance

Figure 6 splits participation in two ways, by whether the protest attended turned violent and by whether the attender regularly went to the venue on days without a protest; Table A21 reports the coefficients.[39] Block groups that sent anyone to a nonviolent protest shifted 0.07 percentage

[37] Within zip codes, the standard deviation of the 2016–2020 change in the Democratic share, weighted by votes, is 2.5 percentage points, so the shift is about a thirtieth of it.

[38] The test has limited power: a 2012–2016 shift half the size of the estimate would not have been detected, and the placebo is not distinguishable from the estimate either (Table A18).

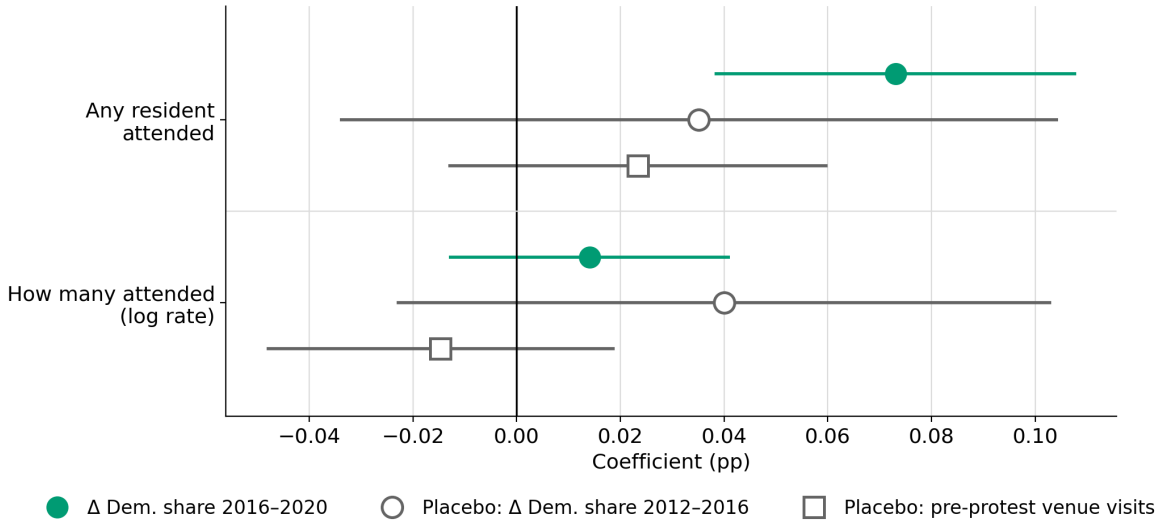
[39] Each split estimates its four terms, two extensive margins and two intensive margins, in one regression, so that each group is estimated holding the other fixed.

Table 2: Block groups where protest participants lived against the rest of their zip code

	Sending mean	Non-sending mean	Diff. (within zip code)
% White	61.6	57.5	0.1
			(0.2)
% Black	12.5	13.1	-0.1
			(0.1)
% Hispanic	16.8	21.9	-0.8***
			(0.1)
% Asian	6.0	4.5	0.7***
			(0.1)
Median age	39.6	40.5	-1.1***
			(0.1)
Median household income	$74,403	$64,511	$3,998***
			($280)
Unemployment rate (%)	5.9	6.7	-0.3***
			(0.0)
% associate or higher	43.3	36.0	2.6***
			(0.2)
% bachelor's or higher	35.0	27.8	2.4***
			(0.1)
% Renter	35.8	36.0	-0.3
			(0.2)
Population	1,668	1,287	508***
			(30)
Density (per km^2)	2,832	2,290	-108***
			(15)
Distance to city center (km)	23.9	25.4	-0.0
			(0.0)
Republican vote share 2016 (%)	40.5	40.5	0.5***
			(0.1)
Democratic vote share 2016 (%)	53.8	54.2	-0.5***
			(0.1)
Δ Republican share 2012–2016 (pre-trend, pp)	**-2.25**	**-0.85**	**-0.04**
			(0.03)
Δ Democratic share 2012–2016 (pre-trend, pp)	**-1.72**	**-2.79**	**0.02**
			(0.03)
Observations	85,399	50,292	135,691
Joint-balance F (within zip code)			73.22
p-value			0.000

Notes: The table compares the block groups where at least one resident was observed attending a George Floyd protest with the other block groups of the participation estimation sample of Section 5.1, 85,399 against 50,292. Columns 1 and 2 are the unweighted means of each characteristic in the two groups. Column 3 is the within-zip-code difference: the coefficient from a regression of the characteristic on the sending indicator with zip code fixed effects, weighted by votes cast in 2020, so that every comparison is between block groups of the same zip code, estimated on the 8,900 zip codes that hold both kinds of block group; standard errors in parentheses, clustered by metropolitan area; 333 clusters. The joint F tests the fifteen characteristics above the two pre-trend rows at once. The two pre-trend rows, set apart by a midline, are the 2012–2016 change in the Republican and in the Democratic share, the outcomes of the placebo tests of Sections 4.2 and 5.2. In these two rows column 3 is conditional on the controls: it is the same regression with the controls of equation (3) added, and without the intensive margin. With the intensive margin in, as in the placebo of Figure 5 and column 6 of Table A22, the Democratic row is 0.035 (0.035). Units: the shares are percentages of residents (or, for renters, of housing units); median household income is in dollars; population is residents and density residents per square kilometer; distance to city center is the distance from the block group to the central business district, in kilometers; the 2016 vote shares and the two 2012–2016 changes are shares of all votes cast for president, in percent and in percentage points. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure 5: Participation and the change in the Democratic vote share



Notes: Each marker is an estimated coefficient and each whisker a 95 percent confidence interval, in percentage points of the Democratic share of all votes cast for president. The upper group is the extensive margin, whether the block group sent anyone to a protest, and the lower group the intensive margin, how much it sent. Filled circles are the estimate; open circles are the same specification on the 2012–2016 change; open squares replace the treatment with attendance at earlier protests at the same venues. The regression is of the change in the Democratic share of all votes cast for president between the 2016 and 2020 elections on the two terms, with the venue-traffic controls, the demographic controls, the neighborhood mobility controls, indicators for deciles of the block group's panel devices, and zip code fixed effects, so that every comparison is between block groups in the same zip code; the regression is weighted by votes cast in 2020. Regulars at protest venues are counted, and the venue-traffic controls hold fixed the residents' presence at the same venues in the three weeks before the first event. The intensive margin is the log of attendances per panel device, zero where nobody attended. The demographic controls are listed in the note to Figure 2; the mobility controls are four measures of how far residents' devices ranged on an ordinary day before the pandemic. The sample is the block groups of metropolitan areas with a protest venue that have vote returns in 2012, 2016 and 2020 and a device panel; $N = 135{,}691$ for the estimate and the first placebo, 85,399 of which sent at least one resident, and 132,482 for the second placebo. Standard errors are clustered by metropolitan area; 333 clusters. The coefficients, standard errors and the tests of each placebo against the estimate are in Table A18.

points more Democratic, driving the main result documented in Section 5.2. For violent protests the estimate is −0.02 percentage points and is not distinguishable from zero.[40] A test that the two are equal rejects at the 5 percent level. Among block groups that sent people, sending more is associated with no further shift for nonviolent protests (0.01, not significant) and with a negative shift for violent ones (−0.09). These also differ on the same test at the 1 percent level.[41]

This result suggests that mobilization takes place mostly in the block groups where participants in nonviolent protests lived, and is not detectable at the typical attendance rate when the protests

[40] Attendance at violent protests is the smaller part: 40,723 attendances from 24,333 block groups, against 261,701 from 79,966 at nonviolent protests, a ratio of 0.16 in attendances. Tables A28 and A32 print one fewer of each, as their regressions drop two block groups without per-capita outcomes.

[41] The extensive-margin difference reaches the 5 percent level at 100, 200 and 400 m but not at 50 m; the intensive-margin difference reaches it at 200 and 400 m only; and at 50 and 100 m the nonviolent intensive margin inverts, to −0.17 and −0.07, while the violent intensive margin is negative and distinguishable from zero at all four radii (Table A28).

turned violent. Moreover, the more residents a block group sent to a violent protest, the further its vote moved away from the Democratic candidate. At the low attendance rates of most block groups that sent anyone to a violent protest, the two coefficients together imply a shift toward the Democrats, which turns toward the Republicans as attendance rises (Table A36).[42] This pattern is consistent with a backlash entering through participation in violent protests: not only did the residents of the block groups close to violent protests react against them, but so did the residents of the block groups that sent people to them. The mobilization at nonviolent protests appears in the extensive margin of participation, while the backlash to violent protests appears in the intensive margin.

The second splits attendance by whether the attender regularly went to the venue on days without a protest.[43] I treat regulars as incidental participation: residents who would have been at the venue regardless of the protest, possibly because they work there or go there often for other reasons; the other attenders are residents who went there specifically to take part in the protest. The shift comes from the attenders who were not regulars: their extensive margin is 0.08 and significant at the 1 percent level, while the regulars' extensive margin is -0.07 and their intensive margin -0.13, both distinguishable from zero with that sign, and the two groups differ on both terms at the 1 percent level.

This means that the mobilization takes place only in the block groups whose participants would not have been at the venue otherwise. The block groups whose residents were there anyway—the incidental participation—reacted negatively to being there and moved away from the Democratic candidate. This is again consistent with a backlash entering through participation: Section 4 studied the residents of the block groups around a venue, while this is a backlash among people who live elsewhere and go to the venue regularly for work or other reasons.[44] An alternative reading of this result that cannot be excluded is that the regulars' shift may be picking up the effect of people who kept traveling to protest venues during the pandemic, such as essential workers, rather than what they saw during the protests.

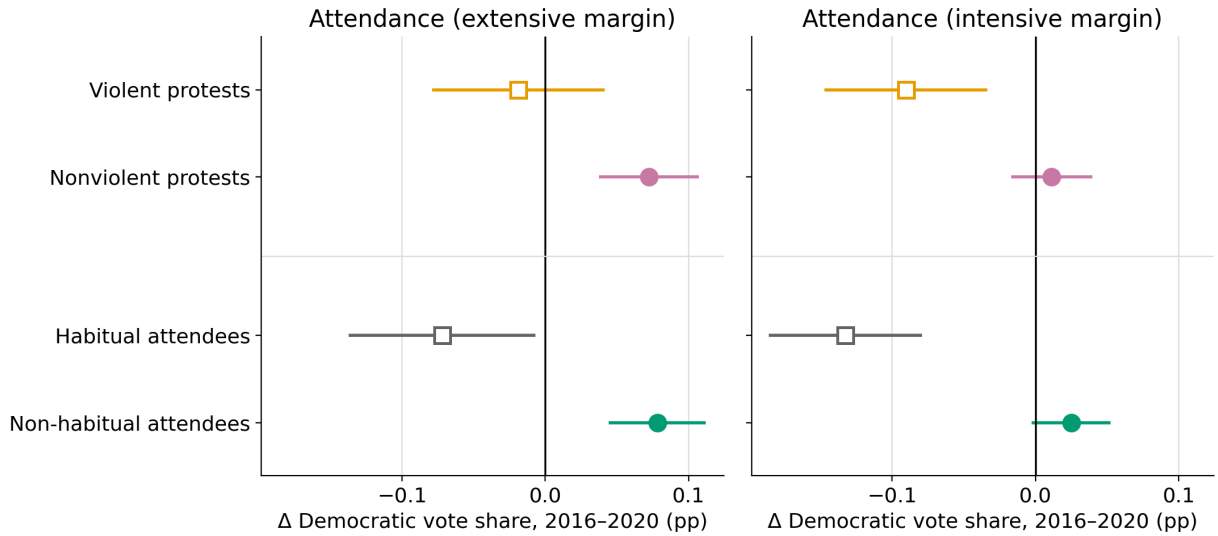
Taken together, the two splits point the same way. The shift toward the Democratic candidate appears in the block groups whose residents went to a nonviolent protest and those whose residents were not regulars at the venue. Where residents went in larger numbers to a protest that turned violent, or were regulars at the venue, the vote moved toward the Republicans. A block group that was already moving toward the Democrats would be expected to move further in that direction whatever protest its residents attended and however often they had been at the venue before, so a trend already under way does not fit these opposite signs. A vote trend that drew residents to

[42] Replacing the log-linear form with ten indicators of the violent attendance rate leaves the total at about 38,000 votes and the signs at both ends, although the path between them is not smooth.

[43] A resident is defined as regularly going to a protest venue if their device was seen at the venue, in the same hours, on at least five days between 4 and 26 May 2020, before the wave.

[44] Holding the exposure rings of Section 4 fixed and dropping the block groups within a kilometer of a venue leaves the regulars' intensive margin and the non-regulars' extensive margin unchanged. The regulars' extensive margin falls to -0.05 and is no longer distinguishable from zero.

Figure 6: Participation split by protest violence and by whether the attender was a regular



Notes: Each marker is an estimated coefficient and each whisker a 95 percent confidence interval, in percentage points of the Democratic share of all votes cast for president; the left panel is the extensive margin and the right panel the intensive margin. The upper two rows come from one regression that splits attendance by whether the protest attended turned violent; the lower two rows come from a second regression that splits it by whether the attender was a regular at that venue. Each group enters as its own extensive margin and its own intensive margin, so a block group whose residents attended both kinds of venue-day carries both pairs of terms; the groups are not mutually exclusive categories. The outcome, the controls, the zip code fixed effects and the weights are those of Figure 5. $N = 135{,}689$ for the violence split and 135,691 for the split by whether a regular. Standard errors are clustered by metropolitan area; 333 clusters. The treatment is split and the venue-traffic controls are not. The coefficients, standard errors and the Wald tests of each split are in Table A21.

the protests would have to produce them too. What the pattern does not exclude is that the block groups where participants in a violent protest lived, or whose residents were regulars, were on a different voting trend from the others. The opposite signs would then reflect who went to each kind of protest rather than what happened there.[45]

5.4 Difference-in-differences approach

As for exposure (Section 4.4), a difference-in-differences regression follows the same block groups across four presidential elections—from 2012 to 2024—and asks whether the block groups where protest participants lived moved differently from the other block groups of the same zip code in each election relative to 2016. The regression splits participation by whether the protest attended turned violent, as in Section 5.3.

[45]Heavier attendance also goes with a shift toward the Republicans among regulars, at earlier left- and right-wing protests and at nonviolent protests within 100 m (Tables A19, A21 and A28).

The estimating equation is

$$Y_{it} = \sum_{t \neq 2016} \beta_{D,t} \left(D_i \times \mathbf{1}[\text{year} = t] \right) + \sum_{t \neq 2016} \beta_{L,t} \left((L_i - c) D_i \times \mathbf{1}[\text{year} = t] \right) + \sum_{t \neq 2016} \gamma_t' \left(C_i \times \mathbf{1}[\text{year} = t] \right) + \alpha_i + \delta_{z(i),t} + \varepsilon_{it}, \tag{4}$$

where Y_{it} is the Democratic share of all votes cast in block group i in election year t, one of 2012, 2016, 2020 and 2024, in percentage points; D_i and $(L_i - c) D_i$ are the extensive and the intensive margin of participation of equation (3); $\mathbf{1}[\text{year} = t]$ is an indicator for election year t; C_i collects the controls of that equation; α_i is a block group fixed effect; and $\delta_{z(i),t}$ is a zip code by election year fixed effect. The coefficients of interest are the $\beta_{D,t}$, the difference between sending and non-sending block groups in year t relative to the same difference in 2016; $\beta_{D,2012}$ is the pre-trend test. The panel is balanced and includes 132,336 block groups with a Democratic share in all four elections and at least one other block group of their zip code in the sample, 529,344 block group by year observations. The regression is weighted by votes cast in 2020 and standard errors are clustered by metropolitan area. The identifying assumption is the same: sending and non-sending block groups of a zip code would have moved alike without the protests.

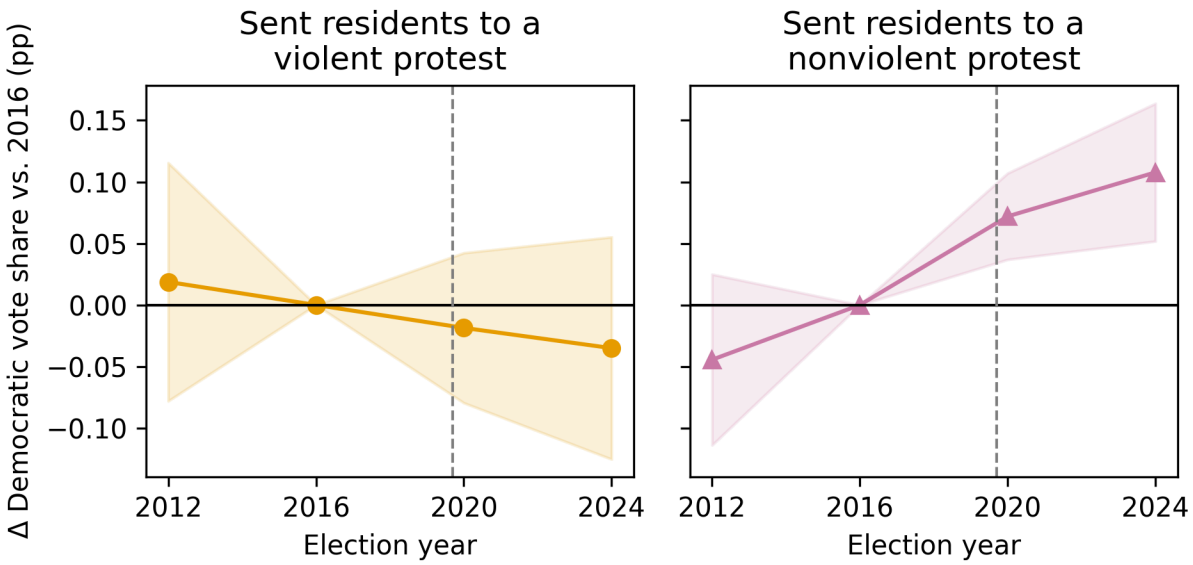
Figure 7 plots the coefficients for participation in a violent and in a nonviolent protest, and Table A24 reports them for every series. The difference-in-differences results show no detectable pre-trend before the protests and a divergence after them: for the block groups where participants in a nonviolent protest lived, the 2012 coefficient is not distinguishable from zero, and the 2020 and 2024 coefficients are positive and significant at the 1 percent level. The 2012 coefficient is not distinguishable from zero in any of the ten series of Table A24, for participation at any protest, for the two groups of the violence split and for regulars and non-regulars, on either margin. For the block groups where participants in a violent protest lived, no coefficient on the extensive margin is distinguishable from zero in any year. A sensitivity analysis following Rambachan and Roth (2023), defined in Section 4.4, uses the 2012 difference between sending and non-sending block groups as its benchmark. For participation in a nonviolent protest, the 2020 coefficient survives departures up to 0.60 of that benchmark and the 2024 coefficient up to 0.45.[46] For participation at any protest the bounds are 0.70 and 0.60.

5.5 Participation across block-group characteristics

Does the shift differ by the kind of block group? The question is asked for the same two reasons as in Section 4.5. For each of six characteristics, block groups are split into four quartiles of the characteristic measured relative to the other block groups of their zip code. The two margins of equation (3) are interacted with the quartiles. Figure A8 reports the extensive-margin coefficient

[46] The multiple is defined in Section 4.4. For participation in a nonviolent protest, 0.60 of the 2012 difference is 0.03 percentage points, about a third of the 2020 coefficient. Every bound here is below one, and with one election before the protests the path cannot distinguish a break in 2020 from a trend that began before 2012.

Figure 7: Event study of the Democratic vote share around participation, 2012–2024



Notes: The figure is an event study of the Democratic vote share in block groups where protest participants lived against all other block groups of their zip code, across four presidential elections. The left panel is participation in a violent protest and the right panel participation in a nonviolent protest; unlike Figure 4, the two series come from one regression that splits participation by whether the protest attended turned violent, each group holding the other fixed as in Figure 6. A marker is the extensive-margin coefficient for an election year relative to 2016, which is normalized to zero, so that a value above zero is a shift toward the Democratic candidate relative to the non-sending block groups of the same zip code, the opposite direction from Figure 4; bands are 95% confidence intervals; the dashed line marks the George Floyd protests. The regression is of the Democratic share of all votes cast in the block group in each of the four elections, in percentage points, on the two margins of participation interacted with an indicator for each election year other than 2016 and on the controls interacted with the same indicators, with block group fixed effects and zip code by election year fixed effects, so that sending and non-sending block groups are compared in the same zip code and election. The controls are those of Figure 5. The panel is balanced: 132,336 block groups with a Democratic share in all four elections and 529,344 block group by year observations. The regression is weighted by votes cast in 2020; standard errors are clustered by metropolitan area; 328 clusters. The 2012 coefficient is the pre-period test. The plotted coefficients are the extensive-margin rows of columns (2) and (3) of Table A24.

by quartile, with its placebo, one panel per characteristic. Whether the four coefficients differ is decided by a joint test of the quartile interactions.

Density is the one characteristic that carries a pattern. The joint test rejects ($p < 0.01$) while its placebo does not (Figure A8): the least dense and the densest quarters of their zip code carry the largest estimates, 0.14 each, the second quarter is negative and the third sits between, so the four are not monotone in either direction.

Race is again the dimension along which selection into participation would likely show, given the nature of this protest movement. If the block groups where protest participants lived were already on a different electoral path for reasons tied to this movement, the estimate could be expected to differ across the Black share more than across any other characteristic and to shrink where a Democratic trend would have to be absent; a trend shared by every quartile would not show in

this test.[47] Figure A8 splits block groups by their Black population share relative to their zip code: the estimate is positive in every quartile, the placebo is not distinguishable from zero in any of them, the most-Black quartile carries the largest point estimate, so the shift does not shrink where a Democratic trend would be weakest, and the joint tests do not reject ($p = 0.37$ for the estimate, 0.27 for the placebo). As in Section 4.5, the figure neither establishes a gradient nor rules one out. Income, education, age and the Hispanic share carry no detectable pattern either (Figure A8): no joint test rejects at the 5 percent level, and allowing the Hispanic share one slope per metropolitan area leaves the estimate at 0.07.[48]

5.6 Persistence to 2024

Does the mobilization persist? Table A25 reports it estimated with three outcome windows, the 2016–2020 change, the 2020–2024 change and the cumulative 2016–2024 change, on the 134,570 block groups that have a vote return in 2024 as well, in 328 metropolitan areas. The comparison stays within zip codes, so a shift in 2024 is relative to the block group's neighbors and not to the swing of the 2024 election. The participation effect did not subside between 2020 and 2024. The 2020–2024 change is positive and significant (0.05), the Republican-share mirror moves with it (-0.05), and the cumulative 2016–2024 shift is 0.12 against 0.07 over 2016–2020, which suggests the effect may have grown rather than merely persisted.[49] Sending more added no detectable shift in any of the three windows. Consistent with the difference-in-differences estimates of Section 5.4, where the 2024 coefficient is positive for participation in a nonviolent protest, the mobilization was still present in 2024.[50] In that regression the violent intensive margin and the regulars' intensive margin are negative in 2024 as in 2020, while the regulars' extensive margin is no longer distinguishable from zero (Table A24).

5.7 Robustness

This section summarizes several robustness exercises that address four potential concerns with the main analysis. Each check is reported in full in Appendix E.

The first is whether the participation measure captures attendance at the protests rather than

[47] In a regression of the outcome on the controls alone, the Black population share carries a negative partial slope of 3.6 percentage points over the share's full range, significant at the 1 percent level, on the 2016–2020 change in the Democratic share within zip codes, at the mean level of education, so a shift carried by a Democratic trend that follows the Black share would be weakest in the most-Black block groups. The slope does not rule out a differential trend within a quartile.

[48] Sharpened q-values, which control the expected share of false discoveries among the coefficients called significant, computed over the 32 coefficients of this section and of the two splits of Section 5.3 change no verdict at the 5 percent level; the joint tests of equality across quartiles are not adjusted.

[49] The increase is treated as suggestive rather than established, because 200 m is the only one of four detection radii at which it reaches the 5 percent level. Cumulatively from 2016 to 2024 the effect is significant at the 5 percent level at all four radii and for both parties, positive for the Democratic share and negative for the Republican share, so persistence is the radius-robust finding; the increase is not.

[50] Growth over two elections is also what residential sorting after the protests, or a habit of voting once formed, would produce. The two designs do not separate these from a persisting reaction.

ordinary activity around their venues. Ordinary presence at the venues on days without a protest, entered in place of attendance, predicts no detectable shift in the 2016–2020 change, although on the 2012–2016 change it carries 0.17, significant at the 1 percent level, so its placebo is not clean; entered beside attendance it leaves the extensive margin at 0.07 (Table A30). That presence is measured during the wave, while the control of the main specification is measured before it. The extensive margin is positive and significant at the 1 percent level across all 256 combinations of eight venue-traffic controls (Table A26), at all four detection radii from 50 to 400 m (Table A27), under five denominators of the participation rate (Table A27, Panel B), and with terms for the neighboring block groups entered (Table A29). It also holds when a block group counts as sending only from two attendances. From five attendances the estimate is larger, but its placebo is not clean (Table A28).[51] The pooled intensive margin is not stable across these choices: it changes sign across detection radii, is negative from two attendances on and is a null only under the denominator the paper uses, so the mobilization result rests on the extensive margin, and the pooled intensive margin is read as a null at 200 m and not as a dose-response in either direction. Split by protest violence, the violent intensive margin is negative and distinguishable from zero at all four radii, while the nonviolent one changes sign (Table A28). No single state or large metropolitan area moves the extensive margin outside its full-sample confidence interval (Figure A7, bottom row).

The second is whether the estimate picks up the kind of block group that sends residents to protests of any kind, rather than participation in the George Floyd protests. The earlier-attendance placebo of Section 5.2 differs from the estimate at the 5 percent level on the extensive margin (Table A18). Entered as a control, overall or by political orientation, it leaves both margins inside their confidence intervals (Table A20).[52]

The third is whether the estimate depends on the controls. The extensive margin falls from 0.19 with zip code fixed effects to 0.11 once the demographic controls enter and to about two-fifths of its within-zip-code size, 0.08, once the device-panel controls enter, and moves little when the venue-traffic controls are added or when the block group's own 2012–2016 change is added.[53] With finer controls for the block group's device count, twenty bins or a slope within each decile, it falls to about 0.06, inside its confidence interval. The pre-period placebo is clean only once the demographic controls enter, so the full specification, whose controls matter for identification, is the one reported (Table A22).[54] Adding Census measures of residential turnover leaves both margins inside their

[51] Split instead into ten groups by participation rate, the placebos are jointly not distinguishable from zero and do not differ across the groups.

[52] Attendance at the earlier protests coded neither left- nor right-wing cannot be separated in the attendance data; those columns are estimated on the 132,482 block groups where earlier attendance is observed. Split by orientation, the placebo's intensive margin is negative and distinguishable from zero for both left-wing and right-wing earlier protests, so on that margin the split placebo is not clean (Table A19), and the right-wing column rests on a thin base, the 6.7 percent of block groups with any such attendance.

[53] A block group with more panel devices is more likely to have any attendance detected at all, so the device-panel deciles absorb a correlation between panel size and the extensive margin.

[54] Following Oster (2019), selection on unobserved characteristics would have to be 1.60 times as strong as selection on the controls to bring the extensive margin to zero; with equally strong selection the extensive margin would be 0.029.

confidence intervals (Table A31). So does allowing the Hispanic share one slope per metropolitan area, or a control for the retail share of jobs in the block group.

The fourth is whether the estimate depends on how the vote is measured. Allocating precinct returns to block groups through the population of their census blocks, rather than of the block group as a whole, leaves the extensive margin inside its confidence interval, with a placebo not distinguishable from zero. Weighting by votes cast in 2016 rather than in 2020, or restricting the sample to the block groups that lie within a single precinct, leaves the extensive margin inside its confidence interval (Table A32). On the single-precinct subsample, which retains a little under a quarter of the block groups and about a fifth of the sending ones, the placebo is imprecise and twice the estimate, and the violent intensive margin falls to 0.01 and is not distinguishable from zero. Adding the eight states and the District of Columbia that lack 2012 returns, for the 2016–2020 estimate alone, leaves the extensive margin inside its confidence interval.

6 Aggregate Net Effect

Sections 4 and 5 estimated exposure and participation separately, but the two treatments apply to the same block groups. A block group can be close to a protest venue and send its residents to a protest at the same time, so it receives two treatments that pull in opposite directions.[55] This section puts the backlash and mobilization effects together. Appendix F holds the joint regressions, the decomposition tables and the counterfactual exercise in full.

6.1 The two treatments in one regression

The joint regression enters the participation terms of equation (3) and the exposure terms of equation (1) together, with the controls of both:

$$\begin{aligned}\Delta Y_i = \beta_D\, D_i + \beta_L\,(L_i - c)\, D_i + \sum_{r=1}^{10} \beta_r\, D_{ir} + \sum_{r=1}^{10} \gamma_r\, P_{ir} \\ + A_i'\theta + X_i'\delta + M_i'\lambda + S_i'\kappa + \alpha_{z(i)} + \varepsilon_i,\end{aligned} \tag{5}$$

where ΔY_i is now the change in the net margin of block group i, the change in the Democratic share of all votes cast minus the change in the Republican share between the 2016 and 2020 elections, in percentage points; D_i and $(L_i - c)\, D_i$ are the extensive and intensive margins of protest participation defined in Section 5.1, with c the mean of L_i among the sending block groups of this sample; D_{ir} are the ten exposure rings of Section 4.1 and P_{ir} the same rings for the protests of January 2017 to May 2020; A_i, X_i, M_i and S_i are the controls of the two sections, and $\alpha_{z(i)}$ the zip code fixed effects. The regression is weighted by votes cast in 2020, clustered by metropolitan area, and estimated on the

[55] Of the 133,406 block groups of the joint sample, 10,530 both had a protest venue within one kilometer and sent at least one resident to a protest, 12.5 percent of the block groups that sent anyone and 82 percent of those with a venue within one kilometer.

133,406 block groups on which both treatments are observed together, in 333 metropolitan areas.

Figure 8 draws three quantities from the estimated coefficients on one scale, for all protests and split by protest violence. The two treatments coexist and pull in opposite directions. Holding exposure fixed, the extensive margin of participation shifts the net margin 0.15 percentage points toward the Democrats, significant at the 1 percent level. Holding participation fixed, a protest venue within one kilometer shifts it 0.36 points toward the Republicans, also at the 1 percent level. Heavier sending among the block groups that sent anyone adds no detectable shift (Table A33). The exposure profile survives. In the pooled fit the shift toward the Republicans is detectable in the first two rings, which jointly reject zero, and from the third ring outward no ring is distinguishable from zero at the 5 percent level (Table A33). Split by protest violence, the violent rings within two kilometers are distinguishable from zero only at the 10 percent level (Table A34).[56] In the same fit the participation shift sits at nonviolent protests and heavier participation at violent protests moves the margin toward the Republicans, as in Sections 4.3 and 5.3. On turnout, the fourth outcome of Tables A33 and A34, no term within three kilometers is distinguishable from zero, so the turnout fall of Section 4.2 is not detectable in the joint sample.

6.2 The size of the two effects

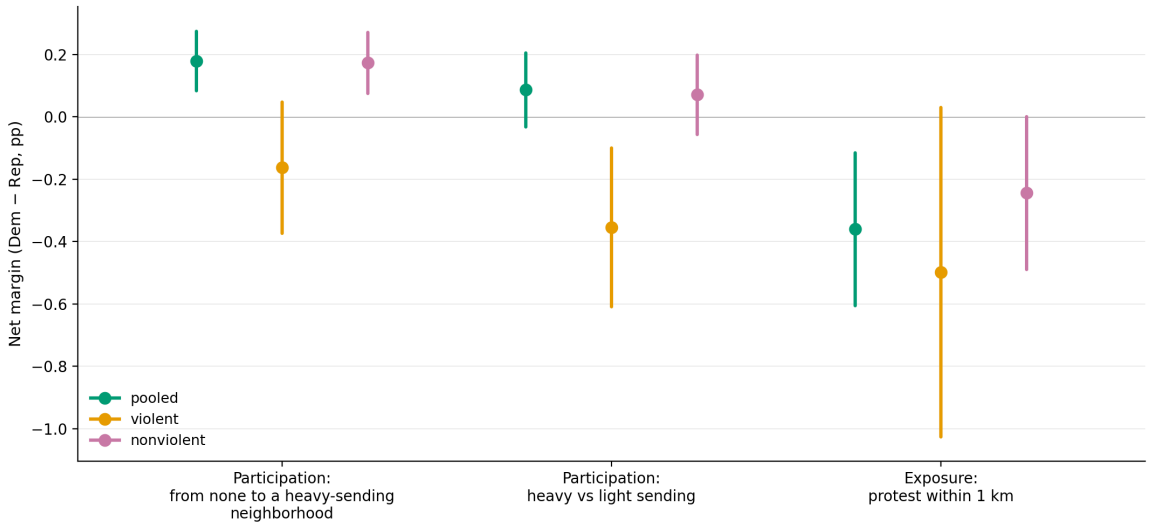
Each coefficient of equation (5) is the shift for one block group that received the treatment, but the two treatments reached very different numbers of block groups. To put them on one footing, each coefficient is converted into an average effect over the block groups that received its treatment. Both averages come from the joint regression split by protest violence (Table A34). For exposure, the average over the block groups with a violent protest within one kilometer is 0.62 percentage points on the Republican share.[57] For participation, the average over the block groups that sent anyone to a protest is 0.09 on the Democratic share (Table A35, Panel A). The backlash effect is the larger and the less precisely estimated. The mobilization effect is smaller and precise.

To give the two effects a common scale, each is expressed as a persuasion rate, the measure used to compare media and campaign interventions that reach different shares of their audience (DellaVigna and Gentzkow, 2010). The persuasion rate is the share of the people a treatment reached, among those not already voting the way it pushes, whose vote it changed. It divides the estimated shift in the vote share by the reach of the treatment and by the share of voters left to persuade. The shifts are the estimates of Sections 4 and 5—the 0–1 km coefficient for exposure and the extensive margin for participation. The reach of exposure is not observed. The rate is therefore reported under three assumptions: that every resident within one kilometer was reached, which gives the lower bound on the rate, that one half were, and that one quarter were. The reach of participation is measured as

[56] Farther out, the violent ring at 4–5 km and the nonviolent rings at 4–5 and 7–8 km reach the 5 percent level. As in Section 4.2, these rings are not read as effects of the protests.

[57] The average for exposure is larger than the 0.34 ring coefficient of Section 4.3 because the block groups with a violent protest within one kilometer usually also had violent protests farther out, nearly half of them within two kilometers, and the average adds the coefficients of those rings as well, out to ten kilometers.

Figure 8: Participation, exposure and the change in the net margin, by protest violence



Notes: Each marker is an estimated shift in the net margin, the change in the Democratic share of all votes cast for president between the 2016 and 2020 elections minus the change in the Republican share, in percentage points; a marker above zero is a shift toward the Democrats. Whiskers are 95 percent confidence intervals. The three groups along the horizontal axis are three quantities: sending at the 90th percentile of the participation rate, over all block groups, against sending nobody; the 90th against the 10th percentile of the rate among the block groups that sent anyone; and at least one protest venue within one kilometer of the block group. Within each group the three series are all protests (pooled), violent protests and nonviolent protests. The first two groups are built from the two margins, so their markers are not the coefficients of Table A33: 0.18, −0.16 and 0.17 in the first group and 0.09, −0.35 and 0.07 in the second. The regression is of the net margin on the participation extensive and intensive margins and on ten indicators for a protest venue in each one-kilometer ring, with the controls below and zip code fixed effects, weighted by votes cast in 2020; the all-protests series comes from the pooled fit of Table A33, where the equation is written out, and the violent and nonviolent series from the split fit of Table A34. The controls are those of Figure 2 and Table A18 together: the earlier-protest ring indicators, which in the split fit are split by protest violence as the exposure rings are, the demographic controls, the venue-traffic controls, the neighborhood mobility controls and the indicators for deciles of the block group's panel devices. The sample is the 133,406 block groups on which both joint fits are estimated. Each whisker is built as in Table A36, on 333 metropolitan-area clusters. On exposure the two ring families are not statistically distinguishable, over all ten rings jointly ($F(10, 332) = 1.41$, $p = 0.174$) or at the one ring drawn ($p = 0.399$), so the gap between the violent and nonviolent markers of the third group is a difference in point estimates.

the number of attendances per voting-age resident of the sending block groups, divided by the share of a protest's crowd that the device panel detects. That share falls as crowds grow (Figure A2), so a single share measures the reach only approximately. The shift it is divided into is measured over the whole block group, attenders and their neighbors alike, so the rate of participation is an upper bound on the persuasion of the attenders themselves only if their neighbors did not move the other way. It is also a rate per attendance, not per person.

Figure 9 sets the lower bounds for exposure and the rate at median coverage for participation beside published ones. The persuasion rate of exposure to any protest is 0.25 percent at the lower bound, 0.50 percent if one half of the residents were reached and 1.00 percent if one quarter were. For

exposure to a violent protest the three rates are 0.41, 0.82 and 1.65 percent.[58] The shift near a protest comes mainly through fewer Democratic votes (Section 4.2), so these rates measure a change in the vote share and not necessarily votes switched. The persuasion rate of participation is 2.19 percent at the panel's median coverage of a crowd and 10.39 percent at the 75th percentile of that coverage. If instead every resident of the sending block groups is counted as reached, the rate is 0.16 percent, below the lower bound of the exposure rates (Table A35). The rates of exposure are of the order of the rate of a candidate's whole television advertising campaign (Spenkuch and Toniatti, 2018) and below every other published rate in the figure, among them the arrival of Fox News on local cable (DellaVigna and Kaplan, 2007) and door-to-door canvassing (Pons, 2018). At the measured reach, the rate of participation is above the rates of exposure, and at median coverage it is below every published rate in the figure except the television campaign's. Their reach differs: a protest within one kilometer is assumed to reach between a quarter and all of a block group's residents, while a sending block group's reach, measured as above, is between 1.5 and 7.2 attendances for every hundred of its voting-age residents (Table A35).

6.3 The net effect in votes

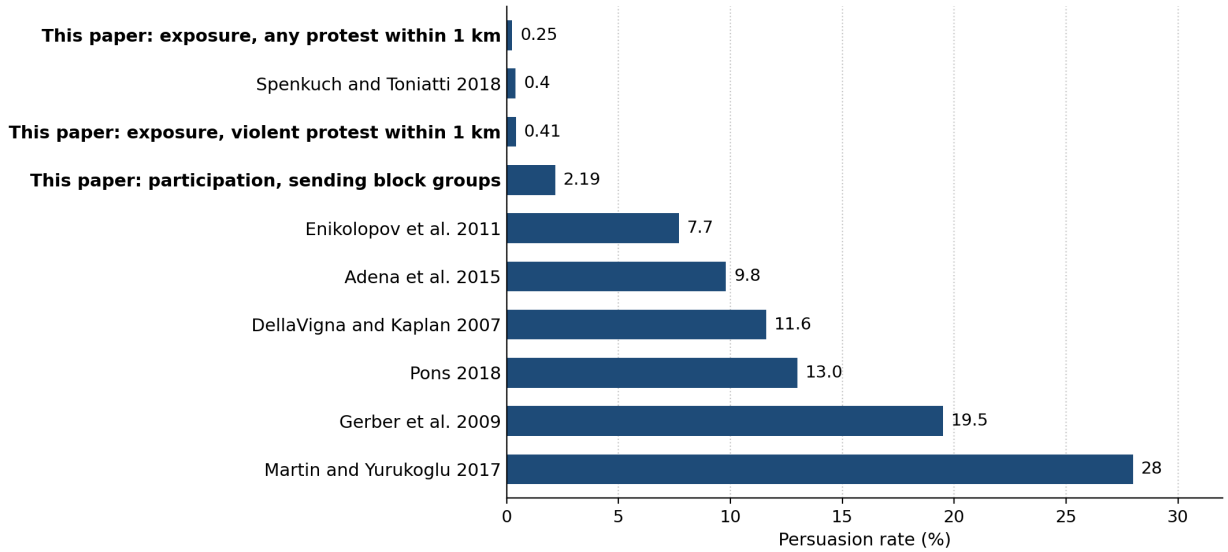
What do the two effects add up to in votes? A back-of-the-envelope calculation applies the coefficients of the joint regression to each treated block group of the sample and sums the implied vote shifts. It therefore speaks about the electorate in the sample and not about the national electorate (Figure 10; Table A36 is the full decomposition, with every interval). Participation moved about 137,000 votes toward the Democrats, with an interval clear of zero. About 38,000 of them come from the block groups where participants in a violent protest lived, with an interval clear of zero, even though both violent participation coefficients are negative. As Section 5.3 noted, most of those block groups had a low participation rate, where the two coefficients together imply a shift toward the Democrats (Table A36). Exposure within three kilometers of a venue moved about 87,000 votes toward the Republicans, also with an interval that excludes zero (Table A36). Beyond three kilometers the exposure contribution is about 54,000 votes toward the Democrats, but it is not distinguishable from zero, and with the rings drawn differently the total exposure contribution changes sign (Table A37). The net of the two is 104,428 votes moved toward the Democrats, which amounts to one tenth of a percentage point of the net margin over the sample's electorate, and it is not distinguishable from zero. Within three kilometers, participation and exposure together moved about 50,000 votes toward the Democrats, also not distinguishable from zero (Table A36).[59]

The net is imprecise because the specification carries the exposure rings out to ten kilometers. Section 4 showed the coefficients beyond three kilometers imprecisely estimated, so the votes those rings imply are imprecise too, while the contribution within three kilometers is precise on its own with

[58] These rates use the 0–1 km coefficients as estimated. Net of the mean under random placement (Section 4.7), the coefficients are 0.11 and 0.23, and the rates fall in the same proportion.

[59] Net of the mean under random placement (Section 4.7) the 87,000 falls to about 29,000 votes and this sum rises to about 108,000, and only the second is distinguishable from zero.

Figure 9: The paper's persuasion rates against published persuasion rates



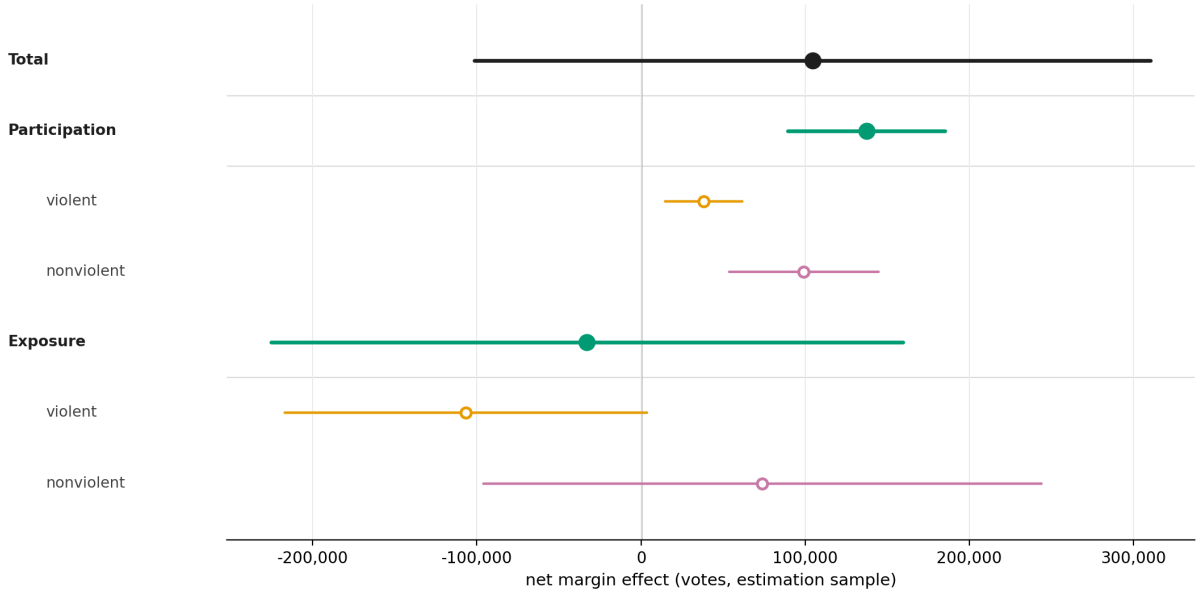
Notes: The figure sets three persuasion rates implied by this paper's estimates, in bold, beside persuasion rates published for media and campaign interventions. Each bar is one rate in percent: among the people a treatment reached who were not already voting the way it pushes, the share whose vote it changed (DellaVigna and Gentzkow, 2010). The rate is $f = 100 \times (y_T - y_C)/[(e_T - e_C)(1 - y_0)]$, where $y_T - y_C$ is the effect on the vote share, $e_T - e_C$ the reach, the share of the treated group that the treatment reached net of the comparison group's, and y_0 the share already voting that way. For exposure, $y_T - y_C$ is the 0–1 km coefficient of Table A8 (any protest) or of Table A9 (violent protest), and y_0 the mean 2016 Republican share of the block groups with such a protest within one kilometer. Who among their residents saw the protest is not observed, so the two exposure bars assume a reach of one, every resident within one kilometer reached, which gives the lower bound on the rate. For participation, $y_T - y_C$ is the extensive margin of Table A18, y_0 the mean 2016 Democratic share of the block groups that sent anyone, and the reach is measured: attendances per voting-age resident of the sending block groups, divided by the median share of a protest's reported crowd that the device panel detects, 0.04 over the 4,211 events with a reported crowd size. Table A35 gives the rates under the other reach assumptions. Each published rate uses its own paper's definition of reach: Spenkuch and Toniatti (2018), the Republican candidate's whole advertising campaign, from advertising impressions; Enikolopov et al. (2011), as listed in Table 1 of DellaVigna and Gentzkow (2010), the audience measured by recall; Adena et al. (2015), the 1933 election, with an assumed number of listeners per radio subscription; DellaVigna and Kaplan (2007), as listed in the same table, the audience measured by viewing diaries; Pons (2018), the second round, the households that opened their door; Gerber et al. (2009), as listed in the same table; and Martin and Yurukoglu (2017), the 2008 election, from a structural model. The bars are point estimates.

this ring structure.[60] Over 2016 to 2024 the same decomposition gives a participation contribution of about 188,000 votes toward the Democrats, with an interval clear of zero, and a total of about 203,000 that is not distinguishable from zero.

Could the direct local effects of the protests have decided the presidential election by themselves? Unlikely, on these estimates. Table A38 applies the same coefficients to the treated block groups of the six closest states that the sample covers, and sets the estimated net shift against each state's final margin. In every state the participation contribution is distinguishable from zero on the national coefficients and the exposure contribution is not. Only in Georgia is the estimated net

[60] With rings two and a half kilometers wide (Table A37) participation barely moves, the exposure contribution changes sign and straddles zero, and the near band of 0–2.5 km no longer excludes zero though its sum with participation does.

Figure 10: The aggregate net effect in votes, by treatment and protest violence



Notes: Each marker is a number of votes and each whisker a 95 percent confidence interval on that number. The outcome is the net margin: votes shifted toward the Democrats minus votes shifted toward the Republicans, so a marker to the right of zero is a net shift toward the Democrats. The aggregate is built by applying the estimated coefficients to the treated block groups of the estimation sample and summing the implied vote shift over them. It is therefore a statement about the electorate in the sample—133,406 urban block groups—and not about the national electorate. The rows are the total and the two treatments, participation and exposure, each split by protest violence. The coefficients are those estimated from the joint regressions of Tables A33 and A34. The dependent variable is the change in the net margin between the 2016 and 2020 elections, in percentage points; the independent variables are the participation extensive and intensive margins and the ten one-kilometer exposure rings, with the controls and zip code fixed effects of Figure 8, weighted by votes cast in 2020. Each interval is built as in Table A36, with standard errors clustered by metropolitan area (333 clusters). The whiskers represent the interval in votes, with the percentage-point endpoints multiplied by the summed 2020 vote weight. The total is not distinguishable from zero whatever its point estimate, and the point total is not evidence that the movement gained votes on net. The difference between the violent and the nonviolent distance profiles is not statistically established on this fit's own test over all ten rings ($F(10, 332) = 1.41$, $p = 0.174$). Table A36 is the full decomposition behind this figure, with the distance bands.

shift distinguishable from zero at the 5 percent level and on the scale of the final margin—5,753 votes against 11,779—though still short of it.[61] In the other five states the estimated shift is well below the margin. Across the three states with the smallest margins it adds up to 7,681 votes against a combined margin of 42,918.

Two limits of this calculation should be kept in mind. First, the sample includes only metropolitan areas, whose electorate leans more Democratic than the nation's (Section 2.3). Second, the votes counted in this analysis are only those moved by direct exposure to protests and by participation in protests, between block groups of the same zip code. Whatever the protests did to the vote through the media, party politics or any other factor that is common to all the block groups of a zip code is absorbed by its fixed effect and is not part of this count. If a protest also reached, through

[61] Net of the mean under random placement (Section 4.7) Arizona, Pennsylvania, North Carolina and Michigan also reach the 5 percent level.

spillover effects, the block groups of its zip code that had none nearby, the estimates measure only the difference between the exposed block groups and their unexposed neighbors, and the aggregate counts none of what the protests did to those neighbors, which could run in either direction. The sum also applies the same coefficient to every block group a treatment reached, adds the ring coefficients for a block group with venues in several rings, and applies the intensive margin at each block group's own attendance rate.

6.4 The cost of the wave's violence

A striking feature of the George Floyd movement compared with the American protest movements that preceded it was its violence. 10.4 percent of its events turned violent, against 2.2 percent of American protests in the three preceding years. The last exercise asks what violence may have cost the movement in votes. It re-runs the aggregate of Section 6.3 under alternative shares of violent protests, to answer three questions: what the net shift would have been had no protest turned violent, what it would have been with only as many violent protests as in earlier movements, and at what share of violent protests the backlash would have outweighed the mobilization gains.

The exercise rests on two strong assumptions. First, participation is held at its observed level in every scenario, since the attendance response to a protest's character is not observed and is not modeled. Second, the geography of protests is unchanged. For a violent share below the observed one, violent protest days are relabeled as nonviolent at random until the share reaches its target. For a share above it, nonviolent protest days are relabeled as violent. The distance-ring exposures are then recomputed under the new labels.

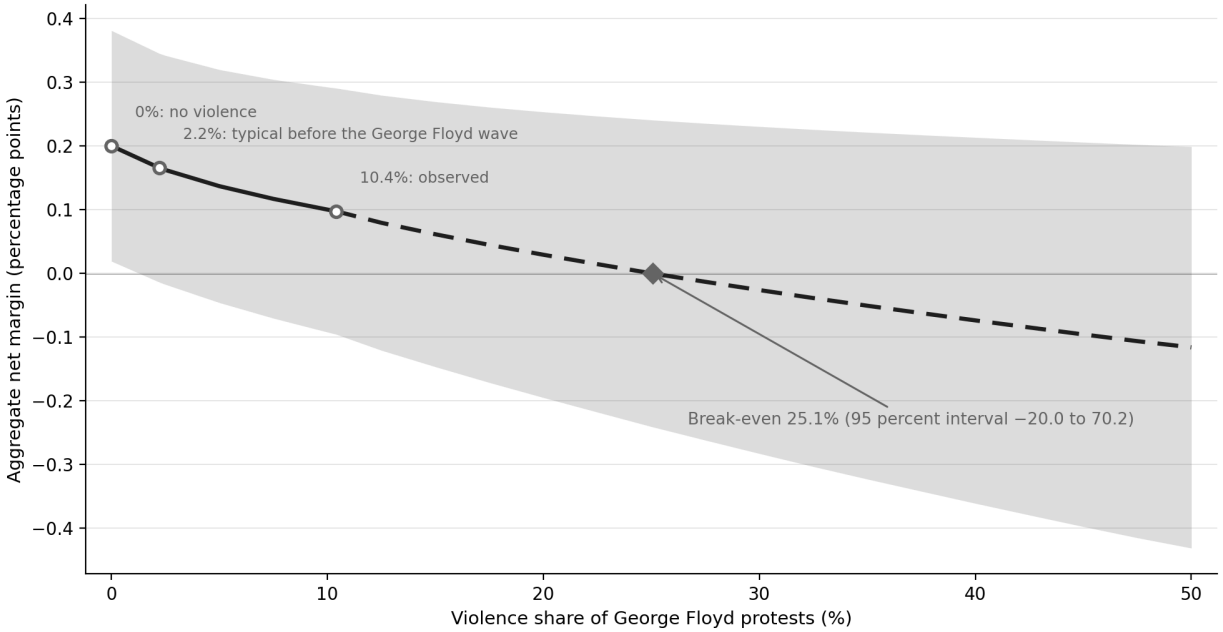
On the point estimates, violence is associated with a loss of Democratic votes. With no violence at all the net shift would have been about twice what was observed, and at the violence rate of earlier movements it would already have been most of the way to that level (Table A39). The difference between the net shift with no violence and the observed net shift is not distinguishable from zero.[62] What the point estimates suggest is that violence may have cost the movement about half of the net shift it would otherwise have produced in the neighborhoods it reached. Figure 11 traces the same exercise over every violent share: the point path crosses zero, the share at which the backlash outweighs the mobilization gains, at 25.1 percent, one protest in four, far above the 10.4 percent that turned violent.

7 Mechanisms: Evidence from the North Carolina Voter Files

Sections 4 and 5 found two shifts in the vote, toward the Republican candidate in block groups near a protest venue and toward the Democratic candidate in block groups whose residents went to one. A vote share alone cannot say how these shifts arrived. The same change in a party's share can

[62] The difference is 0.10 percentage points, with a 95 percent confidence interval from −0.002 to 0.207. It is 0.07 if attendance at violent protests is also relabeled, 0.03 within three kilometers, about zero with both changes, and 0.05 net of the mean under random placement.

Figure 11: The aggregate net effect as the share of violent protests varies



Notes: The figure traces the aggregate net effect as the share of protests that are violent is varied, holding everything else at what was observed. The horizontal axis is that share; the vertical axis is the aggregate net margin in percentage points, on the same construction and the same 133,406 block groups as Figure 10. The line is the point path, solid up to the observed share and dashed beyond it. The shaded band is the 95 percent confidence interval of the net margin at each share, not of the crossing. Landmarks are marked along the axis: the rate observed in the George Floyd wave, the rate typical of American protests in the three years before it, and the share at which the point path crosses zero. This is a model-based reallocation and not a causal claim about where violence occurred or why. It reallocates the exposure channel only: the distance rings are recomputed under counterfactual violence labels. The mobilization channel is held at its observed level at every point on the path: the attendance response to a protest's character is not observed and is not modeled. Every share on the axis above the observed rate is outside the range the data contain, so the dashed part of the path is an extrapolation. The crossing is at a violent share of 25.1 percent, with a 95 percent confidence interval running from −20.0 to 70.2 percent, wider than the whole range of shares the figure draws, so the crossing locates where the central estimate changes sign and establishes nothing about where the sign actually changes; it is a landmark on the point path and not an estimated threshold. The scenarios behind the path are in Table A39.

come from people who were already voting changing their vote, from people turning out at different rates than before, or from a change in who lives and votes in the block group. Sections 4.2 and 4.3 could say only that the votes missing near a protest were Democratic ones.

This section uses the North Carolina voter files described in Section 2.4 and Appendix A.4 to separate three channels: turnout, the composition of the roll, and the movements that change it. North Carolina's files allow each channel to be followed person by person on the block groups where the vote is measured.[63] At the individual level, I sort registrants into three classes by their record before the protests: habitual voters, who voted in both the 2016 general election and the 2018 midterm; occasional voters, who voted in one of the two; and registrants who voted in neither.[64]

[63]Few states publish a file with all of these. Many states do not register voters by party at all, several restrict the file to their own residents, parties or candidates, and almost none keeps the file as it stood at past elections.

[64]Fewer than half of the state's counties hold a sample block group, so the asymptotic standard error is not trusted

The first question is whether the national results appear in this state at all. Table A40 re-runs the exposure specification of Section 4.1 and the participation specification of Section 5.1 on urban North Carolina, beside their national estimates. The shift toward the Republicans near a violent protest appears in the state too. Within one kilometer the Republican share rose 0.99 percentage points more than in block groups of the same zip code without a violent protest in that ring, significant only at the 10 percent level, against 0.34 nationally.[65] The participation shift has no detectable counterpart.

Every channel is estimated in the joint specification of Section 6, with both treatments, its controls other than the four mobility measures, and zip code fixed effects. Table A41 estimates each channel again under each design's own specification, and every estimate distinguishable from zero keeps its sign and rough size there. Two of them, the violent-protest registration estimate and the new-registration extensive margin, are no longer distinguishable from zero there.

The first channel is turnout. Figure A9 plots the change from 2016 to 2020 in the turnout of the people on the roll at both elections, by distance to a violent and to a nonviolent protest. Within one kilometer of a nonviolent protest turnout fell by 1.51 percentage points.[66] Within one kilometer of a violent protest the estimate is 0.34 percentage points, with an interval that covers zero. On the point estimates this is the reverse of the national pattern, where the turnout fall was larger near violent protests (Section 4.2). The fall near nonviolent protests is present for registered Democrats, Republicans and unaffiliated voters alike, each with an interval below zero (the right panel of the figure; Appendix G.2). Which registrants stopped voting? Figure A10 sorts the same people by their voting history and, within the block group, compares each class's change in turnout with that of the habitual voters. Within one kilometer of a nonviolent protest the turnout of occasional voters fell 4.38 percentage points further than the turnout of habitual voters, and the turnout of registrants who had not voted 1.50 further, both distinguishable from zero. Within one kilometer of a violent protest the turnout of occasional voters fell 3.83 points further and the turnout of registrants who had not voted 1.64 further, the first significant only at the 10 percent level and the second not distinguishable from zero. When the block group's controls and earlier-protest rings are allowed to differ by class, none of the four contrasts is distinguishable from zero at the 5 percent level, and those for occasional voters fall to 2.08 and 2.65 points.

The second channel is the composition of the roll: who is registered, and with which party. Near nonviolent protests the Democratic share of registered voters did not detectably change (Table A41). Within one kilometer of a violent protest it fell by 1.00 percentage points, significant only at the 10

on its own: every star on a North Carolina estimate in this section and in Appendix G comes from a wild cluster bootstrap on the county clusters (Cameron et al., 2008), the national column of Table A40 keeps the asymptotic stars of the paper's own estimates, and the tables print the asymptotic standard error beside each coefficient.

[65] Table A40 prints the coefficients, with the joint specification of Section 6 as a third block; its national cells are the paper's own estimates. The North Carolina point estimate is about three times the national one, on a much wider interval; Appendix G.1 reads that comparison.

[66] Among the registrants at the same address in 2016 and 2020 the fall is 2.19 points, and on the whole 2016 roll, counting the registrants removed by 2020 as non-voters, 1.25 points, both distinguishable from zero at the 1 percent level.

percent level. Figure A11 splits that change into the five movements of Section 2.4, each estimated as its own outcome on the same specification, so that the five sum to the total. The whole of the decline comes from new registration. The roll became less Democratic because of who newly joined it, not because anyone already on it left or changed party. The newcomers were 41.8 percent Democratic, above the 31.5 percent of new registrants in the rest of the sample but below the 51.2 percent of the roll they joined, so their arrival lowered its Democratic share.[67]

The worry this raises is that the block groups near the George Floyd protest venues were already registering less Democratic before the protests. Table A42 tests it directly, taking the same five movements over the earlier window from the 2012 to the 2016 election and regressing each on the 2020 treatment.[68] Within one kilometer, the ring that carries the registration result, no movement loads on the 2020 treatment and neither does the total, but there the check is imprecise rather than clean. The intervals on the total and on new registrations include declines of about one percentage point.[69] Every one of the 50 block groups with a violent protest within one kilometer is in the upper half of the sample's distribution of residential turnover over 2012 to 2016, and none is in the lower half, so at this distance proximity to a violent protest cannot be separated from being a high-turnover neighborhood.

Read together, the voter files paint the following picture. Near violent protests, the roll became less Democratic through new registrants while the turnout of the people already registered did not detectably change. Near nonviolent protests turnout fell while the composition of the roll did not detectably change. On the extensive margin the participation shift has no detectable counterpart in the state's vote, and the intensive margin is significant only at the 10 percent level, and only in the joint specification. On the roll measures the attendance terms are not distinguishable from zero at the 5 percent level, except for Democratic new registration in the joint specification (Table A41). With 2.7 percent of the national sample, the state can show how a shift arrives but not how large it is nationally. A fall in turnout does not by itself give the direction of a shift, which also depends on each party's earlier turnout and on how those who stayed home would have voted. Near nonviolent protests, where turnout fell, the state's Republican share did not detectably rise, at -0.14 points within one kilometer. In this state, then, the backlash near violent protests is consistent with a shift that arrived through who registered, a channel that cannot be separated here from the turnover of those neighborhoods. Near nonviolent protests turnout fell with no detectable shift in the vote. Nothing here measures a change in anyone's preferences.

[67] New registration rose in every party near violent protests, among Republicans and unaffiliated voters as well as Democrats, so the Democratic row is part of a general registration surge rather than a partisan shift (this is consistent with the national pattern of rising party registration after the murder of George Floyd documented by Hassell and Holbein, 2026).

[68] The voter history file begins with the 2016 election (Appendix A.4), so the turnout channel has no earlier window on which the same check could be run.

[69] At two to four kilometers the check is not clean, and the estimates at those distances are not read as effects of the 2020 protests (Appendix G.4).

8 Conclusion

This paper studies the direct local effects of the George Floyd protests on the 2020 presidential election. I assemble a new granular dataset at the level of the census block group that combines the geocoded locations of the protests, protest participation measured from smartphone location data, and the returns of the presidential elections of 2012 to 2024. I compare the change in vote shares between the 2016 and 2020 elections in block groups that had a protest nearby with block groups of the same zip code that did not, and in block groups where protest participants lived with block groups of the same zip code where none did. The zip code fixed effects hold fixed what reached every block group through the media and national politics, and isolate the direct local effect of the protests through exposure and participation.

Exposure to protests nearby produced a backlash against the movement, while participation generated a mobilization for it. Block groups within one kilometer of a protest shifted 0.19 percentage points more toward the Republican candidate than block groups of the same zip code with no protest within one kilometer. Block groups whose residents attended a protest shifted 0.07 percentage points more Democratic than block groups of the same zip code that sent none. Estimated in one joint regression, the two effects coexist and pull in opposite directions. The point estimates of the backlash are larger near violent protests, while the mobilization is driven by participation in nonviolent protests. Neighborhoods that sent more residents to violent protests moved away from the Democratic candidate, and so did neighborhoods whose attenders regularly went to the same venues on days without a protest. Participation moved about 137,000 votes toward the Democrats and exposure within three kilometers about 87,000 votes toward the Republicans. The net is not distinguishable from zero and the direct local effects are unlikely to have decided the election. On the point estimates, violence may have cost the movement about half of the local gains it would otherwise have made.

This paper contributes to the ongoing debate on the effects of protests on electoral outcomes in democratic countries. As protests become an increasingly common tool for citizens to express their political preferences outside of elections, it matters how they affect the broader population, which does not take part in them and may sympathize with the cause of the protesters or turn against it. I argue that two relevant dimensions along which a reaction against protests may appear are the residents who live near the protests, and bear the disruption they bring, and violence, which may increase the salience of that disruption. The effects that run through the media and party politics are outside the scope of this paper, and are likely to push the overall effect in either direction depending on the context. What this paper establishes is that, at the neighborhood level, the George Floyd protests moved the vote in two opposite directions at once—toward the movement where its participants lived and against it where its protests took place—and that their net effect is not distinguishable from zero.

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A Data and sample

This appendix describes how the four sources of Section 2 were compiled: the protest record, the device panel and its home rule, the allocation of precinct returns to block groups, and the North Carolina voter file.

A.1 The protest record

The Crowd Counting Consortium records marches, protests, demonstrations, rallies, strikes, riots and similar actions, and not meetings, panel discussions or town halls; its record begins with the Women's March of January 2017. From April 2017 its main input was a daily crawl of more than 3,400 local and national news outlets, run first by the Count Love project and later by the Consortium itself. The crowd size as the sources described it is converted into a low and a high count by fixed rules.

The compiled file for 2017 to 2020 holds 72,181 events, of which 21,049 are dated between 27 May and 2 November 2020. I drop the events held online, the events without a location within the city or without a state, the events whose location is ambiguous, and the duplicate records of one event, which leaves 17,312. The locations are geocoded with the Google Maps Geocoding API, and five events cannot be geocoded. I keep the events whose coordinate is a street address, the center of a named place or street, or a point interpolated along a street, and drop the 3,170 whose coordinate is only approximate, at the level of a town or a district. The earlier protests that serve as controls (Section 2.1) are not filtered in this way: the 5,955 of them located only to a town or a district are kept. Of the rest, 1,611 fall outside every metropolitan area and 4,606 are not related to Black Lives Matter, counter-protests among them, which leaves the 7,920 events of the wave. They took place at 4,815 venues; 871 venues hosted more than one event, and the busiest hosted 154. Of the 823 violent events, arrests are the most common of the five markers, reported at 555 of them.

A.2 The device panel and the home rule

A device's home is the block group where it spent most of its nights in September and October 2020, taken from each night's median location, and a device is homed only if it is observed on at least ten nights in that window with at least half of them in that block group. The rule homes 12.9 million devices, fewer than half of the 27.5 million that a rule based on all location pings homes (Table A2), and the data do not describe the devices it leaves out.

A block group typically covers a few city blocks. The Census Bureau targets between 600 and 3,000 residents when it draws a block group. I use the 2010 census block groups as published in the 2018 TIGER/Line release throughout, although the Census Bureau redrew block groups for the 2020 census, so the 2020 and 2024 returns are allocated with 2010 populations. The few whose identifier differs between the 2010 and the 2018 releases are matched by spatial overlay (Appendix A.3).

The ordinary presence that enters every participation regression as a control is measured at the

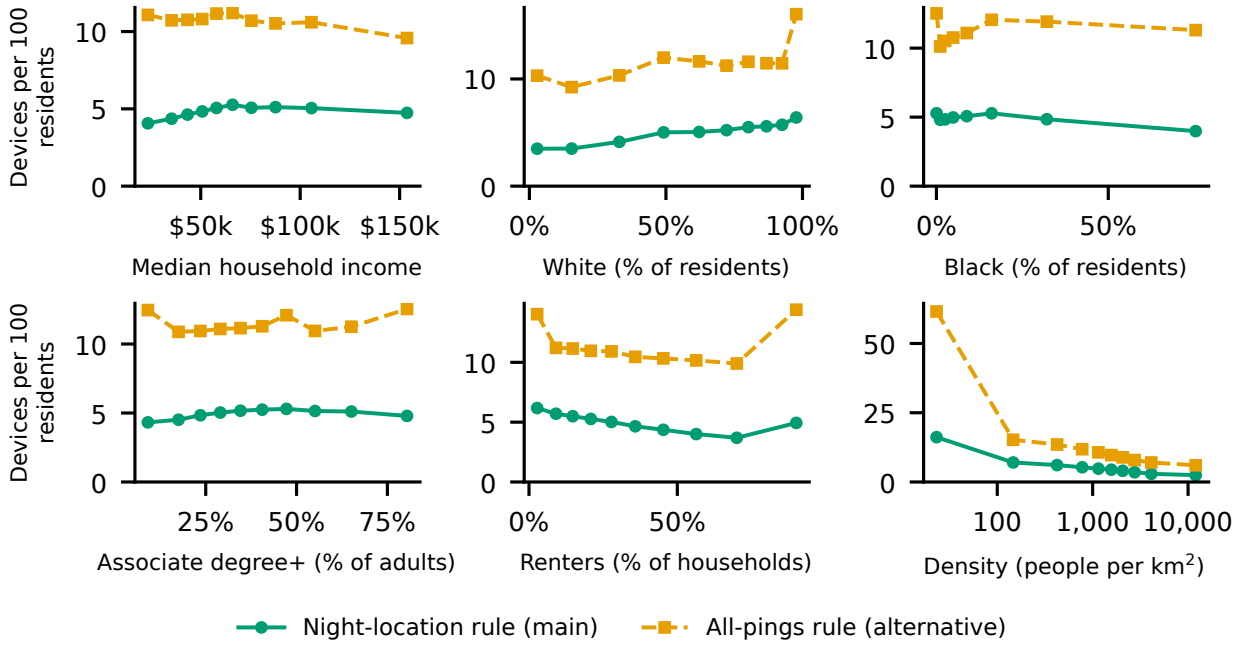
same venues between 4 and 26 May 2020, the three weeks before the first event, on days those venues hosted nothing. The four measures of how far residents' devices ranged on an ordinary day, the mobility controls, are taken over January and February 2020, before the pandemic. The data do not record how many detections the ten-minute rule of Section 2.2 removed, so its effect on the attendance count cannot be reported. The disclosure rules stated in the note to Table A1 apply to every aggregate. Table A1 reports the panel, the attendance it detects and the voting sample; Figure A1 and Table A2 compare the panel's coverage and composition with the resident population.

Table A1: The device panel, detected attendance, the voting sample and the disclosure rules

	Value
Homed devices in the panel	12,867,138
Block groups in the panel	173,544
Metropolitan areas covered	368
Attendances, George Floyd window, by venue detection radius	
50 m	50,185
100 m	137,882
200 m (primary)	385,891
400 m	1,031,983
Block groups with any detected attendance (200 m)	108,679
Voting sample (participation estimation sample)	
Block groups in the sample	135,691
... with any detected attendance (200 m)	85,399
2020 ballots cast in the sample	108,552,069
... as a share of all 2020 ballots cast nationally	68.5%
Disclosure regime (applied before export)	
Block group × day cells built	64,896,796
Day-level small-count suppression	counts of 1–4 withheld (minimum five devices)
Thin-panel rule	block groups with < 10 homed devices withheld
Block group × day rows withheld as thin	1,231,946
Participation events, all hours, export year	587,009
... share sitting in withheld day cells	85.0%
... effect on the attendance measure	none (wave totals published down to 1)

Notes: A metropolitan area is a Census core-based statistical area; the panel covers the 368 that contain at least one protest venue. A device's home is the block group where it spent most of its nights in September and October 2020 (Appendix A.2). An attendance is one homed device observed within the stated distance of a protest venue on that venue's event date, between noon and midnight local time, counted once per device and day; the counts are summed over the George Floyd wave. The block groups with any detected attendance are those with at least one attendance at 200 m, counted over the whole panel in the attendance block and over the voting sample in the voting block. The voting-sample rows describe the participation estimation sample, the block groups on which the participation estimates of Section 5 are fitted, 135,691 in 333 metropolitan areas. The exposure estimates of Section 4 do not require the device panel and use 136,644 block groups in 344 metropolitan areas. The disclosure rows describe the rules applied: counts are built at the block group by day level; a day-level count of one to four is withheld; a block group with fewer than ten homed devices is withheld; a participation event is one device-day of a homed device at a venue on a day it hosted a protest, over all hours of the day and over the whole of the 200 m export's year, 31 December 2019 to 31 December 2020, the disclosure report's own count. Because most of these events sit in withheld day cells, attendance enters the paper only as a total over the wave.

Figure A1: Device coverage per resident by block-group characteristic



Notes: Each panel is a binned scatter of device coverage against one block-group characteristic. Coverage is the number of homed devices per 100 residents; each marker is the mean coverage among the block groups in one bin of the characteristic on the horizontal axis, with bins of equal count. The solid series homes each device to the block group where it spent most of its nights (the night-location rule of Appendix A.2, used throughout the paper); the dashed series homes it to the block group holding most of its location pings (the all-pings rule, shown as an alternative). % associate+ is the share of residents aged 25 and over with an associate degree or higher. The sample is every block group in the 368 metropolitan areas with a protest venue.

Table A2: Composition of the homed-device panel against the Census benchmark

	Census	Night-location rule (main)		All-pings rule (alternative)	
	mean	Device mean	Gap / SD	Device mean	Gap / SD
% White	0.585	0.651	0.21	0.622	0.12
% Black	0.130	0.117	-0.06	0.131	0.00
% Hispanic	0.191	0.152	-0.17	0.163	-0.12
% Asian	0.060	0.049	-0.11	0.051	-0.09
Median age	39.1	39.3	0.02	39.2	0.01
Median household income ($)	71,933	73,742	0.05	71,156	-0.02
Unemployment rate	0.061	0.056	-0.07	0.059	-0.03
% associate degree or higher	0.409	0.417	0.04	0.407	-0.01
% renter	0.364	0.316	-0.18	0.345	-0.07
Log population	7.425	7.475	0.07	7.422	-0.00
Log density	6.822	6.325	-0.28	6.383	-0.25

Notes: The Census column is the population-weighted mean of the characteristic across block groups; the Device mean columns are the device-weighted means, in which each block group counts in proportion to the devices homed there, under the two home rules; Gap/SD is the device-weighted mean minus the population-weighted mean, divided by the standard deviation of the characteristic across block groups. The night-location and all-pings rules are those of Figure A1. Shares are fractions of the block group's population (% White, % Black, % Hispanic, % Asian), of residents aged 25 and over (% associate or higher), of the labor force (unemployment) or of housing units (% renter); log population and log density are natural logarithms of residents and of residents per square kilometer. The sample is every block group in the 368 metropolitan areas with a protest venue; at the homing step, before the thin-panel rule of Table A1 withholds the block groups with fewer than ten homed devices, the night-location rule places 12,890,502 devices in 177,205 of the 177,543 block groups and the all-pings rule 27,458,445 devices in 177,468. A gap within 0.15 standard deviations is read as a match.

A.3 How precinct returns are allocated to block groups

Precincts are the finest unit at which election returns are published, and their boundaries neither follow Census boundaries nor stay fixed between elections. The paper's unit is the block group, so the returns of 2016, 2020 and 2024 are moved from precincts to block groups by the procedure described here; the 2012 returns arrive at the block-group level from another source, described at the end.

The precinct geometries are the 2016 and 2020 files of the Voting and Election Science Team, 177,194 and 170,200 precincts covering the fifty states and the District of Columbia, and for 2024 the precinct file of the New York Times completed with the Redistricting Data Hub files for Alaska, New Hampshire, Rhode Island and Vermont, 162,766 precincts in 48 states and the District of Columbia, with Connecticut and Maine at the township level. The block-group geometries are the 217,739 block groups of the 2018 TIGER/Line release.

Each precinct polygon is intersected with the block-group polygons in an equal-area projection, and every piece of the intersection is given a population: the block group's population in the 2010 census multiplied by the share of the block group's area that the piece covers. A precinct's votes for each candidate are then split across its pieces in proportion to these piece populations, so that each precinct's votes are conserved by construction. The method assumes that population is spread evenly within a block group and that a precinct's vote shares are the same everywhere within the precinct.

Where a precinct's overlapping population is zero, its votes are split by area instead, which happens for 67, 71 and 47 precincts in the three years; a piece with no population inside a precinct with population receives no votes; and a block group that overlaps no precinct receives zero votes, which happens for 278, 298 and 3,790 block groups in 2016, 2020 and 2024. Most of the 2024 ones are in Oklahoma and South Dakota, which have no 2024 returns (Table A3). Precincts whose polygon does not survive the cleaning of the geometries, 768, 1,106 and 239 in the three years, and precincts that overlap no block group, 4, 7 and 1, are dropped before the allocation, and pieces that fall in a different state from their precinct are removed after it. The votes allocated to block groups are 99.976, 99.973 and 99.992 percent of the votes in the three source files; the shortfall is the dropped precincts and the cross-state pieces, and in almost every state it is a few hundred votes or fewer. A block group's presidential total is the sum over all candidates, a party's share is its votes over that total, and a block group with no votes in a year has no share rather than a share of zero.

The 2012 returns are not allocated here. They are the presidential results of the panel of Longuet-Marx (2026), in which the returns of the precincts that reported votes for the House of Representatives are apportioned to the block groups of 2010 through the census blocks inside each precinct, with each block weighted by the share of its area inside the precinct times the block's population. From that file I keep the block groups with at least fifty votes and match them to the 2018 identifiers; all but 51 match by identifier, and those 51, in block groups renumbered between the two releases,

are matched by spatial overlay. The 2012 returns cover the 42 states of Table A3, 174,286 block groups; the eight states and the District of Columbia without them are outside every estimation sample.

Longuet-Marx (2026) works at the level of the census block, as above, and so requires population to be spread evenly only within a block; this paper works at the level of the block group and assumes it there.

As a check, I redid the 2016 and the 2020 allocations with the population of each piece taken from the 2010 census blocks inside it, the rule of Longuet-Marx (2026): over the block groups of the exposure sample the vote shares of the two allocations correlate at 0.998 in each year, the median block group moves by 0.09 percentage points in 2016 and by 0.08 in 2020, about one block group in five moves by more than 0.5 points and one in eight by more than 1 point; the change in the Republican share between the two elections, the outcome of Section 4, correlates at 0.989 across the two allocations, with a median move of 0.07 points and one block group in seven moving by more than 0.5 points.

Table A3: Block-group vote returns by election year, source and coverage

Year	Source	States covered	Block groups covered	% of universe
2012	(Longuet-Marx, 2026)	42 of 51	174,286	80.0%
2016	Voting and Election Science Team	51 of 51	216,784	99.6%
2020	Voting and Election Science Team	51 of 51	216,788	99.6%
2024	New York Times + Redistricting Data Hub	49 of 51	213,161	97.9%

Notes: The 2012 returns are the presidential results of the panel of Longuet-Marx (2026), which reports them at the block-group level. The universe is every block group in the fifty states and the District of Columbia on the 2018 Census boundaries; a block group is covered in a year when that year's returns allocate votes to it, so the uncovered ones also include those that overlap a precinct and receive no votes. The states without returns are: in 2012 Alaska, Arkansas, the District of Columbia, Hawaii, Louisiana, Nebraska, Nevada, South Dakota and Washington; in 2024 Oklahoma and South Dakota. The paper's estimation samples require returns in 2012, 2016 and 2020, so they exclude the eight states and the District of Columbia without 2012 returns; the 2024 returns enter the persistence estimates.

A.4 The North Carolina voter file

The registration file is read as it stood on 6 November 2012, 8 November 2016, 3 November 2020 and 5 November 2024. Besides the fields named in Section 2.4, each record carries the date of registration and the county. The history file, as the State Board of Elections built it on 7 June 2026, is a rolling window that begins with the general election of November 2016, so the elections of 2012 and 2014 are not in it. Registration is being on the roll, as a Democrat, a Republican, unaffiliated, or with another party; a registered voter stays on the roll until removed, and can change the party of registration without voting. Voting is casting a ballot in a given election.

Of the registered voters of 2016, the geocoder places 88.6 percent at an exact address match and 7.6 percent at a non-exact match; for 0.9 percent it ties between candidate addresses and for 2.9 percent it finds no match, and these are the registrants dropped. Counts of registered voters by party in a block group are taken over the geocoded individuals, and a party's registration share is

its registrants over all registered voters in the block group in that year. Turnout over a window is placed in the block group where the person was registered at its start. The five movements that change a party's registration share between two elections are counted person by person: people newly register, leave the roll, move in from elsewhere in the state, move out, or change the party they are registered with; the five add up exactly to the change in the share, and party changes are counted among the people at the same address at both ends. Writing s for the Democratic share of registered voters and N for the registered base, the change from 2016 to 2020 is the sum over the five movements k of $c_k = (\Delta D_k - s_{2016}\,\Delta N_k)/N_{2020}$, where ΔD_k and ΔN_k are the Democrats and all registrants the movement adds (negative for removals and moves out; $\Delta N_k = 0$ for party switching). At the level of the person, the history file sorts voters by their record before the protests: habitual voters, who voted in both the 2016 general election and the 2018 midterm; occasional voters, who voted in one of the two; and registrants who voted in neither.

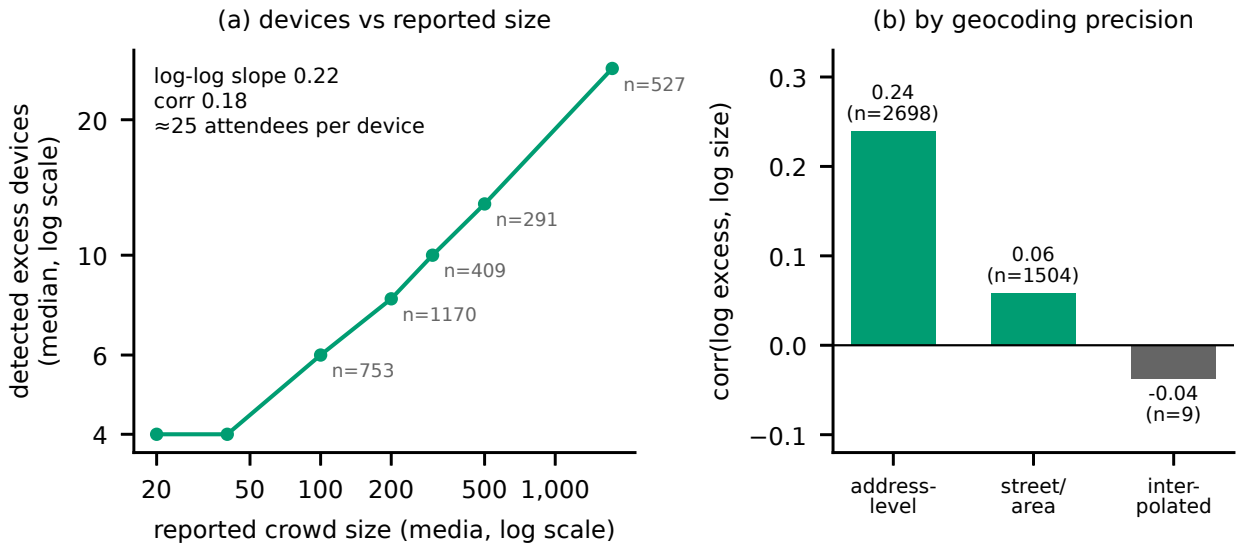
B Validation of the participation measure

This appendix validates the participation measure of Section 2.2.

The measure is built from known dates and venues rather than from raw venue counts.

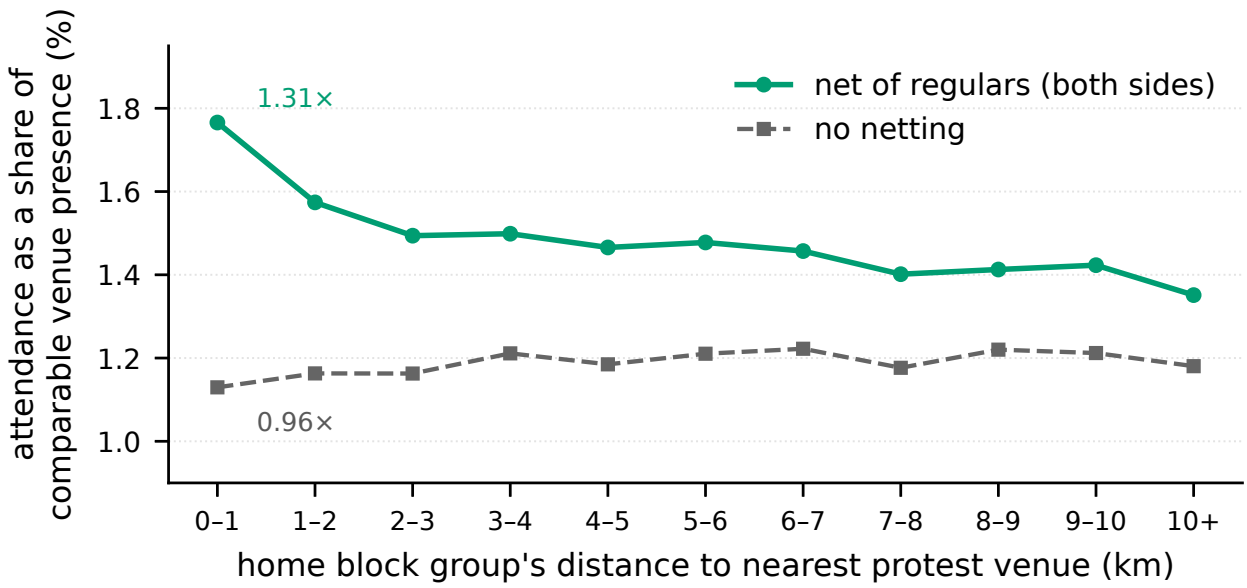
Detected attendance rises with the crowd size reported in the media, with a correlation of 0.18 in logarithms, 0.24 where the venue is geocoded to a street address and 0.06 where it is geocoded to the center of a named place or street, but the panel observes a sample of devices and large crowds spill beyond 200 m, so the measure is a relative index of participation and not a headcount (Figure A2); and net of regulars, attendance is more local than ordinary traffic, a pattern that rests on that netting: without it the nearest and the farthest rings are about equal (Figure A3).

Figure A2: Detected attendance against media-reported crowd size



Notes: Detected excess is the number of distinct devices present, under the rule of Section 2.2, within 200 m of the venue between noon and midnight local time on the event date, minus the venue's median count on its non-event days. Panel (a): events are grouped into bins of reported crowd size; each marker is the median detected excess in the bin, with the number of events in the bin; both axes are logarithmic. The printed slope is from a regression of log(1 + excess) on log(1 + reported size), with negative excess set to zero, and at least a quarter of the events have no detected excess; the attenders-per-device figure is the inverse of the median ratio of detected excess to reported size. Panel (b) is the correlation between the same two logarithms within three groups of events, by the precision of the venue's geocoded coordinate: a street address, the center of a named place or street, and a point interpolated along a street, as in Appendix A.1; the number of events is printed under each bar. Reported crowd size is the midpoint of the Crowd Counting Consortium's low and high estimates. The sample is every event with a reported crowd size, 4,211 events at 3,265 of the 4,815 venues.

Figure A3: Event-day presence relative to ordinary presence, by distance from home to the nearest venue



Notes: The horizontal axis is the distance from the resident's home block group to the nearest protest venue, in one-kilometer rings, with block groups beyond 10 km pooled. The vertical axis is event-day presence by residents of the ring at venues within 200 m, as a share of the same residents' presence at the same venues on non-event days over the wave, in percent; both are totals over the wave, not daily counts. Presence is counted in device-days, one per device and day. For this figure, regulars are devices observed at a venue on at least five non-event days; the rule the estimates use is stated in Section 5.3. The solid series removes regulars from both the event-day count and the ordinary-day count; the dashed series applies no netting. The printed ratio is the nearest ring over the farthest ring for each series. The sample is every homed device in the panel, over the wave window of the presence data, 4 May to 16 November 2020.

C Protest venues and spatial fragmentation

This appendix completes Section 3. Table A4 places the wave beside the earlier and the contemporaneous protests, Table A5 reports the other protests its block groups hosted, and Table A6 the characteristics of the venues' block groups.

Table A4: The George Floyd wave against earlier and contemporaneous protests

	George Floyd (27 May–2 Nov 2020)	Pre-period (1/2017–5/2020)	Non-George Floyd (27 May–2 Nov 2020)
Panel A: Counts			
N events	7,920	36,974	4,606
N unique block groups	4,025	14,058	2,692
Events per block group	1.97	2.63	1.71
Panel B: Event characteristics (event-weighted means)			
Mean crowd size	464	831	328
% violent	10.4	2.2	2.7
% with arrests	7.0	2.0	1.7
% with property damage	4.4	0.2	0.6

Notes: Column 1 is the George Floyd wave; column 2 is every protest event from January 2017 to May 2020; column 3 is every other protest event of the same months. Events are the Crowd Counting Consortium records geocoded to a venue. Panel A counts events and distinct block groups and reports their ratio. Panel B reports event-weighted means: the media-reported crowd size, the midpoint of the Crowd Counting Consortium's low and high estimates; the share of events coded violent in the Crowd Counting Consortium record; the share with arrests; the share with property damage. The violence rate of the wave is the share of its 7,920 events, each event counted once, coded violent.

Table A5: Other protests hosted by the wave's block groups

	All George Floyd block groups	With a violent George Floyd event
Also hosted (% of the column's block groups)		
A pre-period protest (1/2017–5/2020)	44.8	59.5
A violent pre-period protest	5.4	15.7
A non-George Floyd protest, same months	18.6	34.8
A violent non-George Floyd protest	1.2	4.6
A violent George Floyd event	13.5	—

Notes: The table takes two populations of the wave's block groups, all of them and those that hosted a violent George Floyd event, and reports for each the percentage that also hosted another kind of protest: a protest of January 2017 to May 2020, a violent one among those, another protest of the same months, a violent one among those, and a violent event of the wave itself. The last row is not reported in the second column, which is the population defined by having one. Of all the wave's block groups, 51.4 percent did not host a protest in the past or in the same months. The sample is the 4,025 block groups of Table A4.

Table A6: Characteristics of the block groups hosting protests, the wave against earlier and contemporaneous protests

	George Floyd (27 May– 2 Nov 2020)	Pre-period (1 Jan 2017– 26 May 2020)	Other (27 May– 2 Nov 2020)	George Floyd, venue-weighted
Population	1,480	1,501	1,451	1,509
Density (per km^2)	2,846	2,489	2,449	2,380
Republican vote share 2016 (%)	26.0	27.0	28.9	29.1
Democratic vote share 2016 (%)	67.7	66.4	64.6	64.7
Median household income	$65,177	$69,389	$67,038	$63,409
% Black	15.1	11.5	11.7	16.3
% Hispanic	14.1	14.5	13.8	14.9
% associate degree+	50.7	53.3	51.4	46.2
Median age	38.0	37.4	37.7	37.8
Distance to metropolitan-area centroid (km)	28.1	30.0	28.5	28.0
Distance to principal city (km)	17.5	16.9	16.5	18.7
Inside principal city (%)	90.9	92.3	90.8	88.2
Events	7,920	36,974	4,606	—
Distinct venues	4,815	19,085	3,128	4,815
Block groups	4,025	14,058	2,692	4,025

Notes: The table reports event-weighted means of the characteristics of the block group containing each protest venue, for the same three groups of events as Table A4: each event contributes its venue's block group once. Density is residents per square kilometer; the Republican and Democratic vote shares are shares of all votes cast in the 2016 presidential election, in percent; % Black and % Hispanic are population shares; % associate degree+ is the share of residents aged 25 and over with an associate degree or higher; the two distances are from the block group to the metropolitan area's centroid and to the center of its principal city, in kilometers; inside principal city is the share of events whose venue lies in the metropolitan area's principal city. The fourth column weights each venue of the wave once rather than each event; the last three rows count the events, the distinct venues and the block groups behind each column. The sample is the events of Table A4 whose venue falls in a block group with these characteristics available.

D Protest exposure

This appendix reports the exposure estimates of Section 4 in full. Stars in this appendix are from the conventional clustered test; the 0–1 km headline should be judged by the permutation p-values of Table A14 (Section 4.7).

D.1 Sample restrictions and balance

Table A7 compares, within zip codes, the block groups near a violent protest with those near only nonviolent ones on fifteen characteristics. The 2,314 block groups that were the only sample block group of their zip code contribute nothing to the estimates and are dropped, and the block groups that lie in no zip code, 0.4 percent of all block groups, are excluded.

Table A7: Balance between block groups near violent and near only nonviolent protests, within zip codes

	Near-violent mean	Near-nonviolent mean	Diff. (within zip code)
% White	51.4	49.5	0.6
			(0.9)
% Black	17.7	19.2	-1.3
			(0.9)
% Hispanic	18.9	19.5	-0.4
			(0.4)
% Asian	8.2	8.1	1.1***
			(0.3)
Median age	36.1	37.0	-0.3
			(0.3)
Median household income	\$65,953	\$65,774	-\$4,473**
			(\$2,011)
Unemployment rate (%)	7.2	7.4	-0.0
			(0.2)
% associate or higher	50.5	44.6	0.4
			(0.9)
% Renter	67.5	56.7	6.3***
			(1.6)
Population	1,269	1,266	73**
			(32)
Density (per km^2)	10,479	7,049	570*
			(319)
Distance to city center (km)	9.7	16.3	-0.3***
			(0.1)
Δ Republican share 2012–2016 (pre-trend, pp)	-3.77	-2.72	0.21
			(0.25)
Republican vote share 2016 (%)	17.6	23.1	0.0
			(0.3)
Democratic vote share 2016 (%)	76.4	71.4	-0.2
			(0.3)
Observations	3,438	14,679	18,117
Joint-balance F (within zip code)			5.53
p-value			0.000

Notes: The sample is the 18,117 block groups with at least one George Floyd protest venue within 1 km, votes cast in 2020, residents, a zip code and a metropolitan area; the 3,438 near-violent block groups have at least one violent protest within 1 km and the 14,679 block groups near only nonviolent protests have none. This is a wider set than the estimation sample of Section 4.1, which also requires returns in 2012 and 2016 and every control, and holds 13,609 block groups with a protest within one kilometer, 2,494 of them a violent one. Columns 1 and 2 are the unweighted group means. Column 3 is the within-zip-code difference: the coefficient from a regression of the characteristic on an indicator for a near-violent block group with zip code fixed effects, weighted by votes cast in 2020. The racial shares are population shares; the unemployment rate is in percent of the labor force; % renter is the share of renter-occupied housing units; distance to city center is the distance from the block group to the central business district, in kilometers; Δ Republican share 2012–2016 is the change in the Republican share of all votes cast between the 2012 and 2016 elections, in percentage points; the other characteristics are as defined in the note to Table A6. Each difference uses the block groups with the characteristic and another sample block group in their zip code. Standard errors, in parentheses, are clustered by metropolitan area. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The joint-balance F tests all fifteen differences at once, from a within-zip-code regression of the near-violent indicator on the standardized characteristics.

D.2 The ring coefficients, the earlier-protest controls by orientation, and nonviolent exposure

Tables A8 and A9 print the first five ring coefficients of Figures 2 and 3, with Conley standard errors beside the clustered ones; Table A10 splits the earlier-protest controls by political orientation (Section 4.7), and Figure A4 sets the violent and the nonviolent profiles side by side (Section 4.3).

Table A8: Exposure to any protest, ring coefficients for the first five rings

	(1) 0–1 km	(2) 1–2 km	(3) 2–3 km	(4) 3–4 km	(5) 4–5 km
Any protest (0/1)	0.193***	0.166***	0.050	0.040	-0.043
	(0.060)	(0.050)	(0.042)	(0.033)	(0.036)
	[0.063]	[0.056]	[0.041]	[0.034]	[0.035]
Demographic controls	Yes	Yes	Yes	Yes	Yes
Previous protest controls	Yes	Yes	Yes	Yes	Yes
Distance to city center	Yes	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	136,644	136,644	136,644	136,644	136,644
R^2	0.085	0.085	0.085	0.085	0.085

Notes: Each column reports the coefficient on one distance ring from a single regression with ten one-kilometer ring indicators entered jointly; columns (1) to (5) are the first five rings. The specification and sample are those of Figure 2, which plots all ten rings with the 2012–2016 placebo; Observations is N from the fit. The zip codes holding both a block group with a protest in the ring and one without number 2,590, 3,479, 4,086, 4,473 and 4,794 for the five rings shown. Standard errors in parentheses, clustered by metropolitan area; 344 clusters. Conley standard errors (Conley, 1999) in brackets, which allow the errors of any two block groups whose population-weighted centroids lie within 50 km of each other to be correlated, with equal weight at every distance up to 50 km and none beyond; the stars are from the clustered standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A9: Exposure to a violent protest, ring coefficients for the first five rings

	(1) 0–1 km	(2) 1–2 km	(3) 2–3 km	(4) 3–4 km	(5) 4–5 km
Any violent protest (0/1)	0.336**	0.271**	0.164**	0.102	0.098
	(0.135)	(0.121)	(0.078)	(0.083)	(0.074)
	[0.133]	[0.130]	[0.075]	[0.080]	[0.070]
Demographic controls	Yes	Yes	Yes	Yes	Yes
Previous protest controls	Yes	Yes	Yes	Yes	Yes
Distance to city center	Yes	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	136,644	136,644	136,644	136,644	136,644
R^2	0.084	0.084	0.084	0.084	0.084

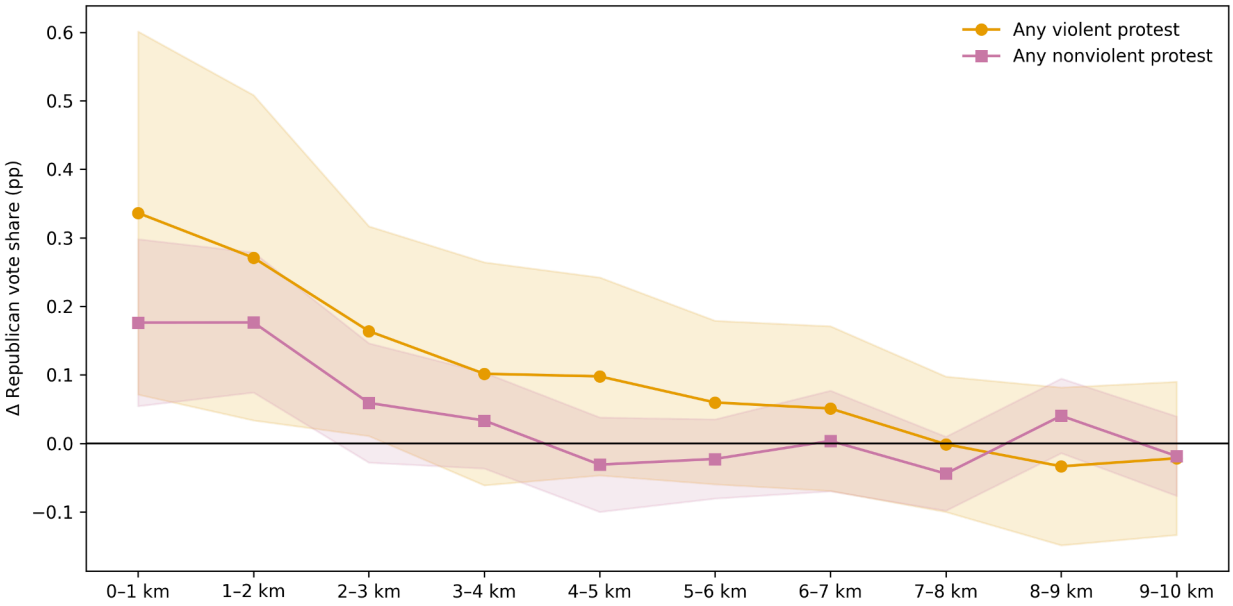
Notes: The ten ring indicators enter one regression jointly and columns (1) to (5) are the first five rings. As in Figure 2, with exposure to a violent protest as the treatment and the earlier-protest ring indicators matched to it: Any violent protest (0/1) is one when at least one violent George Floyd protest had its venue in the ring. Observations is N from the fit. The zip codes holding both a block group with a violent protest in the ring and one without number 497, 860, 1,177, 1,406 and 1,611 for the five rings shown. Standard errors in parentheses, clustered by metropolitan area; 344 clusters. Conley standard errors (Conley, 1999) with the 50 km cutoff of Table A8 in brackets; the stars are from the clustered standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Figure 3 plots all ten rings with the 2012–2016 placebo.

Table A10: Exposure to any and to a violent protest, earlier-protest controls split by political orientation

	Dependent variable: Δ Republican share 2016–2020 (pp)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Previous protest controls	Pooled	Left	Right	Neither	L + R	L + N	R + N	L + R + N
Panel A: Any protest (0/1)								
0–1 km	0.193***	0.195***	0.187***	0.154**	0.191***	0.162***	0.156**	0.162***
	(0.060)	(0.060)	(0.065)	(0.061)	(0.062)	(0.059)	(0.063)	(0.061)
1–2 km	0.166***	0.163***	0.140***	0.110**	0.161***	0.133***	0.114**	0.134***
	(0.050)	(0.050)	(0.054)	(0.046)	(0.051)	(0.044)	(0.048)	(0.046)
2–3 km	0.050	0.046	0.016	-0.006	0.045	0.024	-0.000	0.027
	(0.042)	(0.043)	(0.044)	(0.041)	(0.042)	(0.040)	(0.041)	(0.041)
3–4 km	0.040	0.032	0.005	-0.015	0.032	0.013	-0.009	0.015
	(0.033)	(0.034)	(0.034)	(0.031)	(0.034)	(0.032)	(0.032)	(0.032)
4–5 km	-0.043	-0.052	-0.069*	-0.093***	-0.050	-0.072**	-0.085**	-0.068**
	(0.036)	(0.036)	(0.037)	(0.033)	(0.036)	(0.034)	(0.035)	(0.034)
R^2	0.085	0.085	0.085	0.085	0.085	0.085	0.085	0.085
Panel B: Any violent protest (0/1)								
0–1 km	0.336**	0.330**	0.368***	0.355***	0.330**	0.320**	0.353***	0.321**
	(0.135)	(0.135)	(0.136)	(0.136)	(0.136)	(0.137)	(0.137)	(0.138)
1–2 km	0.271**	0.268**	0.318***	0.300**	0.271**	0.258**	0.301**	0.262**
	(0.121)	(0.121)	(0.120)	(0.118)	(0.123)	(0.121)	(0.120)	(0.123)
2–3 km	0.164**	0.171**	0.197**	0.181**	0.167**	0.155**	0.175**	0.153*
	(0.078)	(0.078)	(0.079)	(0.076)	(0.079)	(0.077)	(0.078)	(0.079)
3–4 km	0.102	0.107	0.119	0.101	0.104	0.091	0.097	0.090
	(0.083)	(0.086)	(0.086)	(0.085)	(0.085)	(0.085)	(0.084)	(0.085)
4–5 km	0.098	0.102	0.112	0.083	0.099	0.078	0.081	0.076
	(0.074)	(0.074)	(0.075)	(0.071)	(0.076)	(0.072)	(0.074)	(0.074)
R^2	0.084	0.084	0.084	0.084	0.084	0.085	0.085	0.085
Demographic controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Distance to city center	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	136,644	136,644	136,644	136,644	136,644	136,644	136,644	136,644

Notes: Each column of each panel is one regression with ten one-kilometer ring indicators entered jointly, of which the first five are shown; Panel A has exposure to any protest as the treatment and Panel B exposure to a violent protest. The columns differ only in the earlier-protest controls, the ring indicators for the protests of January 2017 to May 2020. Column (1) is the specification of Tables A8 and A9, with one indicator per ring for an earlier protest. Columns (2) to (8) replace them with the same indicators built separately for the earlier protests coded left-wing (L), right-wing (R) and neither (N) in the Crowd Counting Consortium's coding of each event's political orientation, entered in the combination named at the head of the column. The indicators are matched to the panel's treatment: in Panel B, in every column, they count violent earlier protests only. The specification is otherwise that of Figure 2; $N = 136{,}644$ in every column. Standard errors in parentheses, clustered by metropolitan area; 344 clusters. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure A4: Exposure to violent and to nonviolent protests, by distance ring



Notes: The figure plots two ring profiles side by side: circles are the coefficients on ten one-kilometer ring indicators for exposure to a violent protest, squares the coefficients for exposure to a nonviolent protest; shaded bands are 95% confidence intervals. The specification is that of Figure 2, with each profile one regression with the ten ring indicators of one family entered jointly, and with the earlier-protest ring indicators matched to the family: they count violent earlier protests for the violent profile and nonviolent earlier protests for the nonviolent one. The nonviolent count in a ring is the number of protests minus the number of violent protests, and the indicator equals one when it is positive. The sample is the same for both profiles, N = 136,644; 344 clusters. The violent coefficients for the first five rings are in Table A9.

D.3 Which votes move, and the controls ladder

Table A11 shows the 0–1 km estimate on five outcomes as each class of control enters, for any protest in Panel A and for violent protests in Panel B (Sections 4.2, 4.3 and 4.7).

Table A11: Exposure to any protest and to a violent protest across five outcomes, as the controls enter

	(1) No fixed effects	(2) Baseline	(3) + Demographic	(4) + Distance to city center	(5) + Previous protest	(6) + Pre-trend control
Panel A. Treatment: any protest within 0–1 km (ten-ring specification)						
Δ Republican vote share (pp)	0.312**	0.442***	0.173***	0.189***	0.193***	0.187***
	(0.157)	(0.090)	(0.064)	(0.063)	(0.060)	(0.062)
Δ Democratic vote share (pp)	-0.159	-0.217**	-0.072	-0.093	-0.129**	-0.125*
	(0.206)	(0.100)	(0.067)	(0.067)	(0.064)	(0.065)
Δ Republican votes (log)	0.014	0.011*	0.005	0.006	0.008*	0.007*
	(0.012)	(0.006)	(0.004)	(0.004)	(0.004)	(0.004)
Δ Democratic votes (log)	-0.024***	-0.021***	-0.008**	-0.008**	-0.007**	-0.007**
	(0.005)	(0.004)	(0.003)	(0.003)	(0.003)	(0.003)
Δ Turnout (pp)	-1.320***	-0.776***	-0.524***	-0.503***	-0.397***	-0.395***
	(0.179)	(0.196)	(0.158)	(0.157)	(0.153)	(0.153)
R^2 (Republican vote share)	0.024	0.003	0.084	0.084	0.085	0.097
Panel B. Treatment: any violent protest within 0–1 km (ten-ring specification)						
Δ Republican vote share (pp)	0.156	0.591***	0.337**	0.372***	0.336**	0.339**
	(0.235)	(0.148)	(0.137)	(0.137)	(0.135)	(0.136)
Δ Democratic vote share (pp)	0.104	-0.282*	-0.179	-0.227	-0.198	-0.200
	(0.279)	(0.170)	(0.139)	(0.138)	(0.141)	(0.141)
Δ Republican votes (log)	-0.001	0.015	0.010	0.011	0.010	0.010
	(0.019)	(0.010)	(0.011)	(0.011)	(0.011)	(0.011)
Δ Democratic votes (log)	-0.018*	-0.030***	-0.021***	-0.021***	-0.019***	-0.019***
	(0.009)	(0.006)	(0.006)	(0.006)	(0.007)	(0.007)
Δ Turnout (pp)	-1.542***	-1.394***	-1.112***	-1.072***	-0.834**	-0.836**
	(0.326)	(0.328)	(0.363)	(0.365)	(0.387)	(0.387)
R^2 (Republican vote share)	0.020	0.001	0.083	0.084	0.084	0.096
Demographic controls	No	No	Yes	Yes	Yes	Yes
Distance to city center	No	No	No	Yes	Yes	Yes
Previous protest controls	No	No	No	No	Yes	Yes
Pre-trend control	No	No	No	No	No	Yes
Zip code fixed effects	No	Yes	Yes	Yes	Yes	Yes
N (vote / log votes)	136,644	136,644	136,644	136,644	136,644	136,644
N (turnout)	124,971	124,971	124,971	124,971	124,971	124,971

Notes: Each row is one outcome and each column adds one class of controls to the regression. As in Figure 2, with the treatment in Panel A any protest (0/1) in the 0–1 km ring, the innermost of the ten ring indicators, the other nine rings entered jointly. Column (1) has no fixed effects and no controls; column (2) adds zip code fixed effects; column (3) adds the demographic controls; column (4) adds distance to city center; column (5) adds the previous protest controls and is the specification of Table A8; column (6) adds the 2012–2016 change in the same outcome as a control (the pre-trend control). Panel B repeats Panel A with any violent protest (0/1) in the 0–1 km ring as the treatment, so that its column (5) is the specification of Table A9. The outcomes are changes between the 2016 and 2020 presidential elections: the Republican and Democratic shares of all votes cast, in percentage points; the natural logarithms of Republican and Democratic votes; and turnout, votes cast per voting-age resident, in percentage points, with the voting-age population from the 2012–2016 American Community Survey for 2016 and from the 2015–2019 one for 2020. N = 136,644 block groups in both panels. The turnout rows use 124,971: the 11,397 block groups where the survey estimate of the voting-age population puts turnout above 100 percent in either year are excluded, with 2 that lack the estimate and 274 that are then alone in their zip code. Standard errors in parentheses, clustered by metropolitan area; 344 clusters. *** p < 0.01, ** p < 0.05, * p < 0.10.

D.4 Difference-in-differences approach

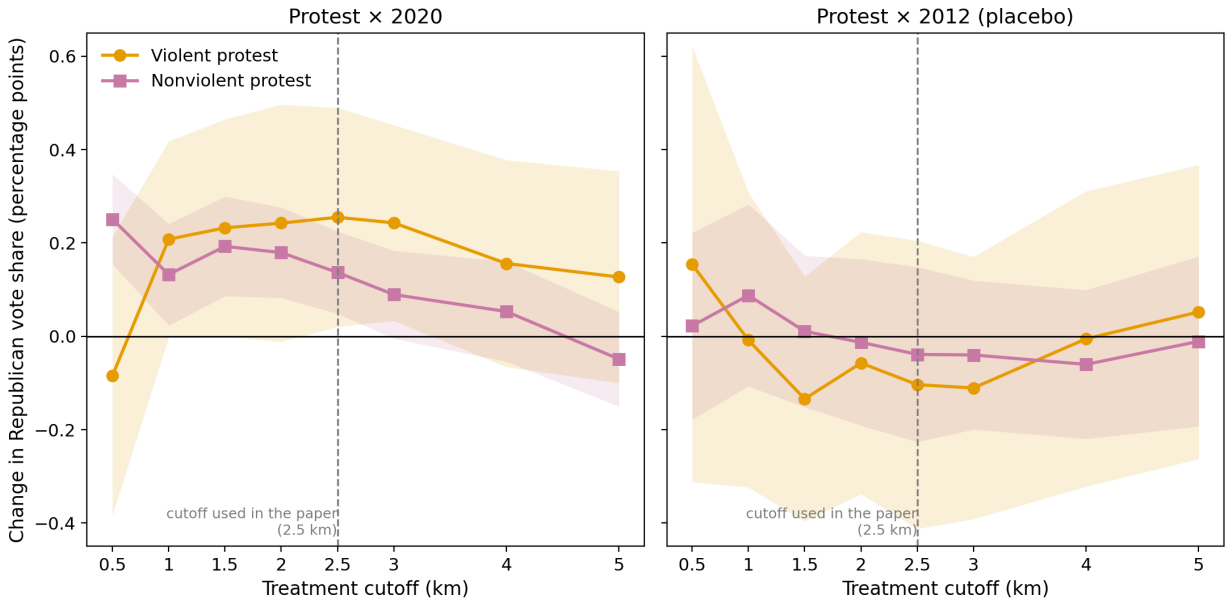
Table A12 reports the event-study coefficients of Figure 4, with two further columns in which the comparison group is restricted, and Figure A5 moves the 2.5 km cutoff (Section 4.4).

Table A12: Difference-in-differences estimates of exposure within 2.5 km, violent and nonviolent

Dependent variable: Republican vote share (pp)	(1)	(2)	(3)	(4)	(5)
	Violent (full sample)	Violent (≤10 km)	Nonviolent (full sample)	Nonviolent (≤10 km)	Violent − nonviolent (one regression)
Treated / control (km)	0–2.5 / all others	0–2.5 / 2.5–10	0–2.5 / all others	0–2.5 / 2.5–10	0–2.5 / all others
Protest × 2020	0.255**	0.154*	0.137***	0.098**	0.119
	(0.119)	(0.087)	(0.045)	(0.040)	(0.122)
Protest × 2024	0.378***	0.233**	0.128**	0.072	0.266**
	(0.106)	(0.091)	(0.063)	(0.062)	(0.118)
Protest × 2012 (placebo)	-0.104	-0.066	-0.039	-0.020	-0.110
	(0.157)	(0.157)	(0.095)	(0.094)	(0.155)
Placebo p (2012)	0.509	0.672	0.684	0.833	0.480
Controls × year	Yes	Yes	Yes	Yes	Yes
Block group fixed effects	Yes	Yes	Yes	Yes	Yes
Zip code × year fixed effects	Yes	Yes	Yes	Yes	Yes
Block groups (treated)	10,267	10,267	41,162	41,162	10,267 / 41,162
Identifying zip codes	898	898	3373	3367	—
Observations (block group × year)	542,288	200,452	542,288	417,484	542,288

Notes: The table reports the event-study coefficients of Figure 4, which plots columns (1) and (3), and two further columns; the specification is that of Figure 4, the rows Protest × 2020, Protest × 2024 and Protest × 2012 being the exposure indicator interacted with the election-year indicators, relative to 2016. Treated block groups have at least one George Floyd protest of the column's family, violent in columns (1) and (2) and nonviolent in columns (3) and (4), within 2.5 km. In columns (1) and (3) the controls are all other block groups; in columns (2) and (4) the control group is restricted to block groups whose nearest protest of the family lies between 2.5 and 10 km. Column (5) is the violent coefficient minus the nonviolent one from a single regression that enters both families together, so it is not column (1) minus column (3). Identifying zip codes is the number of zip codes that contain both a treated and a control block group: the coefficient comes from comparisons within these, while the other zip codes enter only through the year-specific slopes of the controls. Standard errors in parentheses, clustered by metropolitan area; 339, 142, 339 and 321 clusters in columns (1) to (4), and 339 in column (5). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Placebo p (2012) is the p-value of the Protest × 2012 coefficient. These interaction terms are not comparable in magnitude with the cross-sectional tables, which measure exposure in distance rings on a different sample.

Figure A5: Difference-in-differences estimates across treatment cutoffs, 2020 and the 2012 placebo

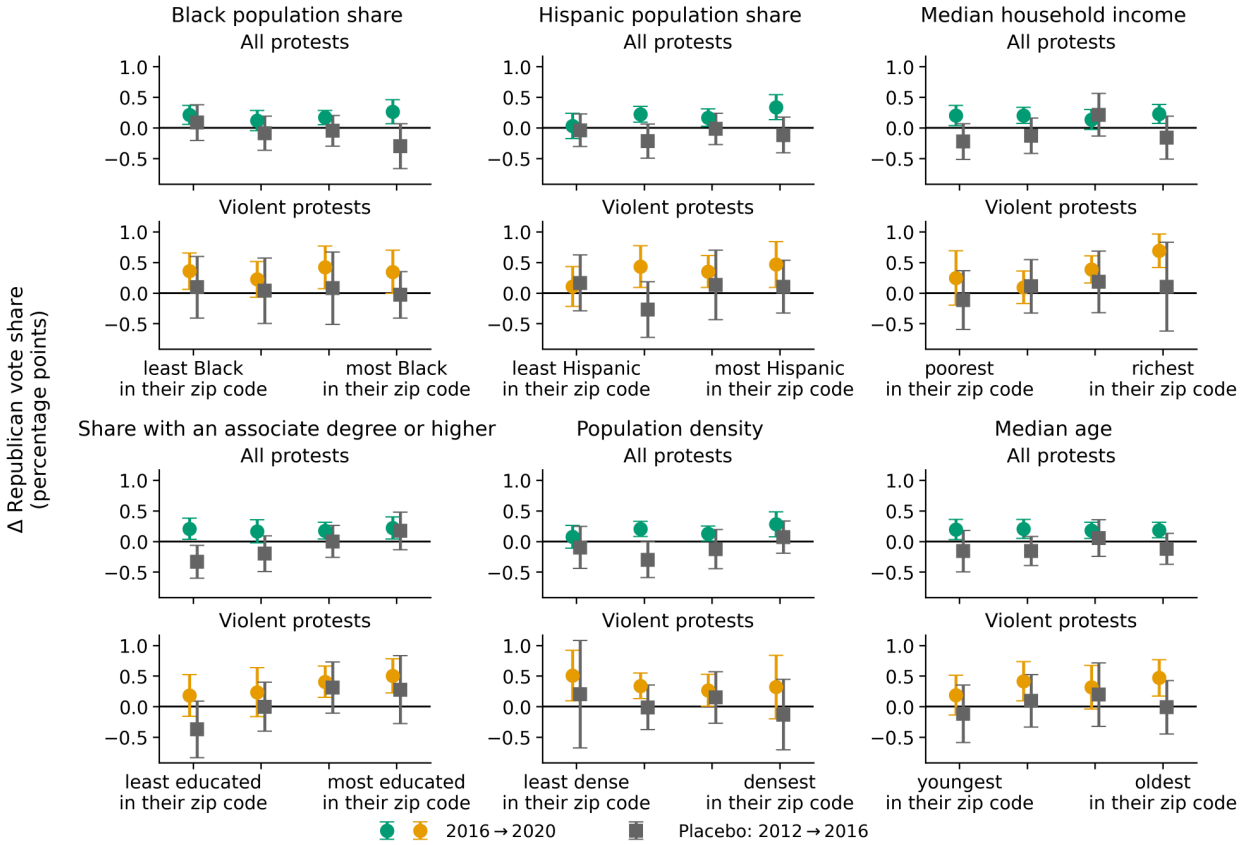


Notes: The figure re-estimates the specification of Figure 4 with the distance that defines exposure moved across eight cutoffs from 0.5 to 5 km; the dashed line marks the 2.5 km cutoff used in that figure. The left panel plots the Protest × 2020 coefficient and the right panel the Protest × 2012 placebo coefficient, at each cutoff, for exposure to a violent protest (circles) and to a nonviolent protest (squares); bands are 95% confidence intervals. At 0.5 km the 2020 coefficient is 0.251 (0.049) for nonviolent and −0.085 (0.151) for violent exposure, with placebos of 0.022 (0.102) and 0.154 (0.237), standard errors in parentheses. At every cutoff the control group is all other block groups; the balanced panel has 542,288 block group by year observations; 339 clusters. Treated block groups lie within the stated cutoff of a protest, and the controls are all other block groups. These interaction terms are not comparable in magnitude with the cross-sectional tables, which measure exposure in distance rings on a different sample.

D.5 Heterogeneity across block-group characteristics

Figure A6 reports the backlash by quartile of six block-group characteristics, one panel per characteristic (Section 4.5).

Figure A6: Exposure by quartile of six block-group characteristics



Notes: The six characteristics are the Black population share, the Hispanic population share, median household income, the share of residents aged 25 and over with an associate degree or higher, population density and median age. In each panel block groups are split into four quartiles of the characteristic measured relative to the other block groups of the same zip code, income and density in logarithms. Within each panel the upper plot is exposure to any protest and the lower plot exposure to a violent protest. Circles are the 0–1 km coefficient for each quartile, from one regression per characteristic and treatment in which the ten ring indicators of Table A8 (any protest) or Table A9 (violent protest) are interacted with the four quartile indicators, with the quartile indicators, the demographic controls of Figure 2 and the earlier-protest ring indicators matched to the family, zip code fixed effects and weights by votes cast in 2020; the outcome is that of Figure 2. Squares are the same coefficients when the 2012–2016 change is the outcome (the placebo). Whiskers are 95% confidence intervals. $N = 136{,}644$ in every fit; standard errors are clustered by metropolitan area; 344 clusters. Whether the four quartile coefficients differ is decided by a joint test of the quartile interactions, whose p-values are, for any protest and for violent protests: Black population share 0.5954 and 0.6208; Hispanic population share 0.1093 and 0.2692; median household income 0.7648 and 0.0008; education 0.8824 and 0.1462; population density 0.2768 and 0.8187; median age 0.9804 and 0.2439. The joint test of the placebo interactions rejects for education under any protest ($p = 0.04$).

D.6 Persistence to 2024

Table A13 re-estimates the ring profiles with the 2020–2024 and the cumulative 2016–2024 change as outcomes (Section 4.6).

Table A13: Ring coefficients for the 2016–2020, 2020–2024 and 2016–2024 changes, nonviolent and violent exposure, first five rings

	Nonviolent protest			Violent protest		
	(1)	(2)	(3)	(4)	(5)	(6)
	2016–2020	Persistence	Cumulative	2016–2020	Persistence	Cumulative
Dependent variable:	$\Delta_{16–20}$	$\Delta_{20–24}$	$\Delta_{16–24}$	$\Delta_{16–20}$	$\Delta_{20–24}$	$\Delta_{16–24}$
0–1 km	0.176***	-0.051	0.126	0.336**	0.182	0.508***
	(0.062)	(0.071)	(0.079)	(0.135)	(0.163)	(0.150)
1–2 km	0.177***	-0.008	0.169**	0.271**	0.124	0.385***
	(0.052)	(0.062)	(0.067)	(0.121)	(0.133)	(0.113)
2–3 km	0.059	0.028	0.091	0.164**	0.154	0.310***
	(0.044)	(0.045)	(0.059)	(0.078)	(0.094)	(0.099)
3–4 km	0.033	0.040	0.077*	0.102	0.052	0.152
	(0.036)	(0.035)	(0.042)	(0.083)	(0.083)	(0.105)
4–5 km	-0.031	-0.024	-0.054	0.098	-0.028	0.070
	(0.035)	(0.034)	(0.043)	(0.074)	(0.082)	(0.076)
Demographic controls	Yes	Yes	Yes	Yes	Yes	Yes
Previous protest controls	Yes	Yes	Yes	Yes	Yes	Yes
Distance to city center	Yes	Yes	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Joint F (inner-5) p	0.004	0.588	0.001	0.222	0.477	0.006
Observations	136,644	135,572	135,572	136,644	135,572	135,572
R^2	0.085	0.082	0.185	0.084	0.083	0.186

Notes: The table reports the ring profiles for nonviolent protests, columns (1) to (3), and violent protests, columns (4) to (6), over three outcome windows, in three columns per family; the first five of ten rings are shown. 2016–2020 is the 2016–2020 change, as in Figure A4 and Table A9; Persistence is the 2020–2024 change; Cumulative is the 2016–2024 change. Every column is the specification of Figure 2, with the outcome window changed, the ten ring indicators of the family entered jointly, and the earlier-protest ring indicators matched to the family: nonviolent earlier protests in columns (1) to (3) and violent ones in columns (4) to (6). Observations is N from the fit: the 2016–2020 columns use the 136,644 block groups of Table A8; the two columns per family that use the 2024 returns use the 135,572 of them with a 2024 return. Standard errors in parentheses, clustered by metropolitan area; 344 clusters in the 2016–2020 columns and 339 in the four columns that use the 2024 returns. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Joint F (inner-5) p is the p-value of a test that the five ring coefficients shown are all zero.

D.7 Robustness of the ring design

This subsection reports the three ring exercises of Section 4.7 in prose, then the tables that section reads directly: the other protests of the same months (Table A15), the residential-turnover controls (Table A16) and the alternative weights and the single-precinct subsample (Table A17).

Random relocation of the venues. In each of 1,000 draws the venues of a metropolitan area are moved to block groups of the same metropolitan area drawn at random, the rings are rebuilt and the specification is re-estimated, which gives the distribution of the 0–1 km coefficient when the venues are as many as observed but placed elsewhere (Table A14). Random placement does not replicate the headline: a 0–1 km coefficient as large as observed arises in 2 of 1,000 draws for any protest and 21 of 1,000 for violent protests, and the observed decay across rings is steeper than the

decay random geometry itself produces.

Equal-area rings. The one-kilometer rings widen with distance, so an outer ring covers far more area, and more protests, than an inner one; with boundaries at $\sqrt{k}$ kilometers the 0–1 km ring is unchanged by construction and only the outer profile is drawn differently. The 0–1 km estimates stay positive and significant for violent and for nonviolent exposure, 0.38 and 0.20.

Single states and metropolitan areas. None of the 72 drops, each of the 42 states and each of the 30 largest metropolitan areas, moves the violent 0–1 km estimate outside the full-sample confidence interval (Figure A7).

Table A14: The 0–1 km estimates and the decay across rings against random relocation of the venues

	(1) Any protest	(2) Violent protest
0–1 km coefficient against random venue relocation (1,000 draws per null; permutation $p = (r+1)/(B+1)$)		
Observed coefficient	0.1931	0.3363
Null mean, relocation within metro	0.0808	0.1080
Permutation p, 0–1 km ring	0.003	0.022
Permutation p, max over all 10 rings	0.003	0.022
Permutation p, population-weighted placement	0.005	0.032
Decay steepness: drop from 0–1 km to the far field (>5 km), in pp; same p convention		
Observed drop	0.2049	0.3254
Null mean	0.0847	0.0823
Permutation p	0.004	0.018
Companion p: slope over inner rings	0.001	0.028
Companion p: inner-vs-outer mean contrast	0.046	0.017
Share of random placements where the conventional t-test fires at 0–1 km ($\lvert t\rvert \geq 1.96$)		
Relocation within metro	48.1%	13.5%
Population-weighted placement	53.4%	11.7%
Positive control: relocation among block groups that hosted a protest	99.9%	66.7%
Density: the 0–1 km coefficient within density deciles		
Same zip code and density decile (fixed effects)	0.1939	0.3478
Standard error	(0.0581)	(0.0943)
Permutation p	0.002	0.012
Effect varying by density decile (exposed-weighted average)	0.0994	0.1792
Standard error	(0.0607)	(0.1344)
Permutation p	0.081	0.164
Observations (each regression)	136,644	136,644

Notes: The table compares the observed 0–1 km coefficients of Tables A8 and A9 with the distribution of the same coefficient when the protest venues are relocated at random, for any protest in column (1) and violent protests in column (2). Each null is built from 1,000 draws: in every draw the venues of a metropolitan area are moved to block groups of the same metropolitan area drawn at random, the ten ring indicators are rebuilt, and the specification of Figure 2, with the earlier-protest ring indicators matched to each column's family, is re-estimated on the same 136,644 block groups. Relocation within metro draws the receiving block groups uniformly; population-weighted placement draws them with probability proportional to population; relocation among block groups that hosted a protest places each event at a random point in a block group of the same metropolitan area that hosted one, so the locations are kept but not each block group's exposure. The permutation p is $(r + 1)/(B + 1)$, where r is the number of draws whose statistic is at least as large as the observed one and $B = 1{,}000$; the max over all 10 rings row takes the largest coefficient across rings in each draw. The decay steepness panel applies the same construction to the drop from the 0–1 km coefficient to the mean coefficient beyond 5 km; the two companion rows test the slope over the inner rings and the contrast between the inner and outer mean. The third panel is the share of draws on which a conventional t-test on the 0–1 km coefficient is significant at the 5% level. The density panel re-estimates the 0–1 km coefficient on the same draws with fixed effects for each zip code and density decile, and with the 0–1 km ring interacted with the density deciles, averaged over the exposed block groups with 2020-vote weights.

Table A15: Exposure to George Floyd protests and to other protests of the same months

Dependent variable:	Change in the Republican share of all votes cast (pp)			
	(1)	(2)	(3)	(4)
	Specification of record	Other protests in place of George Floyd protests	Both families	George Floyd and right-wing protests
Panel A. 2016–2020				
George Floyd protest within 0–1 km, 2016–2020	0.193***		0.188***	0.215***
	(0.060)		(0.060)	(0.063)
Other protest within 0–1 km, 2016–2020		0.092	0.036	
		(0.060)	(0.059)	
Right-wing protest within 0–1 km, 2016–2020				-0.210**
				(0.100)
Joint F, ten George Floyd rings [p]	4.490 [0.000]		4.968 [0.000]	4.147 [0.000]
Joint F, ten other-protest rings [p]		1.885 [0.046]	2.153 [0.020]	
Joint F, ten right-wing rings [p]				1.227 [0.272]
Panel B. 2012–2016 placebo				
George Floyd protest within 0–1 km, 2012–2016	-0.098		-0.096	-0.108
	(0.111)		(0.102)	(0.107)
Other protest within 0–1 km, 2012–2016		-0.016	0.003	
		(0.129)	(0.120)	
Right-wing protest within 0–1 km, 2012–2016				0.119
				(0.219)
Joint F, ten George Floyd rings [p]	1.544 [0.122]		1.308 [0.224]	1.367 [0.194]
Joint F, ten other-protest rings [p]		0.716 [0.709]	0.608 [0.807]	
Joint F, ten right-wing rings [p]				1.281 [0.240]
Observations (block groups)	136,644	136,644	136,644	136,644
Clusters (metropolitan areas)	344	344	344	344
Block groups with a George Floyd protest within 1 km	13,609		13,609	13,609
Block groups with another protest within 1 km		8,359	8,359	
of which also a George Floyd protest within 1 km		4,405	4,405	
Block groups with a right-wing protest within 1 km				2,674
of which also a George Floyd protest within 1 km				1,709
their 2016 Republican share, mean (%)				30.0
their distance to the city center, mean (km)				15.3

Notes: The table enters the other protests of 27 May to 2 November 2020, the 4,606 events of the Crowd Counting Consortium record geocoded to a venue whose claims do not concern Black Lives Matter, racial justice or police violence, in the specification of Figure 2. Column (1) is the estimate of record; column (2) replaces the ten George Floyd ring indicators with ten indicators for another protest in each ring; column (3) enters both sets of ten indicators together. Column (4) enters the George Floyd rings with ten rings for the 1,428 protests of the same months coded right-wing by the Crowd Counting Consortium, 145 of them related to Black Lives Matter. Panel A reports the 0–1 km coefficient of each family for the 2016–2020 change and a joint test over the family's ten rings; Panel B repeats the regressions with the 2012–2016 change as the outcome, the placebo. A test that the ten other-protest ring coefficients are equal to one another gives $p = 0.57$ in column (2) and 0.55 in column (3). The specification is otherwise that of Figure 2; $N = 136{,}644$; standard errors, in parentheses, are clustered by metropolitan area; 344 clusters. The last rows count the block groups with a protest of each family within one kilometer and those with both; the last two describe the block groups near a right-wing protest. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A16: Exposure and the Republican vote share with pre-wave residential-turnover controls

	Dependent variable: Δ Republican share 2016–2020 (pp)			
	(1)	(2)	(3)	(4)
Panel A: Any protest (0/1)				
0–1 km	0.193***	0.193***	0.209***	0.194***
	(0.060)	(0.060)	(0.061)	(0.061)
1–2 km	0.166***	0.167***	0.176***	0.167***
	(0.050)	(0.050)	(0.050)	(0.050)
2–3 km	0.050	0.050	0.055	0.052
	(0.042)	(0.042)	(0.042)	(0.042)
3–4 km	0.040	0.041	0.043	0.041
	(0.033)	(0.033)	(0.033)	(0.033)
4–5 km	-0.043	-0.043	-0.042	-0.042
	(0.036)	(0.036)	(0.036)	(0.036)
R^2	0.085	0.086	0.085	0.086
Panel B: Any violent protest (0/1)				
0–1 km	0.336**	0.339**	0.373***	0.341**
	(0.135)	(0.135)	(0.132)	(0.135)
1–2 km	0.271**	0.271**	0.290**	0.272**
	(0.121)	(0.121)	(0.120)	(0.121)
2–3 km	0.164**	0.162**	0.172**	0.165**
	(0.078)	(0.078)	(0.078)	(0.078)
3–4 km	0.102	0.100	0.107	0.104
	(0.083)	(0.082)	(0.083)	(0.082)
4–5 km	0.098	0.099	0.102	0.101
	(0.074)	(0.074)	(0.074)	(0.074)
R^2	0.084	0.085	0.085	0.085
Population change 2009–13 to 2014–18 (log × 100)	No	Yes	No	No
Moved in the past year, % (tract)	No	No	Yes	No
Householder moved in 2017 or later, %	No	No	No	Yes
Demographic controls	Yes	Yes	Yes	Yes
Previous protest controls	Yes	Yes	Yes	Yes
Distance to city center	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes
Observations	136,644	136,644	136,644	136,644

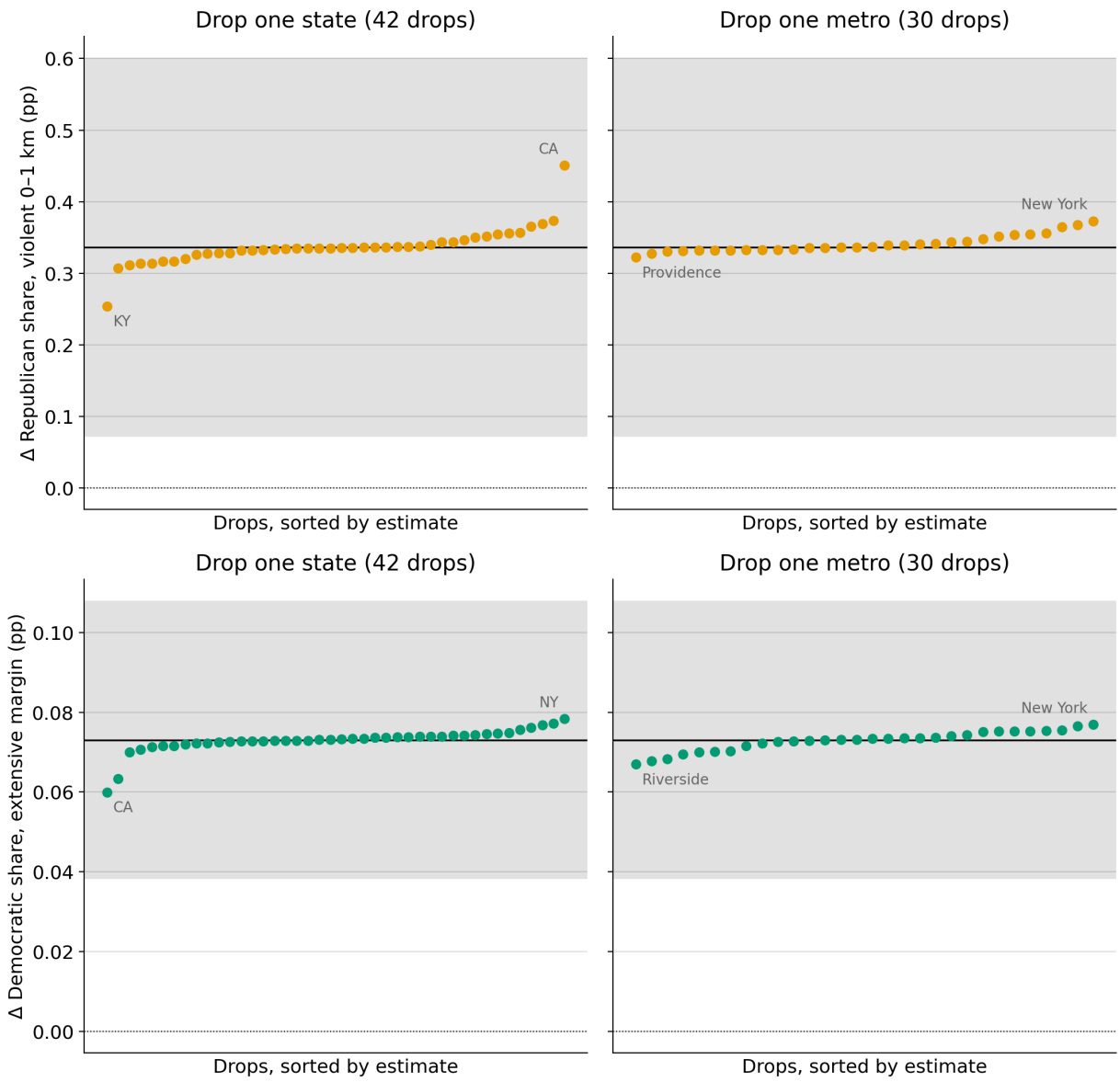
Notes: The table adds Census measures of residential turnover to the specification of Figure 2, one at a time, and reports the first five ring coefficients for exposure to any protest (Panel A) and to a violent protest (Panel B). Column (1) is the estimate of record. Column (2) adds the change in the block group's population between the 2009–2013 and the 2014–2018 American Community Survey five-year estimates, in log points; column (3) the share of residents who moved in the past year, measured at the census-tract level; column (4) the share of householders who moved in 2017 or later. The specification is otherwise that of Figure 2, with the earlier-protest ring indicators matched to each panel's family; standard errors, in parentheses, are clustered by metropolitan area. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A17: Exposure and the Republican vote share under alternative weights and in single-precinct block groups

Dependent variable:	Change in the Republican share of all votes cast (pp)			
	(1) Specification of record	(2) Weights: votes cast in 2016	(3) Single precinct in 2016 and 2020	(4) Arrests-only events recoded nonviolent
Panel A. Any protest within 0–1 km				
0–1 km ring, 2016–2020	0.193***	0.190***	0.185**	0.193***
	(0.060)	(0.057)	(0.083)	(0.060)
0–1 km ring, 2012–2016 placebo	-0.098	-0.106	-0.356	-0.098
	(0.111)	(0.111)	(0.227)	(0.111)
Observations (block groups)	136,644	136,644	29,853	136,644
Clusters (metropolitan areas)	344	344	321	344
Block groups with a protest within 0–1 km	13,609	13,609	2,926	13,609
Panel B. Violent protest within 0–1 km				
0–1 km ring, 2016–2020	0.336**	0.316***	0.076	0.335*
	(0.135)	(0.114)	(0.222)	(0.182)
0–1 km ring, 2012–2016 placebo	0.040	0.009	-0.582*	0.300
	(0.165)	(0.166)	(0.343)	(0.189)
Observations (block groups)	136,644	136,644	29,853	136,644
Clusters (metropolitan areas)	344	344	321	344
Block groups with a violent protest within 0–1 km	2,494	2,494	615	1,775
Panel C. Nonviolent protest within 0–1 km				
0–1 km ring, 2016–2020	0.176***	0.177***	0.189**	0.176***
	(0.062)	(0.059)	(0.078)	(0.060)
0–1 km ring, 2012–2016 placebo	-0.094	-0.107	-0.342	-0.092
	(0.116)	(0.116)	(0.240)	(0.115)
Observations (block groups)	136,644	136,644	29,853	136,644
Clusters (metropolitan areas)	344	344	321	344
Block groups with a nonviolent protest within 0–1 km	12,792	12,792	2,691	13,092

Notes: The table re-estimates the 0–1 km coefficient of Figure 2 under two changes to how the vote is measured, and one to how violence is coded, for exposure to any protest (Panel A), to a violent protest (Panel B) and to a nonviolent protest (Panel C), with the 2012–2016 placebo beneath each. Column (1) is the estimate of record, weighted by votes cast in 2020. Column (2) weights the regression by votes cast in 2016 instead. Column (3) restricts the sample to the block groups that lie within a single precinct in both 2016 and 2020, so that their returns are not allocated across precinct boundaries; the placebo on that subsample uses the 2012 returns, which are reported at the block-group level. Column (4) recodes as nonviolent the 263 violent events whose only marker is arrests, leaving 560, and leaves the earlier-protest controls as in column (1). The specification is otherwise that of Figure 2, with the earlier-protest ring indicators matched to each panel's family; standard errors, in parentheses, are clustered by metropolitan area. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure A7: The violent 0–1 km estimate and the extensive margin of participation with one state or metropolitan area dropped at a time



Notes: The top row re-estimates the violent 0–1 km coefficient of Table A9 with one unit of the sample removed at a time. The bottom row does the same for the extensive margin of participation of Table A18. In each row the left panel drops one of the 42 states and the right panel one of the 30 largest metropolitan areas. Dots are sorted by size. The solid line is the full-sample estimate and the gray band its 95 percent confidence interval. The dotted line is zero. In each panel the drop that moves the estimate furthest in each direction is named. The full sample is $N = 136{,}644$ with 344 clusters in the top row and $N = 135{,}691$ with 333 clusters in the bottom row. A drop removes both the block groups and the clusters of that geography.

E Protest participation

This appendix gives the participation estimates of Section 5 in full.

E.1 The estimates, the two splits and earlier attendance

Tables A18 and A21 print the coefficients of Figures 5 and 6. Table A19 splits the placebo of column (3) of Table A18 by the political orientation of the earlier protests, and Table A20 adds earlier attendance as a control (Sections 5.2 and 5.7). The 2,285 block groups that are the only sample block group of their zip code are absorbed entirely by their fixed effect and contribute nothing to the estimates.

Table A18: Participation and the change in the Democratic vote share, coefficients and placebo tests

	(1)	(2)	(3)
	Δ Dem share 2016–2020	Placebo: Δ Dem share 2012–2016	Placebo: pre-protest attendance, same venues
Any participants (extensive margin)	0.0730***	0.0352	0.0234
	(0.0178)	(0.0353)	(0.0187)
Participants, log (intensive margin)	0.0141	0.0400	-0.0147
	(0.0138)	(0.0322)	(0.0171)
Wald p (extensive margin)		0.3258	0.0268
Wald p (intensive margin)		0.4811	0.1940
Venue-traffic controls	Yes	Yes	Yes
Neighborhood mobility controls	Yes	Yes	Yes
Device-panel size deciles	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes
R^2	0.1089	0.0700	0.1094
Observations	135,691	135,691	132,482

Notes: Each column reports an extensive margin, whether the block group sent anyone to a protest, and an intensive margin, how much it sent. The specification, the controls and the sample are those of Figure 5; $N = 135{,}691$ in columns (1) and (2), 85,399 of which sent at least one resident, and 132,482 in column (3). Standard errors in parentheses, clustered by metropolitan area; 333 clusters. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Column (2) runs the specification on the 2012–2016 change and column (3) replaces the treatment with attendance at earlier protests at the same venues; the Wald rows test each against the column (1) estimate rather than against zero. Column (3)'s test uses the treatment's centering and the column (1) specification re-estimated on the 132,482 block groups of column (3); under the placebo's own centering its p-value is 0.0472.

Table A19: Earlier protest attendance and the change in the Democratic vote share, by political orientation

	Dependent variable: Δ Democratic share 2016–2020 (pp)		
	(1)	(2)	(3)
	Pooled	Left-coded	Right-coded
	Pre-protest attendance, same venues, 2019–May 2020		
Extensive margin	0.0234	0.0243	-0.0557*
	(0.0187)	(0.0180)	(0.0306)
Intensive margin	-0.0147	-0.0603***	-0.1164***
	(0.0171)	(0.0206)	(0.0445)
Block groups with any attendance (%)	46.03	33.46	6.74
Venue-traffic controls	Yes	Yes	Yes
Neighborhood mobility controls	Yes	Yes	Yes
Device-panel size deciles	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes
R^2	0.1094	0.1095	0.1095
Observations	132,482	132,482	132,482

Notes: The table repeats the placebo of column (3) of Table A18, split by the political orientation of the earlier protests. In every column the participation treatment is replaced by the residents' pre-protest attendance: their attendance at the protests of 2019 and of 1 January to 26 May 2020 held at venues that later hosted a George Floyd protest, entered as an extensive and an intensive margin: at all of those protests in column (1), which is column (3) of Table A18; only at those coded left-wing in column (2); and only at those coded right-wing in column (3), in the Crowd Counting Consortium's coding of each event's political orientation. The specification is otherwise that of Table A18, and the sample is that of its column (3), $N = 132{,}482$. Standard errors in parentheses, clustered by metropolitan area; 333 clusters. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A20: Participation and the change in the Democratic vote share, with earlier protest attendance as a control

	Dependent variable: Δ Democratic share 2016–2020 (pp)				
	(1)	(2)	(3)	(4)	(5)
Extensive margin	0.0730***	0.0740***	0.0748***	0.0747***	0.0736***
	(0.0178)	(0.0182)	(0.0182)	(0.0183)	(0.0182)
Intensive margin	0.0141	0.0157	0.0142	0.0163	0.0140
	(0.0138)	(0.0138)	(0.0139)	(0.0139)	(0.0139)
Pre-wave attendance, left-coded	No	Yes	No	Yes	No
Pre-wave attendance, right-coded	No	No	Yes	Yes	No
Pre-wave attendance, any	No	No	No	No	Yes
Venue-traffic controls	Yes	Yes	Yes	Yes	Yes
Neighborhood mobility controls	Yes	Yes	Yes	Yes	Yes
Device-panel size deciles	Yes	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.1089	0.1097	0.1096	0.1098	0.1095
Observations	135,691	132,482	132,482	132,482	132,482

Notes: Column (1) is column (1) of Table A18. Columns (2) to (5) add to that specification the residents' attendance at earlier protests (pre-wave attendance in the table), the attendance of the placebo in column (3) of Table A18, entered in the same two parts as the treatment, an extensive and an intensive margin: attendance at the protests coded left-wing in column (2), at those coded right-wing in column (3), at both, as separate terms, in column (4), and at all earlier protests in column (5), in the Crowd Counting Consortium's coding of each event's political orientation. The coefficients on the earlier attendance are not shown. Attendance at the earlier protests coded neither left- nor right-wing is not available on its own. Columns (2) to (5) are estimated on the 132,482 block groups whose earlier attendance is observed, the sample of column (3) of Table A18; for the other 3,209, part of that count was withheld under the disclosure rules (Table A1). Standard errors in parentheses, clustered by metropolitan area; 333 clusters. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A21: Participation split by protest violence and by whether the attender was a regular, coefficients and Wald tests

Dependent variable:	Change in the Democratic share of all votes cast (pp)			
	(1)	(2)	(3)	(4)
	By violence (one regression)		By whether a regular (one regression)	
	Violent	Nonviolent	Regular	Not a regular
Attendance (extensive margin)	−0.0190	0.0723***	−0.0721**	0.0781***
	(0.0307)	(0.0178)	(0.0332)	(0.0173)
Attendance (intensive margin)	−0.0903***	0.0114	−0.1326***	0.0250*
	(0.0290)	(0.0145)	(0.0274)	(0.0141)
Wald p, extensive margin	0.022		<0.0001	
Wald p, intensive margin	0.003		<0.0001	
Observations	135,689		135,691	

Notes: Each pair of columns is one split of the participation treatment, estimated in one regression with all four resulting terms entered jointly: the first pair splits attendance by whether the protest attended turned violent, the second by whether the attender was a regular at that venue. The first row is each group's extensive margin and the second its intensive margin; the groups are not mutually exclusive categories. The Wald rows test whether the two groups of a split load equally; they are not nested-model tests. The specification is otherwise that of Table A18. $N = 135{,}689$ for the violence split and 135,691 for the split by whether a regular; 333 clusters. Standard errors in parentheses, clustered by metropolitan area. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The treatment is split and the venue-traffic controls are not.

E.2 The controls ladder

Table A22 shows the estimate as each class of control enters (Section 5.7).

Table A22: Participation and the change in the Democratic vote share, as the controls enter

	(1) Treatment only	(2) + Zip code fixed effects	(3) + Demographics except race	(4) + Race	(5) + Device panel	(6) + Venue presence	(7) + Pre-trend control
Any participants (extensive margin)	1.1584***	0.1946***	0.1494***	0.1105***	0.0753***	0.0730***	0.0752***
	(0.2112)	(0.0259)	(0.0209)	(0.0183)	(0.0188)	(0.0178)	(0.0179)
Participants, log (intensive margin)	-0.2939	0.0176	0.0024	-0.0058	0.0154	0.0141	0.0165
	(0.2344)	(0.0182)	(0.0152)	(0.0142)	(0.0146)	(0.0138)	(0.0138)
Placebo: 2012–2016 change, same specification							
Any participants (extensive margin), placebo	1.4297***	0.2861***	0.0118	0.0448	0.0449	0.0352	—
	(0.2155)	(0.0404)	(0.0365)	(0.0373)	(0.0381)	(0.0353)	—
Participants, log (intensive margin), placebo	0.7126***	0.0158	0.0484	0.0567*	0.0485	0.0400	—
	(0.2422)	(0.0342)	(0.0321)	(0.0319)	(0.0318)	(0.0322)	—
Venue-traffic controls	No	No	No	No	No	Yes	Yes
Neighborhood mobility controls	No	No	No	No	Yes	Yes	Yes
Device-panel size deciles	No	No	No	No	Yes	Yes	Yes
Race controls	No	No	No	Yes	Yes	Yes	Yes
Demographic controls (except race)	No	No	Yes	Yes	Yes	Yes	Yes
Zip code fixed effects	No	Yes	Yes	Yes	Yes	Yes	Yes
Pre-trend control	No	No	No	No	No	No	Yes
R^2	0.0163	0.0009	0.0546	0.1072	0.1089	0.1089	0.1199
Observations	135,691	135,691	135,691	135,691	135,691	135,691	135,691

Notes: Each column adds one class of control to the previous one and is otherwise as in Table A18. Column (1) is the treatment alone. Column (2) adds zip code fixed effects. Column (3) adds the demographic controls other than race: median age, median household income, unemployment rate, education, renter share, log population, log density, and distance to city center. Column (4) adds race, which here means the racial-composition shares of the block group and the four products of education with those shares. Column (5) adds the device-panel controls: the mobility controls and the panel-size indicators. Column (6) adds the venue-traffic controls and is the specification of Table A18. Column (7) adds the block group's own 2012–2016 change in the Democratic share as a control. The placebo rows report, for columns (1) to (6), the two margins of the same specification estimated on the 2012–2016 change; column (7) controls for the 2012–2016 change in the Democratic share, which is the placebo's dependent variable, so its placebo coefficient is mechanically zero and is not reported (—). The sample is held at $N = 135{,}691$ in every column; 333 clusters. Standard errors in parentheses, clustered by metropolitan area. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

E.3 Other outcomes

Table A23 estimates the specification on four further outcomes (Section 5.2).

Table A23: Participation across five outcomes: vote shares, turnout and party votes per voting-age person

	(1) Δ Democratic share	(2) Δ Republican share	(3) Δ Turnout	(4) Δ Democratic votes per voting-age person	(5) Δ Republican votes per voting-age person
Any participants (extensive margin)	0.0730***	-0.0664***	-0.0122	0.0353	-0.0374
	(0.0178)	(0.0172)	(0.0957)	(0.1019)	(0.0808)
Participants, log (intensive margin)	0.0141	-0.0114	0.0323	-0.0792	0.0456
	(0.0138)	(0.0132)	(0.0760)	(0.0808)	(0.0653)
Placebo p (extensive / intensive margin)	0.32 / 0.21	0.17 / 0.48	0.75 / 0.59	0.57 / 0.87	0.74 / 0.04
Venue-traffic controls	Yes	Yes	Yes	Yes	Yes
Neighborhood mobility controls	Yes	Yes	Yes	Yes	Yes
Device-panel size deciles	Yes	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.1089	0.0896	0.0044	0.0169	0.0118
Observations	135,689	135,689	121,480	135,689	135,689

Notes: Each column is the same specification with a different outcome. Columns (1) and (2) are vote shares: the change between 2016 and 2020 in the Democratic and in the Republican share of all votes cast for president, in percentage points. Columns (3) to (5) are per-capita outcomes: the change in turnout, in Democratic votes and in Republican votes, each per voting-age person, in percentage points of the voting-age population. The placebo row gives, for each column, the p-value of the same specification run on the 2012–2016 change, for the extensive margin and for the intensive margin in that order. As in Table A18, except that column (3) is estimated on the 121,480 block groups whose turnout does not exceed 100 percent in 2016 or 2020, the turnout definition of Table A11, with the voting-age population from the 2012–2016 American Community Survey for 2016 and from the 2015–2019 one for 2020; the other columns hold $N = 135{,}689$; 333 clusters. Standard errors in parentheses, clustered by metropolitan area. *** p < 0.01, ** p < 0.05, * p < 0.10.

E.4 Difference-in-differences approach

Table A24 reports the event-study coefficients of Figure 7 for every series (Section 5.4).

Table A24: Difference-in-differences estimates of participation, all protests, by violence and by regular attendance

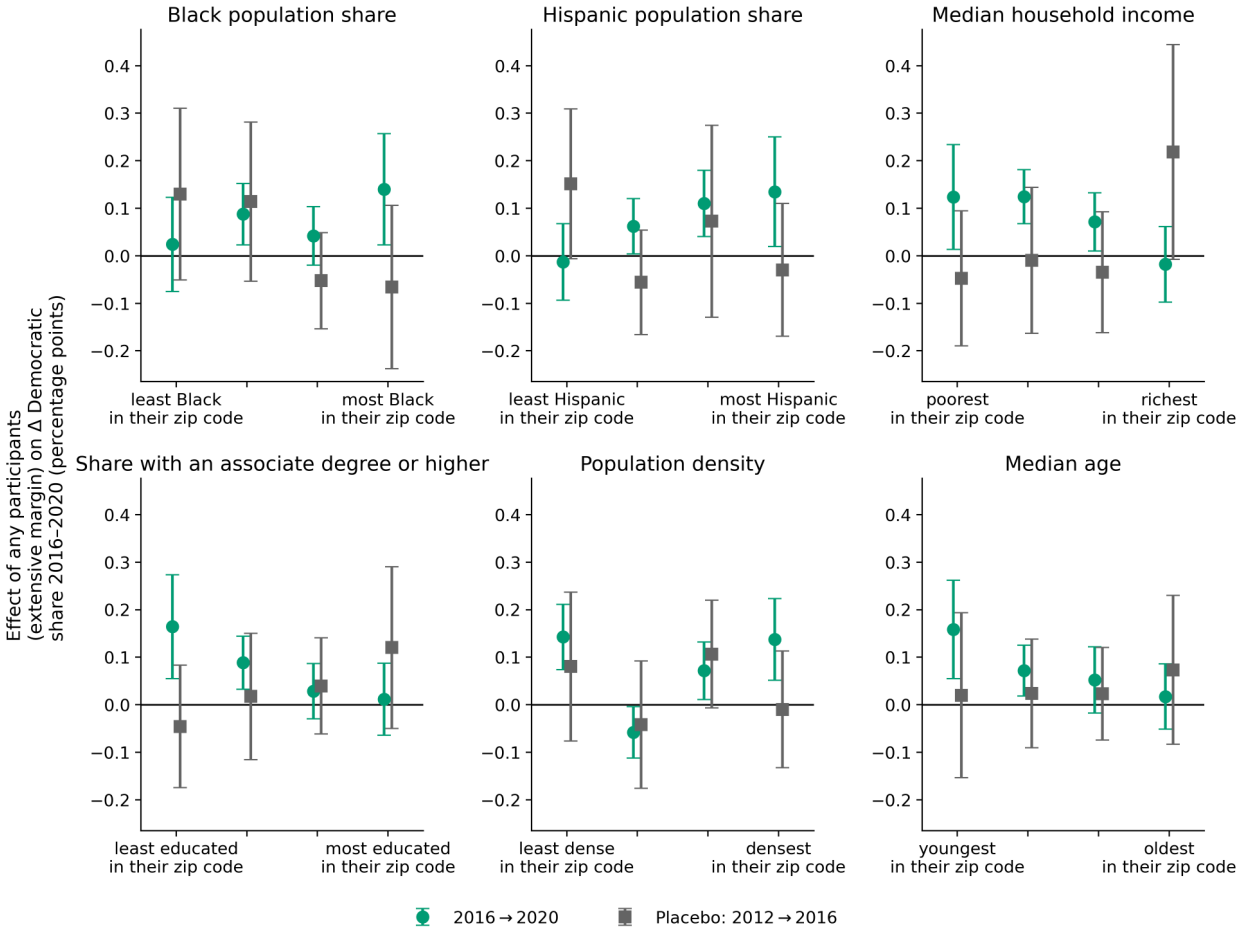
Dependent variable: Democratic vote share (pp)	(1)	(2)	(3)	(4)	(5)
Regression	All protests pooled	Violent	Nonviolent violence split	Regulars	Non-regulars regulars split
Extensive margin × 2020	0.073***	-0.019	0.072***	-0.068**	0.078***
	(0.018)	(0.031)	(0.018)	(0.033)	(0.017)
Extensive margin × 2024	0.122***	-0.035	0.108***	-0.008	0.123***
	(0.028)	(0.046)	(0.028)	(0.039)	(0.028)
Extensive margin × 2012 (placebo)	-0.035	0.019	-0.044	-0.003	-0.035
	(0.035)	(0.049)	(0.035)	(0.058)	(0.038)
Intensive margin × 2020	0.014	-0.089***	0.011	-0.135***	0.025*
	(0.014)	(0.029)	(0.015)	(0.027)	(0.014)
Intensive margin × 2024	0.007	-0.104***	0.017	-0.117**	0.004
	(0.020)	(0.040)	(0.021)	(0.046)	(0.020)
Intensive margin × 2012 (placebo)	-0.041	0.053	-0.043	0.015	-0.044
	(0.032)	(0.050)	(0.029)	(0.039)	(0.036)
Placebo p (2012, extensive margin)	0.319	0.704	0.208	0.960	0.351
Controls × year	Yes	Yes	Yes	Yes	Yes
Block group fixed effects	Yes	Yes	Yes	Yes	Yes
Zip code × year fixed effects	Yes	Yes	Yes	Yes	Yes
Block groups (sending)	83,454	23,690	78,162	15,474	80,663
Identifying zip codes	8,818	6,884	8,947	5,899	8,944
Observations (block group × year)	529,344	529,344	529,344	529,344	529,344
Clusters (metropolitan areas)	328	328	328	328	328

Notes: The table reports the event-study coefficients of Figure 7, which plots the extensive-margin rows of columns (2) and (3), and the coefficients for participation at any protest and for the split by whether the attender was a regular at the venue; the specification is that of Figure 7. The rows Extensive margin × 2020, 2024 and 2012 are the extensive margin interacted with the election-year indicators, relative to 2016, and the Intensive margin rows are the same for the intensive margin. Column (1) is participation at any protest; columns (2) and (3) come from one regression that splits participation by whether the protest attended turned violent; columns (4) and (5) from one regression that splits it by whether the attender was a regular at the venue. Identifying zip codes is the number of zip codes that contain both a sending and a non-sending block group: the coefficient comes from comparisons within these, while the other zip codes enter only through the year-specific slopes of the controls. Standard errors in parentheses, clustered by metropolitan area; 328 clusters. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Placebo p (2012, extensive margin) is the p-value of the Extensive margin × 2012 coefficient.

E.5 Heterogeneity across block-group characteristics

Figure A8 reports the participation estimate by quartile of six block-group characteristics, one panel per characteristic (Section 5.5).

Figure A8: Participation by quartile of six block-group characteristics



Notes: The six characteristics are the Black population share, the Hispanic population share, median household income, the share of residents aged 25 and over with an associate degree or higher, population density and median age. In each panel block groups are split into four quartiles of the characteristic measured relative to the other block groups of the same zip code, income and density in logarithms, as in Figure A6. The specification of Table A18 is estimated with both margins of participation interacted with the four quartile indicators, with the quartile indicators. Circles are the quartile extensive-margin estimates; squares are the same estimates when the 2012–2016 change is the outcome, the placebo; whiskers are 95 percent confidence intervals; the vertical axis is in percentage points of the Democratic share of all votes cast. N = 135,691 in every fit; standard errors are clustered by metropolitan area, 333 clusters. Whether the four quartile estimates differ is decided by a joint test of the quartile interactions, whose p-values are: Black population share 0.370; Hispanic population share 0.135; median household income 0.078; education 0.074; population density < 0.001; median age 0.189. No joint test of the placebo interactions rejects; the smallest p is 0.082, for the Hispanic share.

E.6 Persistence to 2024

Table A25 gives the participation estimate over the 2016–2020, 2020–2024 and 2016–2024 windows (Section 5.6).

Table A25: Participation and the change in the Democratic vote share, 2016–2020, 2020–2024 and 2016–2024, coefficients

	(1) 2016–2020	(2) 2020–2024	(3) 2016–2024 (cumulative)
Panel A. Change in the Democratic share			
Extensive margin	0.0732***	0.0490***	0.1222***
	(0.0179)	(0.0184)	(0.0275)
Intensive margin	0.0139	-0.0070	0.0069
	(0.0139)	(0.0165)	(0.0200)
Panel B. Change in the Republican share			
Extensive margin	-0.0663***	-0.0517***	-0.1180***
	(0.0173)	(0.0189)	(0.0266)
Intensive margin	-0.0113	0.0033	-0.0080
	(0.0133)	(0.0163)	(0.0182)
Venue-traffic controls	Yes	Yes	Yes
Neighborhood mobility controls	Yes	Yes	Yes
Device-panel size deciles	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes
R^2 (Panel A)	0.1097	0.0795	0.1947
Observations	134,570	134,570	134,570

Notes: Each column is the same specification with a different outcome window: the 2016–2020 change, the 2020–2024 change, and the cumulative 2016–2024 change in the Democratic share of all votes cast, in percentage points, in Panel A; Panel B repeats the three columns for the Republican share. Of the four detection radii, the 2020–2024 increase reaches the 5 percent level only at 200 m, and the cumulative 2016–2024 change at all four (Section 5.6). The R^2 row refers to Panel A. As in Table A18, except that the sample is the block groups that have a vote return in 2024 as well, so $N = 134{,}570$ in all three columns; 328 clusters, five fewer than elsewhere in the participation design because five metropolitan areas have no block group with a 2024 return. Standard errors in parentheses, clustered by metropolitan area. *** p < 0.01, ** p < 0.05, * p < 0.10.

E.7 Robustness of the participation measure

This subsection reports the exercises of Section 5.7 on the participation measure (Tables A26 to A30), on residential turnover (Table A31) and on the measurement of the vote (Table A32); the earlier-attendance checks are in Appendix E.1 and the controls ladder in Appendix E.2.

Venue-traffic controls. Eight measurements of ordinary visiting at the same venues, when no protest was happening, can enter in any combination, 256 in all, and each combination was estimated with the 2016–2020 outcome and with the placebo (Table A26). The extensive margin is positive and significant at the 1 percent level in every combination, between 0.06 and 0.08, and the placebo's p-value is below 0.05 in none of them; the intensive margin reaches the 5 percent level in just under half of them. A ninth control, attendance at the protest itself by residents who visited the venue often in 2020, is part of what the treatment counts and is reported alone: 0.10 (0.02) on the extensive margin and 0.07 (0.02) on the intensive margin.

Detection radius. The extensive margin is positive and significant at all four radii of Table A27. It grows with the radius, but a wider radius also admits more ordinary passers-by, so the growth is

not read as a larger effect.

Neighboring block groups. Terms for the neighboring block groups enter in steps (Table A29). The block group's own extensive margin stays positive and significant at the 1 percent level in every column. The neighbor-share term is positive in columns (2) to (4), at the 1 percent level in columns (2) and (3) and the 5 percent level in column (4), but its placebo is not clean in columns (2) and (3), and in the symmetric column it is not distinguishable from zero; the neighbors' mean log participation is negative and significant at the 1 percent level in every column, with a clean placebo. Each column is a different comparison group, and no reading of spillovers is drawn from the table.

Denominators. The extensive margin is positive and significant at the 1 percent level under all five denominators, between 0.07 and 0.09; the intensive margin is positive under all five and significant under four, and is a null only under the denominator the paper uses, the panel devices homed in the block group, the only one that counts the same devices as the attendances (Table A27, Panel B). The reading that heavier participation adds nothing detectable is therefore specific to that rate and is not a property of the participant count itself. One of the ten placebo p-values in that table, the intensive margin's under land area, is 0.01 and not clean.

Single states and metropolitan areas. Each of the 42 states and each of the 30 largest metropolitan areas is dropped in turn (Figure A7, bottom row): every one of the 72 estimates lies inside the full-sample confidence interval and none crosses from significant to not significant.

Table A26: Participation across the 256 combinations of the venue-traffic controls

	Any participants (extensive margin)	Participants, log (intensive margin)
The 256 subsets of the nine venue-traffic controls without attendance by residents who visited the venue often in 2020 (512 regressions)		
Coefficient positive	256 of 256	256 of 256
Range across subsets	0.0556 to 0.0819	0.0004 to 0.0444
Significant at 5%	256 of 256	121 of 256
Significant at 1%	256 of 256	37 of 256
Placebo $p < 0.05$ (Δ 2012–2016)	0 of 256	0 of 256
Subsets the regression could not estimate	0 of 256	0 of 256
Three reference cells inside these subsets		
No venue-traffic controls	0.0753	0.0154
Main specification's control alone	0.0743	0.0179
All eight other controls together	0.0663	0.0364
Separate check: attendance by residents who visited the venue often in 2020, entered alone		
Coefficient	0.0975	0.0691
Standard error	(0.0221)	(0.0180)
Neighborhood mobility controls	Yes	Yes
Device-panel size deciles	Yes	Yes
Demographic controls	Yes	Yes
Zip code fixed effects	Yes	Yes
Observations (each regression)	132,482	132,482

Notes: Eight venue-traffic controls, one of them that of the main specification, can be included or excluded independently, giving 256 combinations; every control enters split into weekdays and weekends, while the main specification enters its one control pooled; each was estimated with the 2016–2020 outcome and again with the 2012–2016 placebo outcome, so the table summarizes 512 regressions. The middle block gives three named cells inside the census: the fit with no venue-traffic control at all, the fit with the main specification's control alone, split here as every control of the census is (Table A18 enters it pooled, on 135,691 block groups), and the fit with all eight entered together. The first is re-estimated on these 132,482 block groups and equals Table A22's column (5), estimated on 135,691, only after rounding. The last block enters the ninth control alone; it is not among the 256 because it is part of what the treatment counts. Every fit is otherwise the specification of Table A18. Each of the nine is built with the treatment's own denominator and functional form. The ninth counts the attendance, noon to midnight on the day the venue hosted, of residents seen there on at least five other days of 2020 or on more than a tenth of the days they were seen anywhere, a broader group than the regulars of Table A21 and a subset of the attendance the treatment counts. The other eight measure ordinary visiting when no protest was happening: visits to the same venue that morning, from 08:00 to 12:00; visits to it overnight, from 02:00 to 06:00; visits to the protest venues on days they hosted nothing, over the protest window; and the same background traffic in five further windows — before the pandemic, during the lockdown, in the weeks just before the protests began, and in the matching protest and pre-protest windows of the previous year. The sample is held fixed at $N = 132{,}482$ in every regression, below the 135,691 of Table A18 because the widest combination needs a window that some block groups lack; 333 clusters. Significance is at the levels marked, on standard errors clustered by metropolitan area.

Table A27: Participation by detection radius and by the denominator of the participation rate

	(1) Extensive margin	(2) Intensive margin	(3) Placebo p (ext.)	(4) Placebo p (int.)
Panel A. Detection radius				
50 m	0.0503***	-0.1585***	0.6832	0.8153
	(0.0188)	(0.0277)		
100 m	0.0624***	-0.0601***	0.3441	0.8396
	(0.0199)	(0.0227)		
200 m	0.0730***	0.0141	0.3206	0.2150
	(0.0178)	(0.0138)		
400 m	0.1044***	0.0226	0.6363	0.1470
	(0.0302)	(0.0144)		
Panel B. Denominator of the participation rate				
Panel devices resident in the block group	0.0742***	0.0151	0.27	0.18
	(0.0178)	(0.0138)		
Resident population	0.0865***	0.0844***	0.23	0.11
	(0.0193)	(0.0169)		
Voting-age population	0.0848***	0.0431***	0.24	0.29
	(0.0187)	(0.0134)		
Land area	0.0683***	0.0264**	0.17	0.01
	(0.0177)	(0.0115)		
None (raw participant count)	0.0921***	0.1001***	0.18	0.06
	(0.0193)	(0.0176)		
Venue-traffic controls	Yes	Yes	Yes	Yes
Neighborhood mobility controls	Yes	Yes	Yes	Yes
Device-panel size deciles	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes
Observations	135,691	135,691	135,691	135,691

Notes: Each row re-estimates the specification of Table A18 with one measurement choice changed, and reports both treatment terms and both placebo p-values from that row's own fit. Panel A changes the detection radius. The radius used everywhere else in the paper, 200 m, is set in bold. Panel B changes the denominator of the participation rate. Its first row is the denominator used everywhere else in the paper, the number of panel devices resident in the block group; the others divide the same participant count by the resident population, by the voting-age population, by the block group's land area, and by nothing at all, which makes the intensive margin a log count. Columns (1) and (2) are the extensive margin and the intensive margin with standard errors in parentheses beneath; columns (3) and (4) are p-values from the same specification run on the 2012–2016 change, one for each term. In Panel B the neighborhood mobility controls enter as their two measures, not split by weekday and weekend into the four of Table A18, which is why its first row differs slightly from the 200 m row of Panel A; and in every row the venue-traffic control is rebuilt on that row's denominator, as the treatment is. The sample is $N = 135{,}691$ in every row; standard errors are clustered by metropolitan area, 333 clusters. *** p < 0.01, ** p < 0.05, * p < 0.10.

Table A28: Participation by protest violence at four detection radii, and by attendance threshold

	Dependent variable: change in the Democratic share of all votes cast, 2016–2020 (pp)			
	(1)	(2)	(3)	(4)
Panel A: Both margins by family, by venue radius				
	50 m	100 m	200 m	400 m
Violent, any participants (extensive margin)	−0.041	−0.040	−0.019	−0.044
	(0.051)	(0.036)	(0.031)	(0.036)
Violent, participants, log (intensive margin)	−0.097**	−0.137***	−0.090***	−0.074***
	(0.046)	(0.038)	(0.029)	(0.022)
Nonviolent, any participants (extensive margin)	0.060***	0.069***	0.072***	0.106***
	(0.021)	(0.019)	(0.018)	(0.032)
Nonviolent, participants, log (intensive margin)	−0.171***	−0.069***	0.011	0.021
	(0.024)	(0.023)	(0.014)	(0.016)
Placebo: 2012–2016 change, same specification				
Violent, any participants (extensive margin), placebo	0.032	0.014	−0.017	0.036
	(0.088)	(0.069)	(0.049)	(0.068)
Violent, participants, log (intensive margin), placebo	0.024	0.013	−0.050	−0.015
	(0.099)	(0.074)	(0.049)	(0.036)
Nonviolent, any participants (extensive margin), placebo	0.003	0.011	0.044	0.039
	(0.035)	(0.031)	(0.035)	(0.047)
Nonviolent, participants, log (intensive margin), placebo	−0.008	0.005	0.042	0.019
	(0.050)	(0.047)	(0.029)	(0.029)
Sending block groups, violent	5,091	11,364	24,332	46,080
Sending block groups, nonviolent	20,788	44,940	79,965	110,868
Centering constant c (pooled)	−4.058	−3.790	−3.344	−2.716
Observations	135,689	135,689	135,689	135,689
Clusters (metropolitan areas)	333	333	333	333
Panel B: Pooled participation at 200 m, sending defined by attendances				
	≥ 1	≥ 2	≥ 5	
Any participants (extensive margin)	0.073***	0.078***	0.110***	
	(0.018)	(0.019)	(0.029)	
Participants, log (intensive margin)	0.014	−0.055***	−0.198***	
	(0.014)	(0.021)	(0.038)	
Placebo: 2012–2016 change, same specification				
Any participants (extensive margin), placebo	0.035	0.052	0.144***	
	(0.035)	(0.050)	(0.048)	
Participants, log (intensive margin), placebo	0.040	0.046	−0.033	
	(0.032)	(0.032)	(0.051)	
Sending block groups	85,399	53,687	16,333	
Centering constant c_k (own mean)	−3.344	−2.984	−2.452	
Observations	135,691	135,691	135,691	
Clusters (metropolitan areas)	333	333	333	

Notes: Panel A re-estimates the violence split of Table A21, both families in one regression, with attendance detected within 50, 100, 200 and 400 m of the venue; the 200 m column is that table's. Tests that the violent and the nonviolent terms are equal give, at 50, 100, 200 and 400 m, $p = 0.098$, 0.011, 0.022 and 0.002 for the extensive margin and $p = 0.152$, 0.099, 0.003 and 0.001 for the intensive margin. Panel B re-estimates the specification of Table A18, column (1), with a block group counted as sending only when its residents recorded at least 1, 2 or 5 attendances over the wave, and the log participation rate centered at the mean of that set of block groups, c_k; block groups with fewer attendances are counted as non-sending, and the ≥ 1 column is Table A18's. The rows marked placebo are the same specification on the 2012–2016 change. The specification is otherwise that of Table A18; standard errors in parentheses, clustered by metropolitan area; 333 clusters. *** p < 0.01, ** p < 0.05, * p < 0.10.

Table A29: Participation as terms for neighboring block groups enter

	(1) Own only	(2) + neighbor participation	(3) + neighbor traffic controls	(4) + neighbor mobility controls	(5) full symmetric mirror
Own: any participants (extensive margin)	0.0730***	0.0867***	0.0847***	0.0633***	0.0602***
	(0.0178)	(0.0183)	(0.0182)	(0.0181)	(0.0183)
Own: participants, log (intensive margin)	0.0141	0.0106	0.0112	-0.0012	0.0011
	(0.0138)	(0.0142)	(0.0142)	(0.0138)	(0.0140)
Neighbor share with any participants	—	0.2454***	0.2146***	0.1194**	0.0472
		(0.0565)	(0.0547)	(0.0543)	(0.0571)
placebo p (Δ 2012–2016)	—	0.0001	0.0008	0.3156	0.2866
Neighbor mean log participants	—	-0.1388***	-0.1340***	-0.2411***	-0.1808***
		(0.0512)	(0.0515)	(0.0462)	(0.0470)
placebo p	—	0.3061	0.7520	0.2659	0.6412
Venue-traffic controls	Yes	Yes	Yes	Yes	Yes
Neighborhood mobility controls	Yes	Yes	Yes	Yes	Yes
Device-panel size deciles	Yes	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.1089	0.1094	0.1095	0.1273	0.1285
Observations	135,691	135,680	135,680	135,673	135,673

Notes: Each column adds one more thing about the surrounding neighborhood to the specification of Table A18, and reports the block group's own extensive and intensive margins above the neighborhood terms. Column (1) is that specification unchanged. Column (2) adds two neighbor terms: the share of neighboring block groups that sent anyone, and the mean log participation among those that did. Column (3) adds the neighbors' venue traffic, column (4) the neighbors' everyday movement measures, and column (5) enters every own-block-group control in a matching form for the neighbors. Each neighbor term carries its own placebo row, the p-value of the same specification run on the 2012–2016 change. Standard errors in parentheses, clustered by metropolitan area; the sample falls slightly as neighbor terms enter, from $N = 135{,}691$ in column (1) to 135,680 in columns (2) and (3) and 135,673 in columns (4) and (5), because a block group with no neighbor in the sample has no neighbor term; 333 clusters throughout. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Column (1) remains the paper's estimate.

Table A30: Participation and presence at the venues on days without a protest

Dependent variable:	Change in the Democratic share of all votes cast (percentage points)		
	(1)	(2)	(3)
	Specification of record	Presence on days without a protest in place of attendance	Both
Panel A. 2016–2020			
Extensive margin	0.073***		0.069***
	(0.018)		(0.017)
Intensive margin	0.014		0.012
	(0.014)		(0.014)
Presence on days without a protest, log rate		0.031	0.016
		(0.026)	(0.026)
Ordinary presence before the protests, any	0.013	0.011	0.011
	(0.047)	(0.046)	(0.046)
Ordinary presence before the protests, log rate	0.004	0.001	-0.002
	(0.012)	(0.011)	(0.011)
Panel B. 2012–2016 placebo			
Extensive margin	0.035		-0.000
	(0.035)		(0.035)
Intensive margin	0.040		0.015
	(0.032)		(0.031)
Presence on days without a protest, log rate		0.167***	0.163***
		(0.045)	(0.044)
Ordinary presence before the protests, any	-0.062	-0.091	-0.089
	(0.072)	(0.074)	(0.074)
Ordinary presence before the protests, log rate	0.029	-0.030	-0.031
	(0.021)	(0.021)	(0.021)
Presence on days without a protest, any	—	dropped	dropped
Observations (block groups)	135,691	135,691	135,691
Clusters (metropolitan areas)	333	333	333
Block groups with presence on days without a protest		135,567	135,567

Notes: The table enters the residents' presence at the protest venues on the days of the wave on which the venue hosted no George Floyd protest, measured at 200 m as attendance is and expressed as a log rate per panel device, in the specification of Table A18. Column (1) is the specification of record. Column (2) replaces the two attendance terms with presence on days without a protest; column (3) enters both. Panel A is the 2016–2020 change and Panel B the same regressions with the 2012–2016 change as the outcome, the placebo. The indicator for any presence on such days is dropped because all but 124 of the 135,691 block groups have some. The ordinary-presence controls of record, measured between 4 and 26 May 2020, stay in every column. Presence on days without a protest is partly measured after the protests began, so the column is a robustness check and not an alternative specification of record. Standard errors in parentheses, clustered by metropolitan area; 333 clusters. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A31: Participation and the Democratic vote share with pre-wave residential-turnover controls

	Dependent variable: change in the Democratic share of all votes cast, 2016–2020 (percentage points)			
	(1)	(2)	(3)	(4)
Extensive margin	0.0730***	0.0733***	0.0710***	0.0721***
	(0.0178)	(0.0178)	(0.0177)	(0.0178)
Intensive margin	0.0141	0.0150	0.0091	0.0131
	(0.0138)	(0.0139)	(0.0137)	(0.0138)
Population change 2009–13 to 2014–18 (log × 100)	No	Yes	No	No
Moved in the past year, % (tract)	No	No	Yes	No
Householder moved in 2017 or later, %	No	No	No	Yes
Venue-traffic controls	Yes	Yes	Yes	Yes
Neighborhood mobility controls	Yes	Yes	Yes	Yes
Device-panel size deciles	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes
R^2	0.1089	0.1096	0.1101	0.1096
Observations	135,691	135,691	135,691	135,691

Notes: The table adds Census measures of residential turnover to the specification of Table A18, one at a time. Column (1) is the estimate of record. Column (2) adds the change in the block group's population between the 2009–2013 and the 2014–2018 American Community Survey five-year estimates, in log points; column (3) the share of residents who moved in the past year, measured at the census-tract level; column (4) the share of householders who moved in 2017 or later. The outcome, the controls, the zip code fixed effects and the weights by votes cast in 2020 are those of Table A18, on its sample; standard errors, in parentheses, are clustered by metropolitan area; 333 clusters. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A32: Participation and the Democratic vote share under alternative weights and in single-precinct block groups

Dependent variable:	Change in the Democratic share of all votes cast (percentage points)		
	(1) Specification of record	(2) Weights: votes cast in 2016	(3) Single precinct in 2016 and 2020
Panel A. All protests			
Extensive margin, 2016–2020	0.073***	0.064***	0.083**
	(0.018)	(0.018)	(0.040)
Intensive margin, 2016–2020	0.014	0.018	-0.013
	(0.014)	(0.014)	(0.034)
Extensive margin, 2012–2016 placebo	0.035	0.037	0.152
	(0.035)	(0.036)	(0.127)
Intensive margin, 2012–2016 placebo	0.040	0.039	-0.009
	(0.032)	(0.029)	(0.071)
Observations (block groups)	135,691	135,691	31,349
Clusters (metropolitan areas)	333	333	328
Block groups sending anyone	85,399	85,399	17,809
Share of the record's treated retained	1.000	1.000	0.209
Panel B. By protest violence, one regression			
Violent, extensive margin, 2016–2020	-0.019	-0.010	0.043
	(0.031)	(0.028)	(0.057)
Violent, intensive margin, 2016–2020	-0.090***	-0.079***	0.008
	(0.029)	(0.024)	(0.043)
Nonviolent, extensive margin, 2016–2020	0.072***	0.065***	0.068
	(0.018)	(0.017)	(0.044)
Nonviolent, intensive margin, 2016–2020	0.011	0.015	-0.019
	(0.014)	(0.014)	(0.035)
Violent, extensive margin, 2012–2016 placebo	-0.017	-0.013	0.063
	(0.049)	(0.046)	(0.087)
Violent, intensive margin, 2012–2016 placebo	-0.050	-0.048	-0.107
	(0.049)	(0.049)	(0.104)
Nonviolent, extensive margin, 2012–2016 placebo	0.044	0.045	0.144
	(0.035)	(0.035)	(0.135)
Nonviolent, intensive margin, 2012–2016 placebo	0.042	0.042	-0.056
	(0.029)	(0.027)	(0.052)
Observations (block groups)	135,689	135,689	31,349
Clusters (metropolitan areas)	333	333	328
Block groups sending anyone to a violent protest	24,332	24,332	5,085
Share of the record's treated retained	1.000	1.000	0.209
Block groups sending anyone to a nonviolent protest	79,965	79,965	16,448
Share of the record's treated retained	1.000	1.000	0.206

Notes: The table re-estimates the two margins of Table A18 under two changes to how the vote is measured, for participation at any protest (Panel A) and split by protest violence in one regression (Panel B), with the 2012–2016 placebo beneath each. Column (1) is the estimate of record, weighted by votes cast in 2020. Column (2) weights the regression by votes cast in 2016 instead. Column (3) restricts the sample to the block groups that lie within a single precinct in both 2016 and 2020, so that their returns are not allocated across precinct boundaries; the placebo on that subsample uses the 2012 returns, which are reported at the block-group level. The specification is otherwise that of Table A18; standard errors, in parentheses, are clustered by metropolitan area. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

F Exposure and participation together

This appendix reports the joint regressions and the aggregate of Section 6 in full.

F.1 The joint regressions in full

Tables A33 and A34 print the joint regressions behind Figure 8, pooled and split by protest violence, every exposure ring with its joint test (Section 6.1).

Table A33: Participation and exposure in one regression, across four outcomes

	(1) Δ Dem (pp)	(2) Δ Rep (pp)	(3) Net margin (Dem − Rep, pp)	(4) Δ Turnout (pp)
Participation (device attendance, two-part)				
Extensive margin	0.075*** (0.018)	-0.072*** (0.017)	0.146*** (0.034)	0.000 (0.092)
Intensive margin (log rate)	0.018 (0.013)	-0.019 (0.013)	0.037 (0.026)	0.052 (0.076)
Exposure: any protest, 1 km rings				
0–1 km	-0.151** (0.065)	0.209*** (0.061)	-0.360*** (0.124)	-0.131 (0.187)
1–2 km	-0.097* (0.051)	0.161*** (0.051)	-0.258** (0.101)	0.039 (0.159)
2–3 km	-0.033 (0.042)	0.039 (0.042)	-0.072 (0.083)	-0.055 (0.138)
3–4 km	-0.015 (0.039)	0.020 (0.036)	-0.034 (0.074)	-0.023 (0.144)
4–5 km	0.042 (0.034)	-0.058* (0.033)	0.099 (0.067)	0.041 (0.130)
5–6 km	0.002 (0.028)	-0.024 (0.026)	0.026 (0.052)	0.083 (0.096)
6–7 km	0.003 (0.038)	-0.002 (0.037)	0.005 (0.073)	0.094 (0.118)
7–8 km	0.043 (0.029)	-0.047* (0.026)	0.090* (0.054)	0.089 (0.118)
8–9 km	-0.014 (0.033)	0.017 (0.032)	-0.031 (0.064)	-0.130 (0.120)
9–10 km	0.020 (0.026)	-0.017 (0.026)	0.037 (0.052)	-0.049 (0.120)
Joint test over the ring family				
0–2 km, F	2.90	7.02	4.76	0.32
p	0.0564	0.0010	0.0092	0.7293
0–5 km, F	1.85	4.53	3.09	0.20
p	0.1028	0.0005	0.0096	0.9621
0–10 km, F	1.64	3.37	2.45	0.54
p	0.0950	0.0003	0.0078	0.8626
Pre-period rings	Yes	Yes	Yes	Yes
Venue-traffic controls	Yes	Yes	Yes	Yes
Neighborhood mobility controls	Yes	Yes	Yes	Yes
Panel-size controls	Yes	Yes	Yes	Yes
Demographics	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes
Observations	133,406	133,406	133,406	122,116
R^2	0.110	0.091	0.101	0.005

Notes: The table reports the regression that carries both of the paper's treatments at once, so each is estimated holding the other fixed. The specification is equation (5) of Section 6.1. The four columns are the change in the Democratic share of all votes cast for president between the 2016 and 2020 elections; the change in the Republican share on the same denominator; the net margin, the first minus the second; and the change in turnout, votes cast per voting-age person. All four are in percentage points. D_i is the participation extensive margin and equals one when at least one resident of the block group attended a George Floyd protest. $(L_i - c)\,D_i$ is the participation intensive margin: the log of attendances per homed device minus c for block groups that sent anyone, and zero for those that sent nobody, as in Section 5.1, with c the mean of that log among the sending block groups of the joint sample. D_{ir} equals one when at least one George Floyd protest had its venue in ring r, one kilometer wide, and P_{ir} is the same indicator for the protests of January 2017 to May 2020, the pre-period rings. A_i, M_i and S_i are the venue-traffic, neighborhood mobility and panel-size controls of the rows near the foot, and X_i the demographics row, all as in Table A18; $\alpha_{z(i)}$ are zip code fixed effects. The regression is weighted by votes cast in 2020. The sample is the 133,406 block groups on which both treatments are observed together, 122,116 for turnout, the block groups whose turnout does not exceed 100 percent in 2016 or 2020, with the 2016 voting-age population from the 2012–2016 American Community Survey. Standard errors in parentheses, clustered by metropolitan area; 333 clusters. *** p < 0.01, ** p < 0.05, * p < 0.10.

Table A34: Participation and exposure in one regression, split by protest violence

	(1) Δ Dem (pp)	(2) Δ Rep (pp)	(3) Net margin (Dem − Rep, pp)	(4) Δ Turnout (pp)
Participation (device attendance, two-part, split by venue violence)				
Violent: Ext. margin	-0.015 (0.032)	0.001 (0.032)	-0.016 (0.063)	0.148 (0.150)
Violent: Int. margin (log rate)	-0.087*** (0.030)	0.072** (0.029)	-0.159*** (0.058)	0.144 (0.141)
Nonviolent: Ext. margin	0.074*** (0.018)	-0.072*** (0.017)	0.146*** (0.034)	-0.093 (0.094)
Nonviolent: Int. margin (log rate)	0.016 (0.014)	-0.014 (0.014)	0.030 (0.027)	0.024 (0.074)
Exposure: violent protest, 1 km rings (overlapping sets; not a decomposition)				
0–1 km	-0.209 (0.144)	0.289** (0.134)	-0.498* (0.269)	-0.697 (0.426)
1–2 km	-0.165 (0.124)	0.243** (0.118)	-0.409* (0.239)	-0.285 (0.388)
2–3 km	-0.132 (0.093)	0.163** (0.080)	-0.295* (0.168)	-0.078 (0.255)
3–4 km	-0.127 (0.088)	0.137 (0.084)	-0.264 (0.169)	-0.367* (0.221)
4–5 km	-0.133* (0.072)	0.148** (0.071)	-0.280** (0.140)	-0.454** (0.198)
5–6 km	-0.102* (0.062)	0.101* (0.059)	-0.202* (0.119)	-0.223 (0.210)
6–7 km	-0.078 (0.068)	0.088 (0.060)	-0.165 (0.126)	-0.244 (0.165)
7–8 km	-0.069 (0.054)	0.037 (0.049)	-0.106 (0.101)	0.038 (0.161)
8–9 km	-0.009 (0.065)	-0.020 (0.063)	0.011 (0.127)	-0.060 (0.234)
9–10 km	-0.013 (0.059)	-0.008 (0.057)	-0.005 (0.115)	0.118 (0.185)
Exposure: nonviolent protest, 1 km rings (overlapping sets; not a decomposition)				
0–1 km	-0.095 (0.066)	0.149** (0.060)	-0.244* (0.125)	0.068 (0.199)
1–2 km	-0.068 (0.051)	0.125** (0.049)	-0.194** (0.098)	0.145 (0.142)
2–3 km	0.000 (0.041)	0.009 (0.041)	-0.009 (0.081)	0.036 (0.144)
3–4 km	0.030 (0.038)	-0.023 (0.035)	0.053 (0.072)	0.107 (0.144)
4–5 km	0.059* (0.032)	-0.074** (0.030)	0.133** (0.061)	0.107 (0.117)
5–6 km	0.029 (0.029)	-0.049* (0.028)	0.077 (0.056)	0.142 (0.093)
6–7 km	0.018 (0.034)	-0.021 (0.032)	0.040 (0.065)	0.098 (0.139)
7–8 km	0.059** (0.030)	-0.058** (0.026)	0.117** (0.055)	0.108 (0.127)
8–9 km	-0.014 (0.035)	0.017 (0.033)	-0.031 (0.067)	-0.101 (0.124)
9–10 km	0.024 (0.028)	-0.017 (0.028)	0.040 (0.055)	-0.062 (0.120)
Pre-period rings (split)	Yes	Yes	Yes	Yes
Venue-traffic controls	Yes	Yes	Yes	Yes
Neighborhood mobility controls	Yes	Yes	Yes	Yes
Panel-size controls	Yes	Yes	Yes	Yes
Demographics	Yes	Yes	Yes	Yes
Zip code fixed effects	Yes	Yes	Yes	Yes
Observations	133,406	133,406	133,406	122,116
R^2	0.111	0.092	0.102	0.005

Notes: The table repeats the regression of Table A33 with both treatments split by whether the protest was violent. Every other element of that specification is unchanged, on the same joint sample of 133,406 block groups, 122,116 for turnout, on the turnout definition of Table A33. Participation enters as four terms rather than two, an extensive margin and an intensive margin for violent protests and an extensive margin and an intensive margin for nonviolent ones, so a block group whose residents attended both kinds carries both; the two intensive margins are centered at the same constant c as in Table A33. Ext. margin = extensive margin; Int. margin = intensive margin. Exposure enters as two families of ten one-kilometer ring indicators, one for violent protests and one for nonviolent ones. The two ring families are overlapping sets and not a decomposition: a block group with a violent and a nonviolent protest in the same ring appears in both families there. The same holds of the participation split. The earlier-protest ring indicators are split by protest violence as the exposure rings are; the other controls are unsplit, and the venue-traffic controls carry no violence split. Standard errors in parentheses, clustered by metropolitan area; 333 clusters. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. On the net-margin column the two participation extensive margins differ between violent and nonviolent protests ($p = 0.046$) and so do the two intensive margins ($p = 0.005$), while the two exposure ring families are not statistically distinguishable, jointly over all ten rings ($F(10, 332) = 1.41$, $p = 0.174$) or at the first ring alone ($p = 0.399$), so the pattern in the exposure rows is a pattern in point estimates.

F.2 The average effects and the persuasion rates

Table A35 gives the two average effects of Section 6.2 and the persuasion rates that Figure 9 draws for this paper.

Table A35: Average effects and persuasion rates of exposure and participation

Panel A. Average effects on the treated					
Exposure, violent protest	Δ Republican vote share, 2016→2020 (pp)	0.623 pp toward Republicans [0.133, 1.112]			
Participation	Δ Democratic vote share, 2016→2020 (pp)	0.091 pp toward Democrats [0.058, 0.124]			
Panel B. Persuasion rates		(1) $y_T - y_C$ (pp)	(2) y_0	(3) $e_T - e_C$	(4) f (%)
Exposure, any protest	Reach assumed	0.193	0.230	1.00	0.25
		(0.060)			(0.08)
	Reach assumed			0.50	0.50
					(0.16)
	Reach assumed			0.25	1.00
					(0.31)
Exposure, violent protest	Reach assumed	0.336	0.183	1.00	0.41
		(0.135)			(0.17)
	Reach assumed			0.50	0.82
					(0.33)
	Reach assumed			0.25	1.65
					(0.66)
Participation	Measured, per voting-age person, coverage-scaled, $s = 0.04$	0.073	0.538	0.0723	2.19
		(0.018)			(0.53)
	Measured, per voting-age person, coverage-scaled, $s = 0.19$			0.0152	10.39
					(2.53)
	Reach assumed			1.00	0.16
					(0.04)

Notes: Panel A gives the two average effects of Section 6.2, with their 95 percent confidence intervals in brackets. Each averages the coefficients of the violence-split joint regression of Table A34 over the block groups that received the treatment. That regression has $N = 133{,}406$, with standard errors clustered by metropolitan area and 333 clusters. The averages are not the coefficients in column (1) of Panel B and do not enter the rates. Panel B gives the persuasion rates that Figure 9 draws for this paper (DellaVigna and Gentzkow, 2010): $f = (y_T - y_C)/[(e_T - e_C)(1 - y_0)]$, in percent with $y_T - y_C$ in percentage points. Column (1), $y_T - y_C$, is the effect on the vote share: for exposure, the 0–1 km coefficient of Table A8 (any protest) or of Table A9 (violent protest), on the change in the Republican share; for participation, the extensive margin of Table A18, on the change in the Democratic share. Column (3), $e_T - e_C$, is the reach, the share of the treated group the treatment reached. For exposure it is assumed: one, every resident within one kilometer, which gives the lower bound on the rate; one half; and one quarter. For participation it is measured: attendances per voting-age resident of the sending block groups, divided by the share of a protest's reported crowd that the device panel detects, written s in the table, at the median of that share over the 4,211 events with a reported crowd size, $s = 0.04$, and at its 75th percentile, $s = 0.19$; the last participation row instead assumes a reach of one, every resident of the sending block groups, and gives the lower bound on the participation rate. The effect, y_0 and their standard error are those of a block's first row and enter every row of the block. Standard errors in parentheses: beneath column (1), the coefficient's own, clustered by metropolitan area, on the samples of the three tables named; beneath column (4), that standard error carried through the formula with y_0 and the reach held fixed.

F.3 The aggregate decomposition

Table A36 is the decomposition behind Figure 10; Table A37 draws the exposure rings 2.5 km wide (Section 6.3).

Table A36: The aggregate decomposition of the net margin, 2016–2020

	Dependent variable: Net margin (Dem − Rep), 2016–2020		
	pp	Votes	95% CI (votes)
Total	0.098	104,428	[-101,587, 310,444]
Participation	0.129	137,320	[89,379, 185,260]
violent	0.036	38,156	[14,581, 61,731]
nonviolent	0.093	99,164	[53,757, 144,571]
Exposure	-0.031	-32,891	[-225,455, 159,672]
0–3 km	-0.081	-86,868	[-164,841, -8,896]
3–10 km	0.051	53,977	[-104,766, 212,719]
violent	-0.100	-106,825	[-217,157, 3,507]
0–3 km	-0.033	-35,204	[-70,456, 48]
3–10 km	-0.067	-71,622	[-166,370, 23,126]
nonviolent	0.069	73,934	[-95,863, 243,731]
0–3 km	-0.048	-51,664	[-121,905, 18,576]
3–10 km	0.118	125,598	[-21,793, 272,990]
Participation + exposure 0–3 km	0.047	50,451	[-44,528, 145,431]
Net of the mean under random placement, exposure terms only (†)			
Total†	0.191	203,790	[-2,225, 409,805]
Participation	0.129	137,320	[89,379, 185,260]
violent	0.036	38,156	[14,581, 61,731]
nonviolent	0.093	99,164	[53,757, 144,571]
Exposure†	0.062	66,470	[-126,093, 259,034]
0–3 km†	-0.027	-29,249	[-107,222, 48,723]
3–10 km†	0.090	95,720	[-63,023, 254,462]
violent†	-0.041	-44,002	[-154,334, 66,330]
0–3 km†	-0.017	-18,149	[-53,401, 17,103]
3–10 km†	-0.024	-25,852	[-120,601, 68,896]
nonviolent†	0.104	110,472	[-59,325, 280,269]
0–3 km†	-0.010	-11,100	[-81,341, 59,141]
3–10 km†	0.114	121,572	[-25,819, 268,964]
Participation + exposure 0–3 km†	0.101	108,070	[13,091, 203,050]
Observations (block groups)		133,406	

Notes: Each row is a contribution to the aggregate, reported in percentage points of the net margin, in votes, and with a 95 percent confidence interval on the votes. The aggregate is built by applying the estimated coefficients of the joint regressions, with the controls and zip code fixed effects of Figure 8, to the treated block groups of the estimation sample and summing the implied vote shift; the percentage-point column is the vote total divided by the votes cast in that electorate and is a mean, not a sum. Exposure is at least one protest with its venue within the stated distance band of the block group. Exposure enters as ten rings one kilometer wide, and the rows report them grouped into a near band, zero to three kilometers, and an outer band, three to ten. The indented rows under each treatment split it by protest violence, and under the violence split by band as well; the rows within a group add to the group and the two treatments add to the total. The quantity is a weighted combination of the fit's own coefficients; its standard error is computed from the fit's own cluster-robust covariance matrix, clustered by metropolitan area, over the whole block of treatment terms, so that the covariances between those terms enter and not only their variances; and the interval is the estimate plus and minus a t critical value on the fit's own clusters less one, 332 degrees of freedom on 333 metropolitan-area clusters. Each is a 95 percent interval for its own row, with no adjustment for the number of rows. The violent participation row is the sum of the fitted contributions of the 24,062 block groups where participants in a violent protest lived, each the sum of its extensive-margin and intensive-margin terms with the intensive margin centered at the pooled constant $c = -3.340$. The sum does not depend on the centering: with each family's intensive margin centered at its own mean, the row is unchanged to the vote. It is positive because the slope on the intensive margin is negative and most violent-sending block groups lie below c: the 18,851 block groups below c carry 41,640 votes, 109 percent of the row, and the 5,211 at or above it carry −3,484. Replacing the log-linear form with ten indicators of the violent attendance rate leaves this row at about 38,000 votes and the signs at both ends, although the path between them is not smooth. In the rows marked † each exposure coefficient is net of its mean under random placement of the venues (Section 4.7); the participation rows repeat unchanged.

Table A37: The aggregate decomposition of the net margin, 2016–2020, with exposure rings 2.5 km wide

	Dependent variable: Net margin (Dem − Rep)		
	pp	Votes	95% CI (votes)
Total	0.155	165,892	[-32,973, 364,757]
Participation	0.126	134,900	[87,072, 182,728]
violent	0.036	38,190	[14,743, 61,637]
nonviolent	0.091	96,710	[51,539, 141,880]
Exposure	0.029	30,992	[-159,352, 221,337]
0–2.5 km	-0.049	-52,702	[-109,743, 4,339]
2.5–10 km	0.078	83,694	[-81,667, 249,056]
violent	-0.111	-118,228	[-248,063, 11,607]
0–2.5 km	-0.030	-32,272	[-63,070, -1,473]
2.5–10 km	-0.081	-85,956	[-199,953, 28,040]
nonviolent	0.140	149,220	[-6,923, 305,363]
0–2.5 km	-0.019	-20,431	[-64,947, 24,086]
2.5–10 km	0.159	169,651	[23,560, 315,742]
Participation + exposure 0–2.5 km	0.077	82,198	[5,010, 159,385]
Observations (block groups)		133,406	

Notes: The table repeats the aggregate decomposition of Table A36 with the exposure rings drawn differently. Here exposure enters as four rings two and a half kilometers wide and the near band is cut at two and a half kilometers; in Table A36 it enters as ten rings one kilometer wide and the near band is cut at three; the earlier-protest ring indicators are redrawn the same way, into four rings two and a half kilometers wide. Everything else is the same. Every interval is built as in Table A36, and each is a 95 percent interval for its own row, with no adjustment for the number of rows.

F.4 The swing-state exercise

Table A38 applies the national coefficients to the six closest states of 2020 that the sample covers (Section 6.3).

Table A38: The aggregate net shift in the swing states against their 2020 margins

	2020 margin (D − R)	Estimated shift (votes)	Participation (votes)	Exposure (votes)
AZ	10,457	1,718 (1,092)	1,911*** (355)	-194 (970)
GA	11,779	5,753*** (1,962)	5,411*** (1,039)	342 (1,459)
WI	20,682	210 (2,889)	2,908*** (516)	-2,697 (2,784)
PA	80,555	6,216 (5,517)	6,801*** (1,244)	-586 (5,160)
NC	-74,483	4,535* (2,390)	5,630*** (990)	-1,095 (2,030)
MI	154,188	4,773 (3,339)	4,117*** (772)	656 (3,093)
AZ+GA+WI	42,918	7,681 (5,704)	10,231*** (1,853)	-2,549 (5,097)
Net of the mean under random placement, exposure terms only (†)				
AZ†	10,457	2,612** (1,092)	1,911*** (355)	701 (970)
GA†	11,779	6,868*** (1,962)	5,411*** (1,039)	1,457 (1,459)
WI†	20,682	3,610 (2,889)	2,908*** (516)	703 (2,784)
PA†	80,555	10,917** (5,517)	6,801*** (1,244)	4,116 (5,160)
NC†	-74,483	6,697*** (2,390)	5,630*** (990)	1,066 (2,030)
MI†	154,188	7,209** (3,339)	4,117*** (772)	3,092 (3,093)
AZ+GA+WI†	42,918	13,091** (5,704)	10,231*** (1,853)	2,860 (5,097)
Observations (block groups, national fit)		133,406	133,406	133,406

Notes: Each row is one state, and the last row is the three closest states taken together. The columns are the state's certified 2020 margin in votes, the Democratic minus the Republican total in the Federal Election Commission's *Federal Elections 2020* (October 2022), the estimated net shift in votes, and the same shift split into the part coming from participation and the part coming from exposure. A positive shift is toward the Democrats. The shift is computed by applying the estimated coefficients to each treated block group in the state and summing, exactly as the national aggregate of Figure 10 is built, with the specification and controls listed in the note to Figure 8. The national coefficients of record are applied to each state's treated block groups, so the per-state split shows where the treated block groups are and how violent that state's protests were, not a state-specific response. The two contributions sum to the net shift by construction. In North Carolina the state's own participation estimate, −0.048 (0.086) in Table A40, is not distinguishable from zero; the row applies the national coefficients. In the rows marked † the exposure contribution, and so the shift, is net of its mean under random placement of the venues (Section 4.7). Standard errors in parentheses, in votes, from the delta method on the fit's cluster-robust covariance; $N = 133{,}406$ block groups in the national fit. Significance stars are from the aggregator's own t statistic on 332 degrees of freedom: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Nevada, one of the closest states in 2020, has no 2012 returns and is outside the estimation sample (Table A3). The vote columns add across states, because the block group sets are disjoint; the standard errors behind the stars are correlated across states and do not.

F.5 The violence scenarios

Table A39 reports the scenarios behind Figure 11 of Section 6.4.

Table A39: The aggregate net effect under three violence scenarios

	Dependent variable: Net margin (Dem − Rep), pp			
	Participation	Exposure	Total	Δ vs baseline
Baseline (observed violence)	0.129	−0.031	0.098	—
Violence at the typical rate (2.2%)	0.129	0.037	0.166	0.068 [−0.001, 0.137]
No violence	0.129	0.072	0.200	0.102 [−0.002, 0.207]
Net of the mean under random placement, exposure terms only (†)				
Baseline (observed violence)†	0.129	0.062	0.191	—
Violence at the typical rate (2.2%)†	0.129	0.093	0.222	0.031 [−0.038, 0.099]
No violence†	0.129	0.108	0.236	0.045 [−0.059, 0.150]
Observations (block groups)	133,406			

Notes: The table reports the aggregate net effect under three assumptions about how violent the protest wave was, and is the exercise Figure 11 traces continuously. Each row is one scenario and each column a contribution to the aggregate net margin in percentage points: the participation contribution, the exposure contribution, their total, and the difference between that total and the observed baseline. The specification is that of Table A36. The first row is what was observed. The second sets the violent share of protests to 2.2 percent, which is the share of protest events coded violent in the three years before the wave, 826 of 36,974 events. The third sets it to zero. The counterfactual is executed by relabeling violent protest days as nonviolent at random until the target share is reached, in 500 seeded draws, and recomputing the distance-ring exposures under the new labels. The participation contribution is held at its observed level in every row. The difference against the baseline is its own weighted combination of the fit's coefficients, so its standard error is that combination's own and not a difference of two separately computed errors; otherwise every interval is built as in Table A36. One source of variation sits outside the interval: which events are relabeled is drawn 500 times, and the spread of the exposure contribution across those draws, 0.002 percentage points, is a separate quantity and is not folded in. In the rows marked † the exposure contribution, and so the total and the difference, is net of its mean under random placement of the venues (Section 4.7).

G North Carolina

This appendix reports the North Carolina estimates of Section 7 in full. The voter files and the measures built from them are described in Section 2.4 and Appendix A.4.

G.1 The paper's specifications in North Carolina

Section 7 reports whether the national results appear in the state; this subsection carries the detail: Table A40 runs the paper's three specifications on urban North Carolina beside their national estimates, and Table A41 estimates each channel under each specification. The North Carolina sample is the state's block groups of the national samples, 3,655 in the exposure specification and 3,649 in the participation and joint specifications (Section 2.4); it is neither the national sample nor a random sample of the state, and block groups outside it are unobserved rather than untreated. The shift toward the Republicans near a violent protest within one kilometer, 0.99 percentage points in North Carolina against 0.34 nationally, is a point estimate about three times the national one on a much wider interval; that the North Carolina estimate is the larger is a comparison of two point estimates from two specifications, the North Carolina one also holding nonviolent protests fixed; dropping any single state from the national sample, North Carolina included, leaves the national violent estimate inside its confidence interval (Section 4.7). The 50 block groups with a violent

protest within one kilometer lie in 9 of the 44 counties. Dropping any one county keeps the sign of every violent estimate in Table A40 and leaves the 0.99 between 0.69 and 1.28. Within the upper half of the sample's distribution of residential turnover over 2012 to 2016, where all 50 lie, the estimate is 0.85 (0.53), not distinguishable from zero on 33 county clusters.

Table A40: Exposure, participation and the joint specification in North Carolina and nationally

	Urban North Carolina	National
Exposure specification: Δ Republican share (pp); national column is the exposure fit, N 136,644		
Violent 0–1 km	0.99* (0.47)	0.34** (0.14)
Violent 1–2 km	0.22 (0.33)	0.27** (0.12)
Violent 2–3 km	0.51 (0.31)	0.16** (0.08)
Violent 3–4 km	0.52** (0.21)	0.10 (0.08)
Participation specification: Δ Democratic share (pp); national column is the participation fit, N 135,691		
Attendance (extensive margin)	−0.048 (0.086)	0.073*** (0.018)
Attendance (intensive margin)	0.068 (0.051)	0.014 (0.014)
Joint specification: Δ net margin (pp); national column is the pooled joint fit, N 133,406		
Violent 0–1 km	−2.074* (0.903)	–
Attendance (extensive margin)	−0.097 (0.168)	0.146*** (0.034)
Attendance (intensive margin)	0.198* (0.104)	0.037 (0.026)
Block groups: exposure	3,655	136,644
Block groups: participation	3,649	135,691
Block groups: joint	3,649	133,406

Notes: The exposure specification's outcome is the change in the Republican share of all votes cast for president between 2016 and 2020, in percentage points. Its treatment differs between the two columns: in the North Carolina column it is twenty one-kilometer ring indicators, ten for a violent and ten for a nonviolent George Floyd protest, entered jointly, of which the first four violent rings are printed; in the national column it is the ten violent-protest ring indicators of Figure 3, of which the same four are printed. The participation specification's outcome is the change in the Democratic share of all votes cast, and its treatment the participation extensive and intensive margins of Table A18. The joint specification carries both treatments at once, with the change in the net margin as its outcome, the Democratic share change minus the Republican share change, so a negative number is a shift toward the Republicans; a vote that passes from one party to the other moves the net margin by about twice the share, so its coefficients run at about twice the scale of the two share outcomes. All three carry the demographic controls and zip code fixed effects; the participation and joint specifications also carry the venue-traffic, mobility and panel-size controls of Table A18; the exposure and joint specifications also carry one-kilometer ring indicators for earlier protests: in the North Carolina column twenty of them, ten for earlier violent and ten for earlier nonviolent protests; in the national column ten, which count violent earlier protests in the exposure specification and any earlier protest in the joint specification. Weights are votes cast in 2020. Each block's second column carries that block's own national estimate, from three national fits and not one. A dash marks a term the national pooled joint fit does not contain; split by protest violence (Table A34), the national violent 0–1 km coefficient is −0.498 (0.269). Standard errors in parentheses are cluster-robust throughout, clustered by North Carolina county in the first column, 44 clusters, and by metropolitan area in the second, 344 clusters for the exposure specification and 333 for the participation and joint specifications. Stars come from each column's own inference of record: for North Carolina the wild cluster bootstrap of Section 7, with Rademacher weights and 9,999 draws, and for the national column the asymptotic test. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A41: North Carolina channel estimates in the joint and the single-treatment specifications

	Joint	Exposure	Participation
Turnout (percentage points)			
Nonviolent 0–1 km	−1.51*** (0.36)	−1.54*** (0.39)	–
Violent 0–1 km	0.34 (1.21)	0.27 (1.21)	–
Attendance (extensive margin)	−0.02 (0.11)	–	−0.08 (0.11)
Democratic registration share (percentage points)			
Nonviolent 0–1 km	−0.09 (0.42)	−0.01 (0.42)	–
Violent 0–1 km	−1.00* (0.48)	−0.93 (0.50)	–
Attendance (extensive margin)	−0.11 (0.10)	–	−0.11 (0.10)
Net Democratic→Republican switching (per 1,000 of the baseline cohort)			
Nonviolent 0–1 km	−0.818 (0.558)	−0.919 (0.563)	–
Violent 0–1 km	−0.319 (0.915)	−0.506 (0.892)	–
Attendance (extensive margin)	0.369* (0.214)	–	0.369* (0.206)
Democratic new registration (per unit of the 2016 registered base)			
Nonviolent 0–1 km	−0.001 (0.004)	0.000 (0.004)	–
Violent 0–1 km	0.042*** (0.012)	0.043*** (0.013)	–
Attendance (extensive margin)	0.002** (0.001)	–	0.002 (0.001)
Block groups	3,649	3,655	3,649

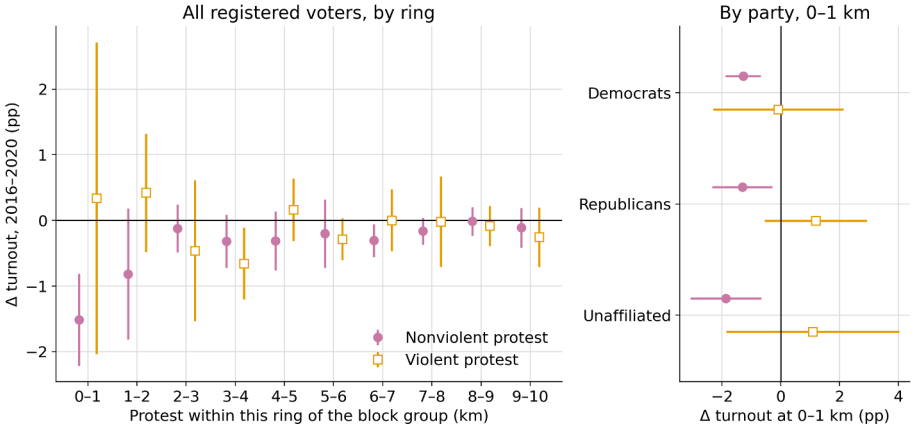
Notes: This table estimates each channel three times: once in the joint specification, once in the exposure specification alone on that design's own sample and control set, and once in the participation specification alone on its own. A dash marks a cell the specification does not contain. Turnout is the change between the 2016 and 2020 general elections in the share of the people on a block group's roll at both elections who cast a ballot, in percentage points, as defined in Section 2.4. Registrants with an inactive status count as on the roll; the state marks a registrant inactive after a period without voting, so that status is partly defined by not voting. The Democratic registration share is the change in the Democratic share of its registered voters, in percentage points. Net Democratic-to-Republican switching counts people already registered at both ends who changed the party they are registered with, as the number who moved from Democratic to Republican minus the number who moved the other way, per thousand of the block group's baseline cohort. Democratic new registration counts people who appear on the roll in 2020 and not on the 2016 roll anywhere in the state and are registered as Democrats, per unit of the 2016 registered base. The three treatments named in the rows are an indicator for at least one nonviolent George Floyd protest within one kilometer of the block group, the same for a violent protest, and the participation extensive margin of Table A18. The participation intensive margin is carried in the joint and the participation columns, beside the extensive margin, and is not reported. All three specifications carry zip code fixed effects, and each carries its own control battery: the joint specification the venue-traffic controls, the device-decile indicators, the demographic controls and twenty one-kilometer ring indicators for earlier protests, ten for violent and ten for nonviolent ones; the exposure specification the demographic controls and the same twenty earlier-protest rings; the participation specification its own venue-traffic controls, the four ranging measures and the device-decile indicators on the same demographic block. Each regression is weighted by its outcome's own baseline denominator. The joint and participation columns are 3,649 block groups and the exposure column 3,655. Standard errors in parentheses are cluster-robust, clustered by North Carolina county, 44 clusters in every column; stars come from the wild cluster bootstrap of Section 7, with Rademacher weights and 9,999 draws. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

G.2 The turnout channel

Figure A9 shows a decline in turnout within one kilometer of nonviolent protests and no detectable change within one kilometer of violent ones, and the joint test that the two families of rings are equal, an asymptotic cluster-robust test with no bootstrap counterpart, $F(10, 43) = 1.02$, $p = 0.44$, does not establish the difference at the 5 percent level. The nonviolent coefficients are negative at every ring out to ten kilometers, though beyond the first ring they are smaller and only one is distinguishable from zero. Registrants who left the state but stay on its roll count as not voting, and most block groups near nonviolent protests have high residential turnover (Figure A11), so part of the fall may be moving away rather than staying home. Its right panel repeats the estimate

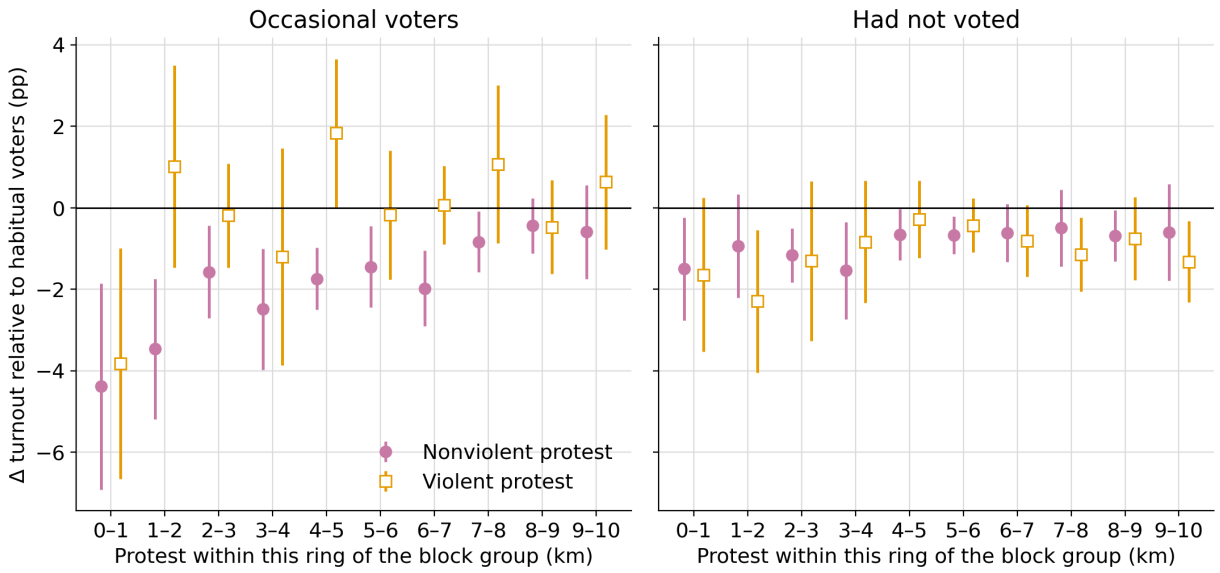
within one kilometer for registered Democrats, Republicans and unaffiliated voters separately: near nonviolent protests turnout fell for each of the three, each with an interval below zero, and near violent protests none of the three intervals excludes zero; the three are separate regressions, so the larger fall among the unaffiliated is a difference of point estimates. Figure A10 sorts the same people by their voting history and compares each class's change in turnout with the habitual voters' of the same block group (Section 7).

Figure A9: Turnout near violent and nonviolent protests in North Carolina, by distance and by party



Notes: The outcome in both panels is the change between the 2016 and 2020 general elections in the share of the people on the block group's roll at both elections who cast a ballot, in percentage points, as defined in Section 2.4. Each marker is an estimated coefficient and each whisker an asymptotic 95 percent confidence interval. The left panel gives the effect for all registered voters together, by distance. Its treatment is two sets of ten indicators, one set for violent and one for nonviolent protests, in one-kilometer rings. The right panel repeats the within-one-kilometer estimate on separate regressions for registered Democrats, Republicans and unaffiliated voters. The specification is otherwise the one used throughout the section: the demographic, venue-traffic and device-decile controls and the twenty earlier-protest rings matched to the family, as in the joint column of Table A40 but without its mobility measures, with zip code fixed effects, together with the two-part participation treatment. Each regression is weighted by the number of people in its own baseline group. The sample in the left panel is 3,649 block groups of urban North Carolina in 325 zip-code cells; in the right panel each party's regression runs on the block groups with registrants of that party, 3,649 for Democrats and 3,648 each for Republicans and the unaffiliated. Standard errors are clustered by North Carolina county, 44 clusters, with a wild cluster bootstrap on those clusters, Rademacher weights and 9,999 draws, as the inference of record.

Figure A10: Turnout of occasional voters and of registrants who had not voted, against habitual voters, by distance

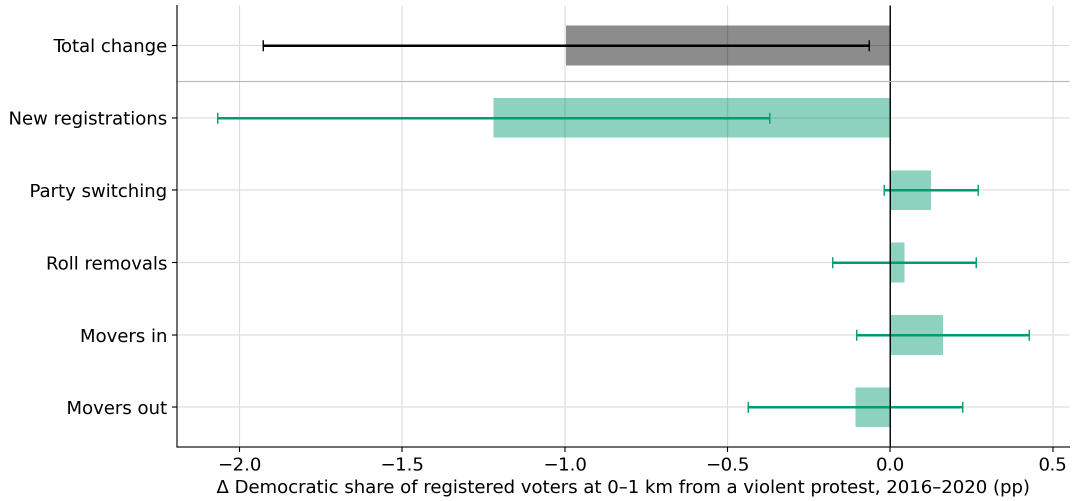


Notes: The unit here is a person and not a block group: the estimation uses the 3,606,754 people registered at both the 2016 and the 2020 snapshots and placed in one of the 3,649 block groups in 2016. The outcome is whether that person voted in the 2020 general election minus whether they voted in 2016. Each person is sorted into one of the three classes of Section 7 on their record before the protests (Appendix A.4). Each panel plots one group against the habitual voters, by distance: the left panel occasional voters, the right panel registrants who had not voted. The markers are estimated coefficients on the interaction of the group indicator with each one-kilometer ring indicator, filled circles for nonviolent and open squares for violent protests, and the whiskers are asymptotic 95 percent confidence intervals. The regression carries block group fixed effects, together with the group indicators themselves and the participation extensive margin interacted with group. Standard errors are clustered by North Carolina county, 44 clusters, with a wild cluster bootstrap on those clusters, Rademacher weights and 9,999 draws, as the inference of record, so the violent estimate at 0–1 km in the left panel is significant only at the 10 percent level although its whisker clears zero. Whether a protest was nearby is a property of the block group, not of the person, so its own effect is absorbed by the block group fixed effect: every coefficient plotted is how much further one group's turnout moved than the habitual voters' in the same block group.

G.3 The registration channel

In the five-movement split of Figure A11, party switching runs the other way from new registration and adds 0.13 percentage points, and removals and moves in and out are each not distinguishable from zero.

Figure A11: The change in the Democratic registration share near violent protests, by component



Notes: The outcome is the change between the 2016 and 2020 general elections in the Democratic share of the block group's registered voters, in percentage points, and the figure reports it for block groups with a violent George Floyd protest within one kilometer. The top bar is that total change. The five bars below it are the five movements of Appendix A.4 each written as its own contribution to the change in the Democratic share over a common denominator, the block group's 2020 registered base, so that the five sum to the total exactly. Each bar is a separate regression of that component on the specification of the section: the demographic, venue-traffic and device-decile controls and the twenty earlier-protest rings matched to the family, as in the joint column of Table A40 but without its mobility measures, with zip code fixed effects, each weighted by the block group's 2016 registered base. The sample is 3,649 block groups of urban North Carolina in 325 zip-code cells. Whiskers are asymptotic 95 percent confidence intervals; standard errors are clustered by North Carolina county, 44 clusters, and the inference of record is a wild cluster bootstrap on those clusters with Rademacher weights and 9,999 draws. Every one of the 50 block groups with a violent protest within one kilometer is in the upper half of the sample's distribution of residential turnover measured over 2012 to 2016, and none is in the lower half, while of the 151 block groups with a nonviolent protest within one kilometer, 136 are in the upper half and 15 in the lower.

G.4 The composition placebo

In the composition placebo of Table A42, at two to four kilometers the check is not clean: moves out, and at one band each new registrations and moves in, load on the 2020 treatment, though the total row is not distinguishable from zero at either distance, so whatever the 2016-to-2020 estimates pick up at those distances was already under way in the earlier window, and they are not read as effects of the 2020 protests. This covers Table A40's 0.52 at three to four kilometers, the only violent ring beyond the first that is distinguishable from zero.

Table A42: Components of the North Carolina registration-share change over 2012–2016, on the 2020 violent-protest rings

Component (2012–2016 change)	Violent 0–1 km	Violent 2–3 km	Violent 3–4 km
Total Democratic share change	−0.20 (0.58)	−0.45 (0.21)	−0.63 (0.32)
New registrations	−0.42 (0.33)	−0.42** (0.18)	−0.26 (0.24)
Party switching	0.13 (0.07)	−0.04 (0.05)	−0.05 (0.05)
Roll removals	0.13 (0.14)	−0.05 (0.09)	0.04 (0.09)
Movers in	0.10 (0.32)	−0.11 (0.14)	−0.49** (0.10)
Movers out	−0.13 (0.17)	0.17* (0.08)	0.13* (0.06)

Notes: The rows are the total change in the Democratic share of a block group's registered voters over 2012 to 2016, in percentage points, and the five movements of Appendix A.4 that sum to it exactly. The columns are three of the violent-protest ring indicators of 2020, at zero to one, two to three and three to four kilometers. The specification is the one used throughout the section: the demographic, venue-traffic and device-decile controls and the twenty earlier-protest rings matched to the family, as in the joint column of Table A40 but without its mobility measures, with zip code fixed effects, weighted by the 2012 registered base. The sample is the one used throughout the section, 3,649 block groups of urban North Carolina; each component is one regression, the three printed rings are three terms of that regression, and the total row is its own regression on the same basis as the components and not their sum. Standard errors in parentheses are cluster-robust, clustered by North Carolina county, 44 clusters; stars come from the wild cluster bootstrap of Section 7, with Rademacher weights and 9,999 draws. At two to three and at three to four kilometers the total row's coefficient is about twice its asymptotic standard error, so its reading as not distinguishable from zero at either distance rests on the bootstrap. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.